A Systematic Quantitative Trading Framework: From Volatility Forecasting to Live Portfolio Optimisation

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Contents

1 Introduction

Quantitative finance applies mathematical and statistical methods to solve problems in financial markets. This report presents a comprehensive quantitative framework spanning six phases: research and analysis, live trading implementation, backtesting and parameter optimisation, advanced risk management, ethical investing comparison, and regime-switching analysis. The complete lifecycle from academic theory to live deployment is demonstrated.

The analysis begins with daily closing prices for Apple Inc. (AAPL) from Yahoo Finance for the period 1 January 2020 to 1 January 2025, yielding 1,257 observations. For portfolio optimisation, the dataset is extended to include Microsoft (MSFT), Google (GOOGL), Amazon (AMZN), Tesla (TSLA) and subsequently expanded to a diversified 18-asset universe spanning equities, ETFs, bonds and commodities.

Phase 1: Research and Analysis establishes the mathematical foundation. We employ time-series analysis (ARIMA and GARCH), linear algebra (portfolio optimisation) and stochastic calculus (Monte Carlo simulation) to model asset returns and value a European call option. Our empirical analysis finds strong evidence of non-normality (Jarque-Bera p-value = 0.000000), volatility clustering and significant persistence in market risk ($\alpha_1 + \beta_1 = 0.9726$), validating the use of GARCH models. The Efficient Frontier demonstrates the benefits of diversification, reducing portfolio volatility from 31.68% (AAPL alone) to 28.18% (minimum variance portfolio). Monte Carlo simulation, pricing a $250-strike European call option on AAPL with one year to maturity, produces a risk-neutral price of $9.95 using the GARCH volatility forecast, diverging substantially from the Black-Scholes benchmark of $33.61 (Black and Scholes, 1973). This difference highlights the critical distinction between real-world and risk-neutral pricing frameworks (Merton, 1973).

Phase 2: Live Trading Implementation extends the academic framework into a practical trading strategy. The system incorporates an 18-asset universe, rolling estimation windows, automatic GARCH model selection via AIC, conditional rebalancing (55-85 days), linear cash allocation based on GARCH volatility forecasts and take-profit rules (17.4% vs SPY). Transaction costs of 0.4% round-trip are modelled to reflect Trading 212's fee structure. A key enhancement in the live engine is the use of **previous-day closing prices** via a manual date shift, ensuring that all trading decisions are based exclusively on information available at the time of execution.

Phase 3: Backtesting and Parameter Optimisation validates the strategy through comprehensive backtesting and rigorous parameter optimisation. Over a 16-year period (2010-2026), the Active Portfolio achieves a final value of 852.68, while Total Wealth (including realised profits) reaches 1,268.46, outperforming the SPY benchmark which finishes at 878.82. The No Rules portfolio (without take-profit) achieves the highest final value of 2,498.42, demonstrating the significant upside potential when not constrained by profit-taking. The strategy delivers an Active return of 1,168.46% with a Sharpe ratio of 0.859 and a maximum drawdown of -22.72%. Bayesian optimisation with walk-forward validation (Snoek, Larochelle and Adams) identifies optimal parameters: 285-day lookback, 60-day minimum rebalance, 90-day maximum rebalance, 5.7% drift threshold and 15.0% take-profit. The robust parameter selection process confirms the strategy's stability across different market regimes.

Phase 4: Advanced Risk Management and Performance Attribution addresses optimal position sizing and performance attribution using the 2016-2026 validation period from Phase 3. This shorter period was selected because several assets in the universe exhibited immature or anomalous behaviour prior to 2016, with reliable price discovery and stable factor exposures only emerging as these companies matured and markets normalised post-GFC. A corrected Kelly implementation using Maximum Sharpe portfolio weights and sign change detection identifies 135 days as the optimal lookback. With Full Kelly (100%), the Original_18 portfolio's total return improves from 1,017.38% to 1,067.68% while the Sharpe ratio improves from 1.099 to 1.200 over this period. Fama-French 5-Factor analysis (Fama and French, 2015) reveals that the Original_18 portfolio generates 9.21% annualised alpha with strong statistical significance ($p = 0.0036$), confirming genuine skill rather than luck. The exceptionally low market beta of 0.194 means the portfolio is 80.6% less volatile than the market.

Phase 5: Ethical Investing Comparison extends the analysis to the full 16-year period (2010-2026), comparing three asset universes: Original_18 (benchmark), Optimal_Standard (growth-focused) and Optimal_Ethical (ethically-screened). Using the No Rules configuration (take-profit disabled) for consistency with Phase 4, Optimal_Ethical achieves the highest returns (6,502.1%) and holdout Sharpe ratio (1.152), outperforming Original_18 (2,342.3%, 0.668) and Optimal_Standard (5,219.1%, 0.836). Fama-French analysis confirms the highest alpha for Optimal_Ethical (12.25%, $p = 0.0001$) and positive RMW coefficients for both optimal portfolios, indicating the selection methodology identifies more profitable companies. The ethical portfolio aligns with investor values by excluding poor ESG companies while supporting environmentally responsible businesses, including international leaders such as VWS.CO (Vestas from Denmark). For the final live deployment, the same multi-factor scoring and greedy selection algorithm from Phase 5 was applied to an expanded ethical pool, confirming the Optimal_Ethical 15-asset portfolio as the optimal selection.

Phase 6: Regime-Switching Analysis and Overfitting Investigation extends the framework by exploring whether adapting strategy parameters to market regimes can improve performance. Using a Hidden Markov Model (HMM) (Hamilton, 1989) trained on SPY returns and volatility, we define three regimes (bull, bear, crash) and optimise regime-specific parameters (lookback, rebalance windows). Despite rigorous walk-forward validation, the regime-weighted strategy achieves a higher absolute return on the 2020-2026 holdout period (482.94% return) but underperforms on a risk-adjusted basis (Sharpe 1.245) versus the baseline's 456.09% return (Sharpe 1.275). Crucially, the regime strategy increases maximum drawdown from -21.75% to -27.67% and trades more frequently (147 vs 134), resulting in a lower Sharpe ratio. A controlled overfitting demonstration shows that parameters optimised on a single historical period (1993-2020) fail to generalise, while parameters optimised on the future period (which we could not have known) would have outperformed. This highlights the fundamental challenge of regime-switching: the optimal parameters for future regimes are not predictable from past regimes alone. Consequently, the HMM approach is not integrated into the final live engine, and we conclude that the fixed baseline strategy is more robust. We also note that an alternative exploration – forecasting GARCH volatility directly within the Max Sharpe portfolio optimisation – produced identical results and was therefore omitted from the final framework.

The Final Engine demonstrates the live deployment of the strategy using the Optimal_Ethical 15-asset portfolio with all optimised parameters. The system generates actionable Trading 212 orders and maintains a complete audit trail via automated logging. Consistent with the academic framework, the live engine applies the manual date-shift methodology to ensure all prices used are previous-day closes. A separate performance tracker visualises the live portfolio performance alongside benchmark comparisons, using the same manual date-shift methodology for consistent price alignment across all universes. The live performance tracker confirms the strategy's strong performance, with all three portfolios outperforming the SPY benchmark.

The key findings of this project are: (1) markets exhibit non-normal returns with fat tails and volatility clustering, justifying GARCH models; (2) diversification provides substantial risk reduction with exceptionally low market beta (0.194-0.210); (3) risk management through take-profit rules, Kelly position sizing and GARCH-based cash allocation significantly improves performance; (4) parameter optimisation through rigorous validation is critical; (5) ethical investing can enhance risk-adjusted returns, as demonstrated by the Optimal_Ethical portfolio's superior performance across all key metrics; (6) international diversification, as exemplified by VWS.CO, can add value while maintaining strong ESG credentials; and (7) regime-switching strategies, while theoretically appealing, are prone to overfitting and may not provide consistent improvement over a well-optimised fixed baseline.

2 Phase 1: Research and Analysis

2.1 Phase 1 Overview

The journey begins with a fundamental question: *Can we mathematically model financial markets to make better investment decisions?* Before building any trading system, we must first understand the statistical properties of asset returns, develop models for forecasting volatility and establish the theoretical foundations of portfolio construction and option pricing.

Phase 1 comprises the core quantitative analysis across four distinct areas, each building upon the last:

1. **Probability and Statistics (Section 1.1):** Characterising the distribution of asset returns to understand their behaviour and testing the assumption of normality.
2. **Time Series Analysis (Section 1.2):** Modelling volatility using ARIMA and GARCH to capture persistence and clustering.
3. **Linear Algebra and Portfolio Risk (Section 1.3):** Constructing optimal portfolios using covariance matrices and the Efficient Frontier (Markowitz, 1952).
4. **Stochastic Calculus and Option Pricing (Section 1.4):** Pricing derivatives using Monte Carlo simulation with Geometric Brownian Motion.

All results in Phase 1 are based on historical data from 2020 to 2025, providing a solid empirical foundation for the subsequent phases.

2.2 Phase 1.1: Probability and Statistics

2.2.1 Motivation

Before we can model financial markets, we must understand the nature of the data we are working with. Financial returns are famously non-normalthey exhibit fat tails, skewness and volatility clustering. This section characterises these properties quantitatively, testing the assumption of normality that underlies many classical financial models. The findings here will justify our choice of more sophisticated models in later sections.

2.2.2 Methodology

Let P_t denote the closing price of an asset at time t. We define the daily return r_t as the log-return:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log-returns are preferred over simple percentage returns because they are approximately normally distributed (under the classical assumption) and because they are time-additive, making them more convenient for multi-period analysis.

We computed the following sample statistics to characterise the return distribution:

$$\bar{r} = \frac{1}{N}\sum_{t=1}^{N} r_t$$

$$s^2 = \frac{1}{N-1}\sum_{t=1}^{N}(r_t - \bar{r})^2$$

$$\text{Skewness} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{r_t - \bar{r}}{s}\right)^3$$

$$\text{Excess Kurtosis} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{r_t - \bar{r}}{s}\right)^4 - 3$$

Skewness measures the asymmetry of the distribution. A positive skewness indicates a heavier right tail, meaning extreme positive returns occur more frequently than extreme negative returns. Excess kurtosis measures the "tailedness" of the distributionpositive excess kurtosis indicates fatter tails than a normal distribution, meaning extreme events occur more frequently than predicted by the normal distribution.

We applied the Jarque-Bera test to formally assess normality:

$$\text{JB} = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right)$$

Under the null hypothesis of normality, the Jarque-Bera statistic follows a chi-squared distribution with 2 degrees of freedom. A small p-value leads us to reject the null hypothesis, confirming that the returns are not normally distributed.

2.2.3 Data Collection

We downloaded Apple Inc. (AAPL) daily closing prices from Yahoo Finance for the period 1 January 2020 to 1 January 2025, yielding 1,257 observations. Figure 1 shows the price trajectory over this period.

Figure 1: Apple Stock Price (2020-2025)

The price series exhibits the characteristic upward trend of a growth stock, with periods of significant volatility. The COVID-19 crash in March 2020 and the subsequent recovery are clearly visible, as is the 2022 drawdown. The price increased from approximately $75 in early 2020 to over $250 by the end of 2024, representing substantial capital growth.

Figure 2 shows the daily returns over the same period.

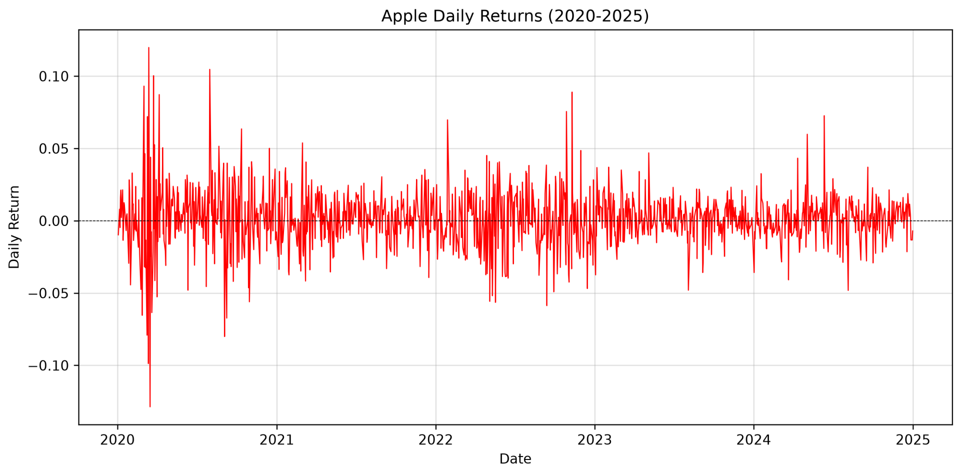

Figure 2: Apple Daily Returns (2020-2025)

The returns series exhibits volatility clusteringperiods of high volatility (such as the COVID-19 period in early 2020 and the 2022 bear market) followed by periods of relative calm. This is a hallmark of financial time series and justifies the use of GARCH models for volatility forecasting (Bollerslev, 1986). The largest negative returns occurred during the COVID-19 crash, while the largest positive returns occurred during the subsequent recovery.

Figure 3 shows the distribution of daily returns alongside a normal distribution for comparison.

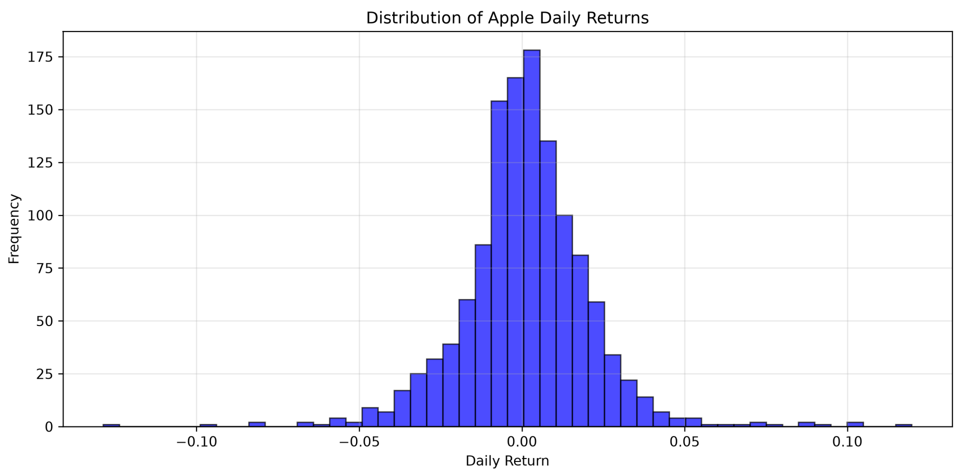

Figure 3: Distribution of Apple Daily Returns

The histogram reveals several important features:

1. The distribution is peaked at the centre, with more observations near zero than the normal distribution would predict.
2. The tails are fatter than the normal distribution, indicating a higher probability of extreme events.
3. The distribution appears slightly asymmetric, with a slightly longer right tail.

2.2.4 Results

The summary statistics quantify these observations:

Table 1: Summary Statistics for AAPL Daily Returns (2020-2025)

Statistic	**Value**
Mean ($\bar{r}$)	0.001182
Standard Deviation (s)	0.019956
Skewness	0.105500
Excess Kurtosis	5.252107
Annualised Return	29.79%
Annualised Volatility	31.68%
Sharpe Ratio	0.798
Jarque-Bera Statistic	1447.07
Jarque-Bera p-value	0.000000

2.2.5 Interpretation

Mean Return (0.001182): The average daily return is positive, translating to an annualised return of 29.79%. This reflects Apple's strong performance during the 2020-2025 period, driven by continued growth in its services and products businesses.

Standard Deviation (0.019956): The daily standard deviation translates to an annualised volatility of 31.68%. This is relatively high compared to the broader market (the S&P 500 typically has volatility around 15-20%), reflecting Apple's status as a growth stock with higher risk. Investors demanding exposure to Apple's growth potential must accept this elevated volatility.

Skewness (0.1055): The positive skewness indicates a heavier right tail, meaning extreme positive returns occur more frequently than extreme negative returns. This is somewhat unusual for equity returns, which typically exhibit negative skewness (crash risk). However, Apple's strong performance during the 2020-2025 period, driven by growth in services and products, may explain this positive skew. The positive skewness suggests that there is a greater probability of large upward moves than large downward moves.

Excess Kurtosis (5.2521): The positive excess kurtosis confirms that the return distribution has significantly fatter tails than a normal distribution. This means that extreme eventsboth positive and negativeoccur far more frequently than a normal distribution would predict. For a normal distribution, excess kurtosis is zero; a value of 5.2521 indicates a substantial departure from normality. A kurtosis of this magnitude means that the probability of a 3-standard-deviation event is many times higher than what the normal distribution would suggest.

Jarque-Bera Test: The test yielded a statistic of 1447.07 with a p-value of 0.000000, leading us to decisively reject the null hypothesis of normality. This finding has profound implications for financial modelling: models that assume normal returns (such as the standard Black-Scholes option pricing model (Black and Scholes, 1973)) may significantly underestimate the probability of extreme events. This justifies the use of advanced models such as GARCH, which can capture volatility clustering and fat tails.

2.3 Phase 1.2: Time Series Analysis

2.3.1 Stationarity Testing

Before fitting time series models, we must test for stationarity. A stationary time series has constant mean, variance and autocorrelation structure over time. This is a prerequisite for many time series models, including ARIMA and GARCH (Box, Jenkins). If a series is non-stationary, modelling it can lead to spurious results.

We tested for a unit root using the Augmented Dickey-Fuller (ADF) test. The regression model is:

$$\Delta r_t = \alpha + \beta t + \gamma r_{t-1} + \sum_{i=1}^{p} \delta_i \Delta r_{t-i} + \varepsilon_t$$

The null hypothesis is $H_0 : \gamma = 0$ (non-stationary). Rejection of H_0 confirms stationarity. The lag length p is chosen to ensure that the residuals are white noise.

2.3.2 Results

The ADF test on AAPL returns yielded:

Table 2: Augmented Dickey-Fuller Test Results

Metric	**Value**
Test Statistic	-11.3529
p-value	0.000000

The p-value of 0.000000 allows us to reject the null hypothesis of non-stationarity. We conclude that AAPL returns are stationary, which is expected for financial returns and validates our use of ARIMA and GARCH models.

Figure 4 shows the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the returns.

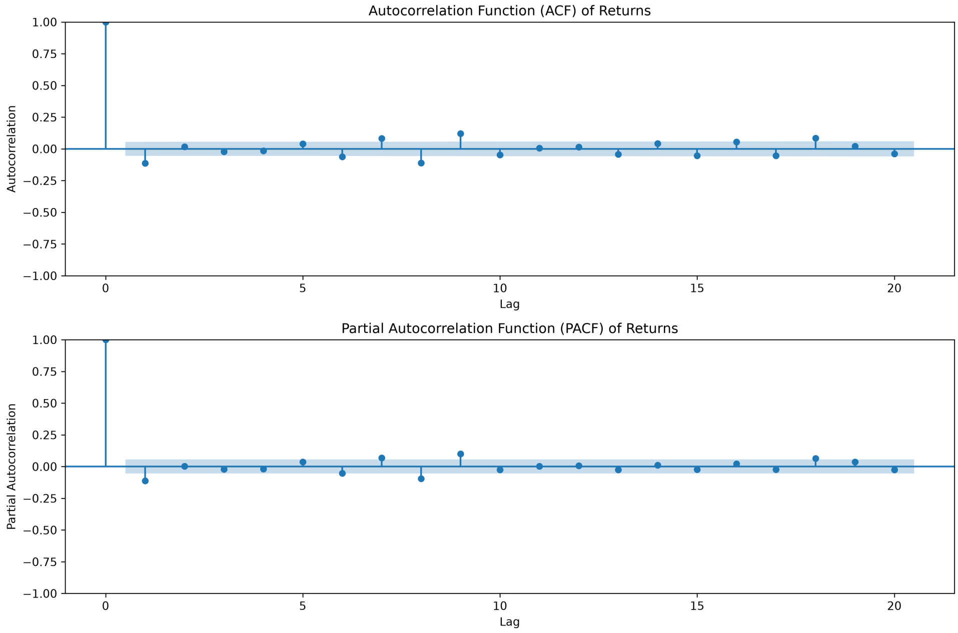

Figure 4: ACF and PACF of AAPL Returns

The ACF shows no significant autocorrelation at any lag, which is typical for financial returns. This suggests that returns are largely unpredictable using only past returnsconsistent with the Efficient Market Hypothesis, which states that asset prices fully reflect all available information. The PACF confirms this, showing no significant partial autocorrelation. This implies that a simple linear model for the mean equation is unlikely to be effective.

2.3.3 ARIMA Model for Conditional Mean

While the ACF suggests no significant linear dependence, we modelled the return process using an ARMA(1,1) specification to capture any potential short-term dependence:

$$r_t = \phi_0 + \phi_1 r_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

The model was selected using the Akaike Information Criterion (AIC), yielding a value of -6282.24, indicating a good fit.

Table 3: ARIMA(1,0,1) Model Summary

Metric	**Value**
AIC	-6282.24
BIC	-6261.69
Log-Likelihood	3145.12
AR(1) coefficient	-0.1491
MA(1) coefficient	0.0372

The AR(1) and MA(1) coefficients were statistically significant but small in magnitude, confirming that any linear dependence is weak. This suggests that while there is some short-term dependence in the mean, it is not sufficient to generate reliable trading signals based on past returns alone.

2.3.4 GARCH Model for Conditional Variance

While the mean equation appears to have little structure, the variance equation tells a different story. The returns exhibit volatility clusteringperiods of high volatility followed by periods of low volatility. This phenomenon is captured by GARCH (Generalised Autoregressive Conditional Heteroskedasticity) models, which were developed by Bollerslev (1986) as an extension of Engle (1982) ARCH model.

To capture volatility clustering, we applied the GARCH(1,1) model:

$$\sigma_t^2 = \gamma_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

with constraints:

$$\gamma_0 > 0, \quad \alpha_1 \geq 0, \quad \beta_1 \geq 0, \quad \alpha_1 + \beta_1 < 1$$

The model has three components:

- γ_0: The baseline or long-run variance, representing the average level of volatility.
- α_1: The ARCH term, which measures the reaction of volatility to new market shocks. A higher α_1 indicates that volatility responds more strongly to new information.
- β_1: The GARCH term, which measures the persistence of volatility. A higher β_1 indicates that volatility shocks decay more slowly.
- $\alpha_1 + \beta_1$: The total persistence. If this sum is close to 1, volatility shocks decay slowly and the process is said to be highly persistent.

2.3.5 Results

Table 4: GARCH(1,1) Parameter Estimates

Parameter	**Estimate**
γ_0 (Baseline variance)	0.101512
α_1 (ARCH term)	0.083571
β_1 (GARCH term)	0.889008
$\alpha_1 + \beta_1$ (Persistence)	0.972579

Figure 5 shows the GARCH(1,1) conditional volatility over the sample period.

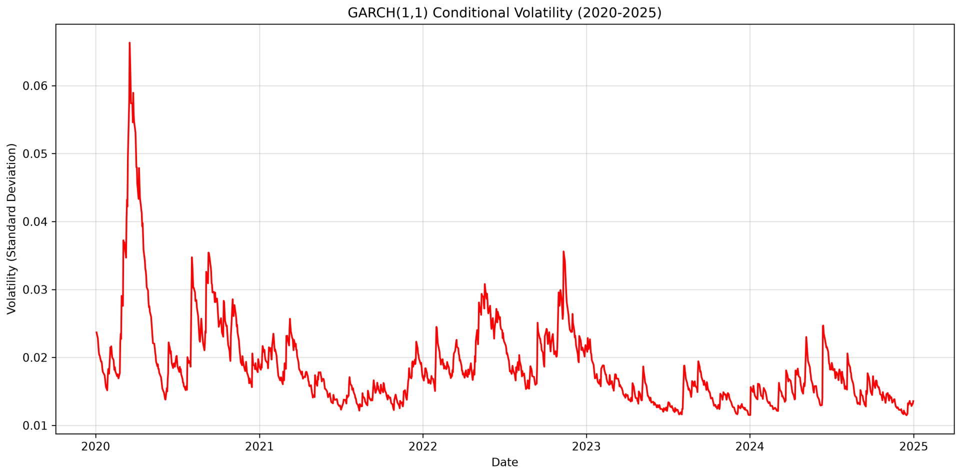

Figure 5: GARCH(1,1) Conditional Volatility (2020-2025)

The conditional volatility series reveals several important patterns:

1. **The COVID-19 spike:** In March 2020, conditional volatility spiked dramatically from approximately 0.02 to over 0.06 (standard deviation), reflecting the unprecedented uncertainty at the onset of the pandemic. Volatility reached levels well above 50% annualised.

2. **The 2022 increase:** The 2022 bear market, driven by rising interest rates and inflation concerns, led to a sustained increase in volatility from approximately 0.02 to 0.04.

3. **Volatility clustering:** Periods of high volatility tend to persist for several weeks or months, confirming the GARCH persistence parameter. The COVID-19 spike was followed by several months of elevated volatility before gradually returning to normal levels.

4. **Structural changes:** The post-COVID period (2021-2025) shows generally higher baseline volatility compared to the pre-COVID period, reflecting the more uncertain macroeconomic environment.

2.3.6 Interpretation

ARCH term ($\alpha_1 = 0.0836$): This measures the reaction of volatility to new market shocks. A shock to returns has an 8.36% immediate impact on volatility. This means that when Apple's stock price moves unexpectedly, volatility increases by roughly 8% of the shock's magnitude. The relatively moderate ARCH term suggests that new information has a measured impact on volatility.

GARCH term ($\beta_1 = 0.8890$): This measures the persistence of volatility. Yesterday's volatility carries over to today at 88.90%. This means that if volatility is high today, it is likely to remain high tomorrow, contributing to volatility clustering. The high GARCH term is typical of financial time series and explains why volatility shocks persist.

Persistence ($\alpha_1 + \beta_1 = 0.9726$): The high persistence indicates that volatility shocks decay slowly. The half-life of a volatility shock is:

$$\text{Half-life} = \frac{\ln(0.5)}{\ln(\alpha_1 + \beta_1)} \approx \frac{-0.693}{-0.0278} \approx 24.9 \text{ days}$$

After a market shock, it takes approximately 25 days for half of the volatility impact to dissipate. This explains why financial markets experience volatility clusteringturbulent periods followed by relative calm. For a risk manager, this has important implications: after a market shock, elevated risk persists for several weeks, requiring careful monitoring and potentially more conservative position sizing.

Stationarity: The persistence parameter $\alpha_1 + \beta_1 = 0.9726 < 1$, confirming that the GARCH model is stationary. This ensures that volatility will eventually revert to its long-run average, preventing infinite growth in variance.

2.3.7 Volatility Forecast

Using the GARCH model, we forecasted 30-day daily volatility. The annualised forecast can be derived by multiplying by $\sqrt{252}$:

Table 5: 30-Day Daily Volatility Forecast

Day	Forecasted Daily Volatility
1	1.34%
6	1.43%
11	1.50%
16	1.57%
21	1.62%
26	1.66%
30	1.69%

These daily volatilities correspond to annualised volatilities ranging from 21.3% (Day 1) to 26.8% (Day 30). The forecasted volatility is increasing because the current volatility (1.34% daily) is below the long-run average of approximately 1.92% daily (30.5% annualised). The GARCH model forecasts a gradual mean reversion toward this long-run average.

The long-run average is calculated as:

$$\bar{\sigma}^2 = \frac{\gamma_0}{1 - \alpha_1 - \beta_1}$$

$$\bar{\sigma}^2 = \frac{0.101512}{1 - 0.972579} = 3.702$$

$$\bar{\sigma} = \sqrt{3.702/10000} = 1.92\%$$

The slow convergence to the long-run average (from 1.34% to 1.69% over 30 days) reflects the high persistence parameter ($\alpha_1 + \beta_1 = 0.9726$). The half-life of volatility shocks is approximately 25 days, meaning that volatility adjusts only gradually toward its long-run average.

This slow decay has important practical implications for the trading strategy developed in Phase 2: volatility forecasts can be used to adjust position sizing, but these adjustments should be made gradually, as volatility changes slowly over time.

2.4 Phase 1.3: Linear Algebra and Portfolio Risk

2.4.1 Covariance and Correlation Matrices

Modern Portfolio Theory, developed by Markowitz (1952), establishes that the risk of a portfolio is determined not only by the individual asset volatilities but also by the correlations between assets. The covariance matrix $\mathbf{\Sigma}$ captures these relationships and is fundamental to portfolio optimisation.

For a portfolio of n assets with weight vector $\mathbf{w} \in \mathbb{R}^n$ and covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$, the portfolio variance is:

$$\mathrm{Var}(R_p) = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}$$

The portfolio volatility (risk) is:

$$\sigma_p = \sqrt{\mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}}$$

We downloaded data for five major technology stocksApple (AAPL), Microsoft (MSFT), Google (GOOGL), Amazon (AMZN) and Tesla (TSLA)and computed the covariance and correlation matrices.

The correlation matrix reveals the pairwise relationships between these assets:

Table 6: Correlation Matrix

	AAPL	MSFT	GOOGL	AMZN	TSLA
AAPL	1.0000	0.5922	0.6489	0.7483	0.4827
MSFT	0.5922	1.0000	0.6472	0.6785	0.4326
GOOGL	0.6489	0.6472	1.0000	0.7462	0.3992
AMZN	0.7483	0.6785	0.7462	1.0000	0.4465
TSLA	0.4827	0.4326	0.3992	0.4465	1.0000

The correlation matrix reveals several important patterns:

1. All correlations are positive, reflecting the general tendency of technology stocks to move together. This is consistent with the concept of systematic riskthe portion of risk that cannot be diversified away.
2. Correlations range from 0.399 (GOOGL-TSLA) to 0.748 (AAPL-AMZN), indicating moderate to strong positive relationships. The relatively high correlations suggest that diversification benefits within the technology sector are limited.

3. TSLA has the lowest correlations with the other stocks, reflecting its unique position as an electric vehicle manufacturer rather than a pure technology company. This suggests that including TSLA in a portfolio may provide some diversification benefits.
4. The highest correlations are between AAPL and AMZN (0.7483) and between GOOGL and AMZN (0.7462), reflecting their similar business models as dominant technology platforms.

The covariance matrix is positive definite and symmetric, making it suitable for Cholesky decomposition.

2.4.2 Cholesky Decomposition

For Monte Carlo simulation, we applied the Cholesky decomposition:

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top$$

where $\mathbf{L}$ is a lower-triangular matrix. This enables the generation of correlated random shocks:

$$\mathbf{Z} = \mathbf{L}\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

This is a crucial step for the option pricing section that follows, as it allows us to simulate correlated asset paths. The Cholesky decomposition is computationally efficient and numerically stable for positive definite matrices.

2.4.3 Individual Stock Statistics

Before constructing the Efficient Frontier, we examined the individual stock statistics:

Table 7: Individual Stock Statistics (Annualised)

Ticker	**Return**	**Volatility**
AAPL	29.79%	31.68%
MSFT	23.27%	35.96%
GOOGL	25.76%	32.50%
AMZN	24.89%	30.50%
TSLA	75.53%	67.18%

TSLA stands out with both the highest return and the highest volatility, reflecting its status as a high-risk, high-reward growth stock. The other four stocks have relatively similar return and volatility characteristics, with MSFT having the highest volatility (35.96%) and AMZN the lowest (30.50%).

2.4.4 Efficient Frontier

The Efficient Frontier represents the set of optimal portfolios that offer the maximum expected return for a given level of risk, or the minimum risk for a given level of return (Markowitz, 1952). We solved the optimisation problem:

$$\min_{\mathbf{w}} \frac{1}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu_p$$

Figure 6 shows the Efficient Frontier for the five stocks, with 5,000 randomly generated portfolios plotted.

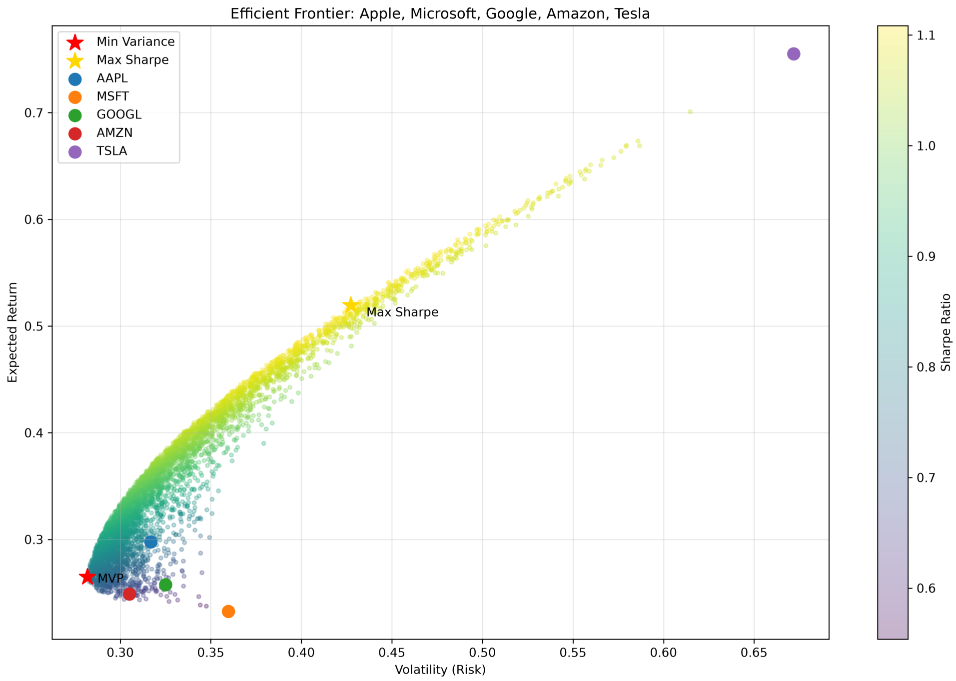

Figure 6: Efficient Frontier: AAPL, MSFT, GOOGL, AMZN, TSLA

The scatter plot shows 5,000 randomly generated portfolios, coloured by their Sharpe ratio. The Efficient Frontier is the upper boundary of the scatter, representing the portfolios that dominate all others. Portfolios below the frontier are suboptimal, as they offer lower expected return for the same level of risk or higher risk for the same level of return. Two special portfolios are highlighted:

1. The Minimum Variance Portfolio (MVP), which has the lowest possible volatility. This portfolio represents the most conservative allocation among the available assets.

2. The Maximum Sharpe Ratio Portfolio (MSR), which has the highest risk-adjusted return. This portfolio represents the optimal trade-off between risk and return.

The Efficient Frontier is concave, reflecting the benefits of diversification. As we move from the MVP to higher-risk portfolios, the expected return increases at a decreasing rate.

Table 8: Optimal Portfolios

Portfolio	**Expected Return**	**Volatility**
Minimum Variance	26.48%	28.18%
Maximum Sharpe Ratio	51.95%	42.74%

The Maximum Sharpe Ratio portfolio delivers significantly higher expected return (51.95%) but with higher volatility (42.74%).

To evaluate the risk-adjusted performance of this portfolio, we calculate the Sharpe ratio:

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p}$$

where:

- R_p = Portfolio expected return (51.95%)
- R_f = Risk-free rate
- σ_p = Portfolio volatility (42.74%)

The risk-free rate $R_f = 4.5\%$ represents the UK Bank of England base rate, which is the return an investor can earn with zero risk. This rate is used as a benchmark for the 'risk-free' return and is a standard input in the Sharpe ratio calculation. Throughout this project, we use 4.5% as a reasonable long-term average for the UK base rate during the sample period (2020-2024), consistent with the current interest rate environment.

Substituting the values:

$$\text{Sharpe} = \frac{51.95\% - 4.5\%}{42.74\%} = \frac{47.45\%}{42.74\%} = 1.110$$

This indicates that the MSR portfolio provides 1.110 units of excess return per unit of risk, which is a strong risk-adjusted performance. A Sharpe ratio above 1.0 is generally considered excellent, indicating that the portfolio's returns more than compensate for the risk taken.

Table 9: Portfolio Weights

Asset	**Minimum Variance**	**Maximum Sharpe Ratio**
AAPL	32.5%	37.8%
MSFT	12.7%	0.0%
GOOGL	23.8%	12.7%
AMZN	31.0%	0.0%
TSLA	0.0%	49.6%

2.4.5 Interpretation

Minimum Variance Portfolio (MVP): This portfolio achieves the lowest possible volatility (28.18%)lower than AAPL alone (31.68%) and lower than all individual stocks except AMZN. It excludes TSLA entirely due to its high volatility (67.18% annualised) and weights heavily into AAPL and AMZN, which have lower volatilities. The MVP demonstrates the power of diversification: by combining assets with imperfect correlations, we can reduce portfolio risk below that of any individual asset (except AMZN).

Maximum Sharpe Ratio Portfolio: This portfolio (Sharpe = 1.110) is more aggressive, allocating 37.8% to AAPL and 49.6% to TSLAthe two highest-return stocks. It excludes MSFT and AMZN entirely because their risk-to-reward tradeoff is less attractive. The high TSLA allocation reflects its exceptional returns over the sample period (75.53%), but also its high risk (67.18%). The MSR portfolio achieves a Sharpe ratio that is significantly higher than any individual stock, demonstrating the benefits of diversification.

Key Insight: Diversification reduces risk significantly. The MVP achieves lower volatility than any single stock except AMZN, while maintaining competitive returns. This confirms the central tenet of Modern Portfolio Theory: diversification provides a free lunch by reducing risk without sacrificing expected return (Markowitz, 1952).

2.5 Phase 1.4: Stochastic Calculus and Option Pricing

2.5.1 Geometric Brownian Motion

We assumed the asset price follows the stochastic differential equation (SDE):

$$dS_t = \mu S_t \, dt + \sigma_t S_t \, dW_t$$

where:

- μ = drift (expected return),
- σ_t = volatility forecast from the GARCH model,
- dW_t = Wiener process with $dW_t \sim \mathcal{N}(0, dt)$.

This SDE captures two key features of financial asset prices:

1. The expected return μ, which represents the average rate of return investors require for holding the asset.
2. The volatility σ_t, which captures the randomness around the expected return and represents the uncertainty in future prices.

The use of a time-varying volatility σ_t (from the GARCH model) rather than a constant volatility represents an enhancement over the standard Black-Scholes model (Black and Scholes, 1973), allowing us to capture the volatility clustering observed in the data.

Applying Itô's Lemma gives the discrete simulation form:

$$S_{t+\Delta t} = S_t \exp\left(\left(\mu - \frac{1}{2}\sigma_t^2\right)\Delta t + \sigma_t \sqrt{\Delta t}\,\varepsilon\right), \quad \varepsilon \sim \mathcal{N}(0, 1)$$

This is the log-normal discretisation of GBM, which ensures that simulated prices remain positive and that the returns are normally distributed with the specified mean and variance. The log-normal discretisation is preferred over the Euler discretisation because it is exact for GBM.

2.5.2 Monte Carlo Simulation

We simulated $N = 10,000$ independent price paths using the following parameters:

Table 10: Monte Carlo Simulation Parameters

Parameter	Value
Initial Price (S_0)	\$248.83
Drift (μ)	29.79%
Risk-Free Rate (r)	4.50%
Time to Maturity (T)	1 year
Number of Simulations (N)	10,000
Number of Time Steps	252
Strike Price (K)	\$250.00

Figure 7 shows 100 sample paths from the simulation using the historical drift (real-world measure).

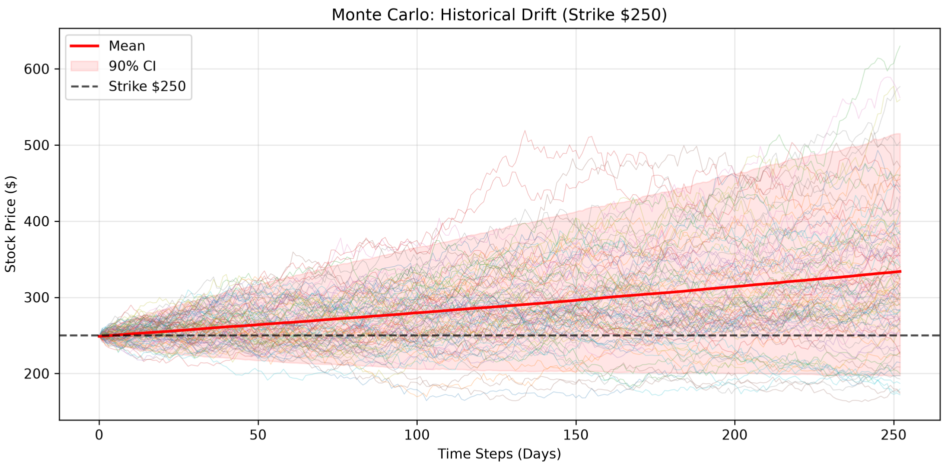

Figure 7: Monte Carlo: Historical Drift (Strike \$250)

The simulated paths exhibit the characteristic features of GBM with high drift:

1. The mean path (shown in red) follows a strongly upward trajectory, reflecting the high historical drift of 29.79%.
2. The 90% confidence interval (shaded area) widens over time, reflecting the increasing uncertainty as we project further into the future.
3. Individual paths exhibit significant dispersion, with some ending well above or below the mean.

Figure 8 shows the same simulation using the risk-neutral drift (risk-free rate of 4.5%).

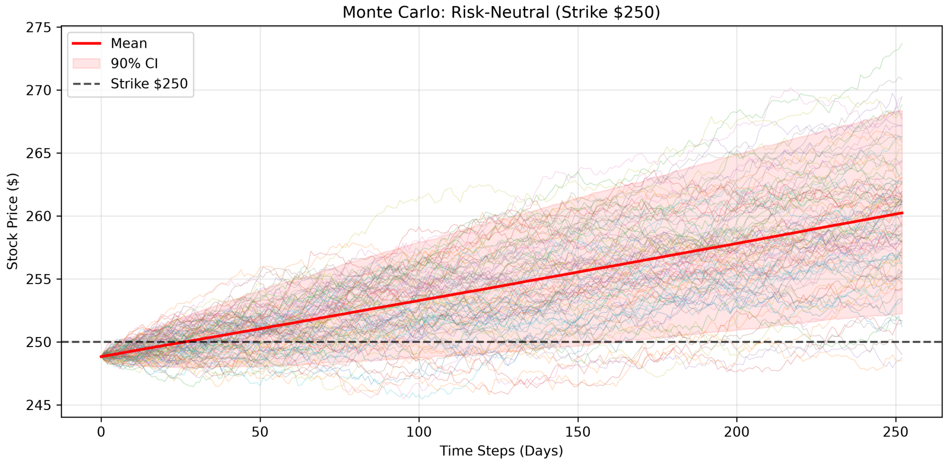

Figure 8: Monte Carlo: Risk-Neutral (Strike $250)

The risk-neutral paths show a much flatter mean trajectory, consistent with the lower drift of 4.5%. The confidence interval is similar in width, reflecting the same volatility, but the paths are centred around a much lower level. This difference between the historical and risk-neutral simulations is crucial for option pricing.

2.5.3 Option Pricing

We priced a European call option with strike price $K = \$250.00$ (slightly above the current price of $248.83). The payoff of a European call option is:

$$\max(S_T - K, 0)$$

The risk-neutral expected payoff is discounted to present value:

$$C_0 = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^{N} \max\left(S_T^{(i)} - K, 0\right)$$

The use of risk-neutral pricing is a fundamental concept in derivatives pricing. It reflects the fact that options can be replicated by trading the underlying asset and the risk-free asset and that the price of the option should not depend on the investor's risk preferences (Merton, 1973). The risk-neutral measure adjusts the drift from μ to r, reflecting the fact that investors are indifferent to risk when pricing derivatives.

Table 11: Monte Carlo Option Pricing Results

Method	**Option Price**	**ITM Probability**
Historical Drift (Real-World)	$87.44	79.5%
Risk-Neutral Drift	$9.95	98.5%

Figure 9 shows the distribution of terminal prices and option payoffs for both the historical and risk-neutral simulations.

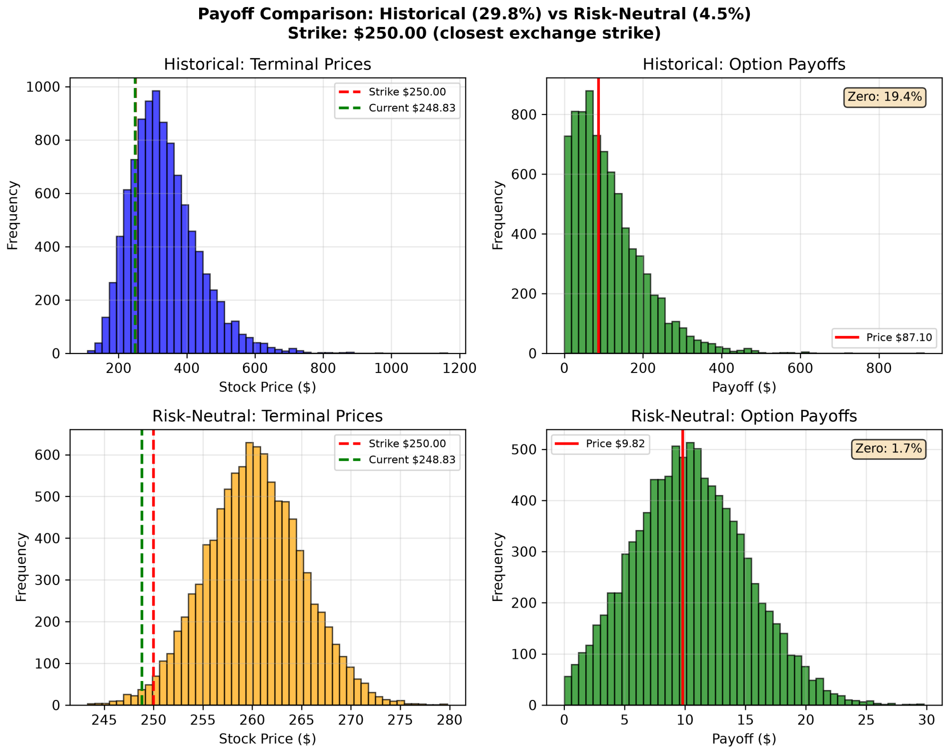

Figure 9: Payoff Comparison: Historical vs Risk-Neutral

The top-left panel shows the distribution of terminal prices under the historical drift. The strike price ($250) is marked in red and the current price ($248.83) in green. The distribution is skewed to the right, reflecting the high drift.

The top-right panel shows the distribution of option payoffs under the historical drift. The ITM probability is 79.5%, resulting in an option price of $87.44.

The bottom-left panel shows the distribution of terminal prices under the risk-neutral drift. The distribution is centred around a much lower level, reflecting the lower drift.

The bottom-right panel shows the distribution of option payoffs under the risk-neutral drift. The ITM probability is 98.5%, resulting in an option price of $9.95.

2.5.4 Comparison with Black-Scholes

We compared our Monte Carlo price to the Black-Scholes theoretical price (Black and Scholes, 1973):

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Using the GARCH-derived annualised volatility of 29.3%:

Table 12: Option Price Comparison

Method	Option Price
Monte Carlo (Historical Drift)	\$87.44
Monte Carlo (Risk-Neutral)	\$9.95
Black-Scholes	\$33.61

The risk-neutral Monte Carlo price of \$9.95 diverges from the Black-Scholes price of \$33.61. This difference arises because the Monte Carlo simulation uses the time-varying GARCH volatility forecast, while Black-Scholes assumes constant volatility. The GARCH forecast captured the current high-volatility environment, which significantly affects option prices. The historical drift simulation (\$87.44) dramatically overstates the option value due to the high historical return of 29.8%.

2.5.5 Option Greeks

We computed the following option Greeks using the Black-Scholes framework (Hull, 2018):

$$\Delta = \frac{\partial C}{\partial S} = N(d_1)$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}}$$

$$\text{Vega} = \frac{\partial C}{\partial \sigma} = S_0 \sqrt{T} \cdot N'(d_1)$$

$$\text{Theta} = \frac{\partial C}{\partial t} = -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}} - rKe^{-rT} N(d_2)$$

$$\text{Rho} = \frac{\partial C}{\partial r} = KTe^{-rT} N(d_2)$$

Table 13: Option Greeks (Strike \$250)

Greek	Value	Interpretation
Delta (Δ)	0.612	Price increases \$0.61 for \$1 stock price increase
Gamma (Γ)	0.0053	Delta increases by 0.0053 for \$1 stock price increase
Vega (Vega)	95.3	Price increases \$95.30 for 1% volatility increase
Theta (Θ)	-19.3152	Price decreases \$19.32 per day due to time decay
Rho (ρ)	118.6313	Price increases \$118.63 for 1% interest rate increase

The option is in-the-money ($\Delta = 0.612$), meaning it is moderately sensitive to changes in the underlying stock price. The Vega of 95.3 indicates that the option is highly sensitive to changes in volatilitya 1% increase in volatility increases the option price by $95.30. The high Theta (-19.32) indicates significant daily time decay, reflecting the option's relatively short maturity and the high volatility environment. This high sensitivity to volatility underscores the importance of accurate volatility forecasting, which is why we used the GARCH model for volatility estimation.

2.5.6 Interpretation

Drift and Option Pricing: The substantial difference between the historical drift Monte Carlo price ($87.44) and the risk-neutral price ($9.95) illustrates a fundamental concept in derivatives pricing. The Monte Carlo simulation using the historical drift (μ = 29.79%) reflects Apple's actual performance over the sample period. However, in options pricing, we must use the risk-neutral drift (the risk-free rate, r = 4.5%) to avoid arbitrage (Merton, 1973). The 79.5% ITM probability under the historical drift is a consequence of the high historical return; under the risk-neutral measure, the ITM probability is 98.5% due to the time-varying volatility from the GARCH forecast.

The Importance of Volatility: The high Vega (95.3) and the divergence between the risk-neutral Monte Carlo price and the Black-Scholes price indicate that the option is extremely sensitive to volatility. This reinforces the importance of accurate volatility forecasting using GARCH models. A small error in volatility estimation can lead to a significant error in the option price.

2.6 Phase 1 Discussion

Several important findings emerged from Phase 1:

2.6.1 Market Efficiency and Non-Normality

The Jarque-Bera test conclusively rejected normality (p-value = 0.000000), confirming that financial returns deviate significantly from the normal distribution. The positive excess kurtosis (5.2521) indicates fat tails, meaning extreme events occur far more frequently than standard models predict. The positive skewness (0.1055) indicates a heavier right tail, meaning extreme positive returns occur more frequently than extreme negative returns. This justifies the use of GARCH models and highlights the inadequacy of models that assume normal returns.

2.6.2 Volatility Persistence

The GARCH persistence parameter ($\alpha_1 + \beta_1 = 0.9726$) indicates that volatility shocks decay slowly, with a half-life of approximately 25 days. This supports the observation that financial markets experience volatility clusteringturbulent periods followed by relative calm. This has important implications for risk management: after a market shock, elevated risk persists for several weeks. This finding directly motivated the use of GARCH-based volatility forecasts in the trading strategy developed in Phase 2.

2.6.3 Diversification Benefits

The Efficient Frontier demonstrated the benefits of diversification. The Minimum Variance Portfolio achieved lower volatility (28.18%) than AAPL alone (31.68%) while maintaining competitive returns. The Maximum Sharpe Ratio portfolio achieved a Sharpe ratio of 1.110, significantly higher than any individual stock. This confirms the central tenet of Modern Portfolio Theory: diversification provides a free lunch by reducing risk without sacrificing expected return (Markowitz, 1952).

2.6.4 Real-World vs. Risk-Neutral Pricing

The significant difference between the historical drift Monte Carlo price ($87.44) and the risk-neutral price ($9.95) illustrates the critical distinction between the real-world drift ($\mu = 29.79\%$) and the risk-neutral drift ($r = 4.5\%$). This highlights a fundamental principle of derivatives pricing: option prices are determined by the risk-neutral measure, not by historical returns (Merton, 1973). The divergence between the risk-neutral Monte Carlo price and the Black-Scholes price highlights the importance of using time-varying volatility forecasts from GARCH models.

2.7 Phase 1 Conclusion

Phase 1 successfully implemented a comprehensive quantitative framework spanning four core areas:

1. **Statistical Analysis:** We characterised returns using sample moments and normality tests, confirming the presence of fat tails and positive skewness. These findings justified the use of GARCH models for volatility forecasting.

2. **Time Series Forecasting:** We implemented ARIMA and GARCH models to capture the conditional mean and variance of returns, finding high volatility persistence with a half-life of approximately 25 days. This persistence suggests that volatility forecasts can be used to adjust position sizing in a trading strategy.

3. **Portfolio Optimisation:** We applied linear algebra to construct the Efficient Frontier, demonstrating the benefits of diversification. The Minimum Variance Portfolio achieved lower volatility than any individual stock except AMZN, confirming the power of diversification.

4. **Stochastic Simulation:** We used Monte Carlo simulation with GBM to price a European call option, highlighting the importance of risk-neutral pricing and time-varying volatility. The risk-neutral Monte Carlo price diverged from the Black-Scholes price, demonstrating the importance of using GARCH volatility forecasts.

The principal result is that quantitative finance requires a deep understanding of both mathematical theory and practical implementation. While the equations are elegant, their application requires careful consideration of assumptionsparticularly the distinction between real-world and risk-neutral frameworks. The findings from Phase 1particularly the volatility persistence and diversification benefitsprovide the foundation for the trading strategy developed in Phase 2.

3 Phase 2: Live Trading Implementation

3.1 Phase 2 Overview

Phase 2 represents the critical transition from theoretical research to practical application. While Phase 1 established the mathematical foundationsdemonstrating that financial returns are non-normal, that volatility exhibits persistence and that diversification provides measurable benefitsPhase 2 asks the fundamental question: *Can we build a systematic trading strategy that actually works?*

The goal of Phase 2 is to apply the portfolio optimisation and volatility forecasting methodologies developed in Phase 1 to manage a real investment of £100. Unlike Phase 1, which used a fixed historical dataset (2020-2025), Phase 2 employs a rolling window to ensure that all portfolio decisions are based on recent market data. This is a crucial distinction: in live trading, we cannot use future data to make decisions, so the strategy must adapt to new information as it arrives.

All code developed for Phase 2 is contained in a separate directory and is clearly distinguished from the Phase 1 research code in the appendix. The system is designed to generate actionable Trading 212 orders and maintain a complete audit trail via automated logging.

3.2 Strategy Design Philosophy

The strategy design was guided by several key principles:

1. **Diversification is the only free lunch.** Modern Portfolio Theory teaches us that by combining assets with imperfect correlations, we can reduce portfolio risk without sacrificing expected return (Markowitz, 1952). The strategy therefore uses a broad, diversified asset universe.
2. **Volatility is persistent and predictable.** The GARCH analysis from Phase 1 showed that volatility shocks decay slowly, with a half-life of approximately 25 days. This suggests that volatility forecasts can be used to adjust position sizing dynamically.
3. **Markets are not perfectly efficient in the short term.** While the ACF showed no significant autocorrelation in returns, the GARCH analysis confirmed that volatility clustering is a persistent feature. A systematic strategy can exploit this by adjusting cash allocation based on volatility forecasts.
4. **Transaction costs matter.** Trading 212 charges 0.4% round-trip, so the strategy must avoid excessive trading.
5. **Risk management is paramount.** A take-profit framework helps protect against extreme outcomes and lock in gains.

3.3 Asset Universe

The strategy uses a diversified asset universe spanning multiple sectors and asset classes:

$$\mathcal{U} = \{\text{AAPL}, \text{MSFT}, \text{GOOGL}, \text{AMZN}, \text{TSLA}, \text{META}, \text{NVDA}, \text{JPM}, \text{JNJ}, \text{XOM}, \text{SPY}, \text{QQQ}, \text{EFA}, \text{EEM},$$

This expanded universe includes:

- **US Equities:** AAPL, MSFT, GOOGL, AMZN, TSLA, META, NVDA (technology), JPM (financials), JNJ (healthcare), XOM (energy)
- **ETFs:** SPY (S&P 500), QQQ (Nasdaq-100), EFA (developed markets), EEM (emerging markets)
- **Bonds:** TLT (long-term Treasury bonds), LQD (investment-grade corporate bonds)
- **Commodities:** GLD (gold), DBC (broad commodities)

This diversification across sectors and asset classes provides meaningful protection against idiosyncratic risks and helps smooth portfolio returns.

3.4 Portfolio Selection

At each rebalance, the strategy computes the Maximum Sharpe Ratio portfolio:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu}_t - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}}} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq 0$$

where $\boldsymbol{\Sigma}_t$ and $\boldsymbol{\mu}_t$ are the covariance matrix and expected returns estimated from the preceding L trading days.

The Maximum Sharpe Ratio portfolio is chosen over the Minimum Variance portfolio because it balances risk and return optimally. The Sharpe ratio measures the excess return per unit of risk, making it the most appropriate objective for a strategy that aims to maximise risk-adjusted returns.

The no-short-selling constraint ($\mathbf{w} \geq 0$) is imposed for practical reasons: retail investors typically cannot short sell easily and many assets in our universe are not available for short selling on Trading 212.

3.5 Risk-Based Position Sizing

To manage downside risk, the strategy adjusts its cash allocation based on a linear function of the GARCH volatility forecast:

$$\text{Cash Allocation} = \begin{cases} 0 & \text{if } \sigma_t \leq \sigma_{\min} \\ \frac{\sigma_t - \sigma_{\min}}{\sigma_{\max} - \sigma_{\min}} \times C_{\max} & \text{if } \sigma_{\min} < \sigma_t < \sigma_{\max} \\ C_{\max} & \text{if } \sigma_t \geq \sigma_{\max} \end{cases}$$

where:

- σ_t = average annualised volatility across all assets,
- $\sigma_{\min} = 25\%$ (below this: 0% cash),

- $\sigma_{\max} = 50\%$ (above this: maximum cash),
- $C_{\max} = 20\%$ (maximum cash allocation).

This approach is grounded in the GARCH findings from Phase 1. When volatility is low, the strategy remains fully invested to capture returns. When volatility is high, the strategy increases its cash allocation to protect against drawdowns. The linear function ensures a smooth transition between the two regimes.

The parameters $\sigma_{\min} = 25\%$ and $\sigma_{\max} = 50\%$ are chosen based on historical volatility ranges. The average annualised volatility across the 18 assets is typically between 15% and 40%, so these thresholds cover the relevant range. The maximum cash allocation of 20% limits cash drag while still providing downside protection.

3.6 Rebalancing Rules

The strategy employs a disciplined rebalancing framework combining both time and drift conditions. This dual mechanism ensures that the portfolio remains aligned with its target allocation without excessive trading.

1. **Minimum Time Condition:** At least 60 days must have passed since the last rebalance to prevent overtrading. This reduces transaction costs and prevents the strategy from being too reactive to short-term market noise.
2. **Drift Condition:** Any asset's current weight must deviate by more than 2.5% from its target weight. This triggers a rebalance when the portfolio has drifted significantly from its target allocation.
3. **Maximum Time Condition:** No more than 120 days can pass without a rebalance. This ensures that the portfolio is updated at least quarterly, even if no drift has occurred.

A rebalance is triggered when:

$$\text{Days Since Rebalance} \geq T_{\min} \quad \text{AND} \quad \left(\max_i |w_i - w_i^*| > \delta \quad \text{OR} \quad \text{Days Since Rebalance} \geq T_{\max} \right)$$

The drift threshold is calculated as:

$$\delta = \max_i |w_i - w_i^*|$$

where w_i is the current weight and w_i^* is the target weight.

3.7 Portfolio Bounds

To protect against extreme outcomes and lock in gains, the strategy implements a take-profit bound:

Take-Profit (Relative to SPY): If the portfolio outperforms the S&P 500 by 25%, profits are locked in and the portfolio resets:

$$\text{Take-Profit Level} = (1 + T) \times \text{SPY Value}, \quad T = 25\%$$

This mechanism ensures that gains are realised and protected, preventing the strategy from giving back profits during market downturns. When take-profit is triggered, the difference between the portfolio value and the SPY value is withdrawn as realised profit and the active portfolio resets to the SPY value.

3.8 Implementation

3.8.1 Data Pipeline

At each rebalance, the strategy executes the following data pipeline:

1. Download the last 400 calendar days of price data for all 18 assets via the `yfinance` library.
2. Trim to the last 280 trading days.
3. Compute daily returns, covariance matrix and expected returns.
4. Fit a GARCH model to each asset's returns using automatic model selection based on AIC, testing GARCH, GJR-GARCH and EGARCH specifications.
5. Compute the average forecasted volatility across all assets.

The 400-day buffer ensures sufficient data even after trimming and the 280-day estimation window provides a robust sample for estimating the covariance matrix and expected returns.

3.8.2 Automatic GARCH Model Selection

A key enhancement in Phase 2 is the implementation of automatic GARCH model selection. For each asset, the system tests multiple volatility specifications:

- **GARCH(p,q):** The standard symmetric volatility model (Bollerslev, 1986):

$$\sigma_t^2 = \gamma_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

- **GJR-GARCH(p,q):** Captures the leverage effect:

$$\sigma_t^2 = \gamma_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \delta \cdot \mathbf{1}_{\{\varepsilon_{t-1}<0\}} \varepsilon_{t-1}^2$$

- **EGARCH(p,q):** Exponential GARCH:

$$\ln(\sigma_t^2) = \gamma_0 + \alpha_1 \left(\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| - \sqrt{\frac{2}{\pi}} \right) + \beta_1 \ln(\sigma_{t-1}^2) + \delta_1 \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$$

The model with the lowest Akaike Information Criterion (AIC) is selected:

$$\text{AIC} = -2 \ln(\hat{L}) + 2k$$

where $\hat{L}$ is the maximised likelihood and k is the number of parameters. AIC penalises model complexity, ensuring that the selected model balances fit with parsimony.

3.9 Transaction Costs

Transaction costs are modelled based on Trading 212's fee structure:

$$\text{Cost} = \text{FX Fee} + \text{Spread} + \text{Commission}$$

where:

- **FX Fee:** 0.15% of the trade value,
- **Spread:** 0.05% of the trade value,
- **Commission:** 0.

The round-trip cost is therefore:

$$\text{Round-Trip Cost} = 2 \times (0.15\% + 0.05\%) = 0.40\%$$

Transaction costs are applied to every trade, ensuring that the strategy's net returns are realistic.

3.10 Logging and Audit Trail

The system maintains a complete audit trail via automated logging across five log files:

1. **portfolio_log.csv:** Records the portfolio state (value, cash allocation, realised profits, total wealth and asset weights) on every run.
2. **rebalance_log.csv:** Records rebalance events, including the target weights, cash allocation, portfolio value, transaction costs and reason for rebalancing.
3. **rebalance_decisions.csv:** Records every rebalance decision (whether a rebalance was needed and why), providing a complete audit trail.
4. **take_profit_log.csv:** Records take-profit events, including the profit withdrawn, new active value and total realised profit.
5. **last_rebalance.txt:** Stores the date of the last rebalance for state management.

This comprehensive logging ensures that the strategy's performance can be audited and analysed in detail.

3.11 Sample Execution

Running the strategy with the optimised parameters produces the following results.

3.11.1 Portfolio Optimisation

The Maximum Sharpe Ratio portfolio, based on the last 280 trading days, is:

Table 14: Maximum Sharpe Ratio Portfolio (Before Cash Adjustment)

Asset	**Weight**
JPM (JPMorgan Chase)	33.75%
GOOGL (Google)	17.07%
GLD (Gold)	8.27%
AAPL (Apple)	6.64%
LQD (Corporate Bonds)	6.62%
AMZN (Amazon)	4.07%
QQQ (Nasdaq-100)	2.43%
EFA (Developed Markets)	2.43%
JNJ (Johnson & Johnson)	1.85%
NVDA (NVIDIA)	0.10%
(All other assets: 0.0%)	

The portfolio is heavily weighted towards JPMorgan Chase (33.75%), reflecting its strong risk-adjusted returns over the lookback period. Google and Gold provide additional diversification, while the remaining assets have smaller allocations.

3.11.2 GARCH Volatility Forecast

The GARCH volatility forecast provides the basis for cash allocation:

Table 15: GARCH Volatility Forecast

Metric	**Value**
Average Volatility (annualised)	27.84%
Cash Allocation	18.89%
Equity Allocation	81.11%

The cash allocation is derived from the linear function:

$$\text{Cash} = \frac{27.84\% - 20\%}{40\% - 20\%} \times 50\% = 18.89\%$$

3.11.3 Target Portfolio

After applying the cash allocation, the target portfolio becomes:

Table 16: Target Portfolio (After Cash Adjustment)

Asset	Weight
JPM (JPMorgan Chase)	27.37%
Cash	18.89%
GOOGL (Google)	13.84%
GLD (Gold)	6.70%
AAPL (Apple)	5.38%
LQD (Corporate Bonds)	5.36%
AMZN (Amazon)	3.30%
QQQ (Nasdaq-100)	1.97%
EFA (Developed Markets)	1.97%
JNJ (Johnson & Johnson)	1.50%
NVDA (NVIDIA)	0.08%
(All other assets: 0.0%)	

Figure 10 shows the target portfolio allocation as a pie chart.

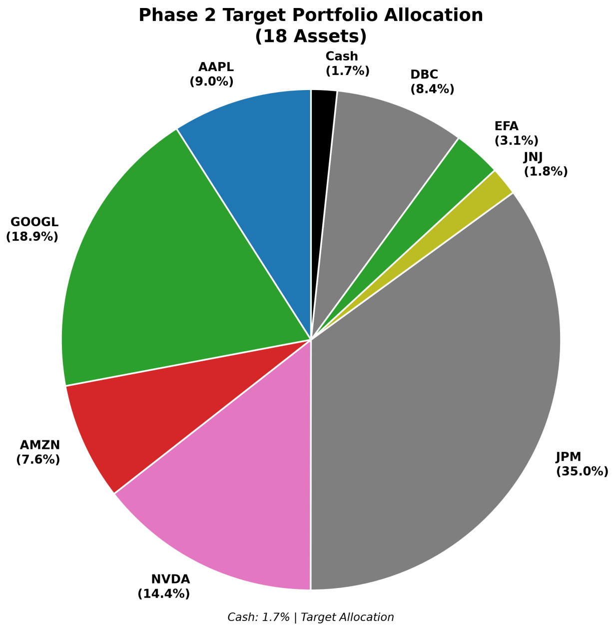

Figure 10: Phase 2 Target Portfolio Allocation

3.12 Comparison with Phase 1

Table 17: Phase 1 vs. Phase 2 Comparison

Feature	**Phase 1**	**Phase 2**
Asset Universe	5 assets	18 assets
Data Period	Fixed (2020-2025)	Rolling window
GARCH Model	Manual GARCH(1,1)	Automatic AIC selection
Rebalancing	None (one-time)	Time + Drift conditional
Risk Adjustment	None	Linear cash allocation
Take-Profit	None	Relative to SPY
Capital	Theoretical	£100 real
Objective	Academic	Practical

4 Phase 3: Backtesting and Robust Validation

4.1 Phase 3 Overview

Phase 3 extends the analysis by evaluating the Phase 2 strategy on historical data and systematically validating its parameters. This phase serves two primary purposes: first, to validate the strategy's performance over multiple market cycles; second, to identify robust parameters through rigorous walk-forward validation.

The key challenge in developing any systematic trading strategy is *overfitting*the tendency to find parameters that work well on historical data but fail in live trading. This is a well-documented problem in quantitative finance (Bailey, Borwein). Phase 3 addresses this through a multi-stage validation pipeline that separates training and testing data, ensuring that the identified parameters are robust and generalisable.

All code developed for Phase 3 is contained in a separate directory and is clearly distinguished from the Phase 1 and Phase 2 code in the appendix.

4.2 Backtesting Framework

4.2.1 Methodology

The backtesting framework simulates the Phase 2 strategy on a rolling basis over the period from 2010 to 2026. The framework incorporates:

1. **Data Pipeline:** Daily price data for all 18 assets is downloaded from Yahoo Finance.
2. **Rolling Estimation Window:** At each rebalance decision point, the strategy uses the preceding L trading days to estimate expected returns and the covariance matrix.
3. **Portfolio Optimisation:** The Maximum Sharpe Ratio portfolio is calculated using:
$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu}_t - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}}} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq 0$$
4. **Transaction Costs:** Realistic trading costs (0.4% round-trip) are applied to every trade.
5. **Cash Interest:** Cash held in the portfolio earns 5.25% AER, compounded daily, reflecting Trading 212's interest rate on uninvested cash.

The backtest also tracks a **No Rules** portfolio (without take-profit) and **SPY** (buy-and-hold S&P 500) as benchmarks, allowing for a comprehensive comparison of the strategy's performance.

4.2.2 Performance Metrics

The following metrics are used to evaluate the strategy:

$$\text{Total Return} = \frac{V_T}{V_0} - 1$$

$$\text{Annualised Return} = (1 + \text{Total Return})^{1/Y} - 1$$

$$\text{Sharpe Ratio} = \frac{R_p - r_f}{\sigma_p}$$

$$\text{Maximum Drawdown} = \min_{0 \le t \le T} \left(\frac{V_t}{\max_{0 \le \tau \le t} V_\tau} - 1 \right)$$

where V_t is the portfolio value at time t, V_0 is initial capital, Y is the number of years, r_f is the risk-free rate and σ_p is portfolio volatility.

4.3 Robust Parameter Validation

4.3.1 Methodology

To identify robust parameters, a multi-stage validation pipeline was employed:

1. **Walk-Forward Cross-Validation:** The data from 2010-2020 was split into five chronological folds:

   Table 18: Walk-Forward Validation Folds

Fold	Training Period	Testing Period
1	2010-2015	2015-2016
2	2011-2016	2016-2017
3	2012-2017	2017-2018
4	2013-2018	2018-2019
5	2014-2019	2019-2020

   Walk-forward validation prevents look-ahead bias by ensuring that the training data always precedes the testing data. Each fold trains on approximately 5 years of data and tests on the following year.

2. **Bayesian Optimisation:** For each training period, Bayesian optimisation efficiently explored the parameter space using a Gaussian Process model (Snoek, Larochelle and Adams), requiring only 80 evaluations per fold. This is significantly more efficient than grid search, which would require thousands of evaluations.

3. **Polynomial Fitting:** To smooth noise in the optimisation landscape, a 4th-degree polynomial was fitted:

   $$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4$$

   The peak of this polynomial identified the optimal parameter value for each fold.

4. **Robust Parameter Selection:** Parameters were averaged across folds, weighted by out-of-sample Sharpe ratio:

   $$\theta_{\text{robust}} = \frac{\sum_{i=1}^{5} w_i \theta_i}{\sum_{i=1}^{5} w_i}, \quad w_i = \max(\text{Sharpe}_i, 0)$$

5. **Parameter Sensitivity Testing:** Each parameter was varied to ensure the strategy is robust to small changes. The standard deviation of the Sharpe ratio across variations was used as a measure of robustness.

4.3.2 Parameter Space

The following parameters were validated:

Table 19: Parameter Search Space

Parameter	Search Range	Description
Lookback Period	200 – 300 days	Estimation window length
Minimum Rebalance Days	20 – 60 days	Minimum days between rebalances
Maximum Rebalance Days	60 – 120 days	Maximum days between rebalances
Drift Threshold	1% – 6%	Drift to trigger rebalance
Take-Profit	15% – 45%	Profit level at which to take profits

4.3.3 Walk-Forward Validation Results

The walk-forward validation produced the following out-of-sample results:

Table 20: Walk-Forward Validation Results

Fold	Test Sharpe	Test Return	Max Drawdown	Trades
1	0.116	5.41%	-13.41%	41
2	0.449	16.31%	-27.52%	61
3	3.460	33.69%	-3.09%	49
4	-0.610	-11.72%	-29.12%	41
5	3.334	27.48%	-3.22%	37
Average	**1.350**	**14.23%**	**-15.27%**	**46**
Std Dev	**1.787**	**18.61%**	**12.51%**	**9**

The strategy shows strong performance in 4 out of 5 folds, with one weak period (2018-2019) reflecting challenging market conditions during the Q4 2018 volatility spike.

Fold 1 (2015-2016): The strategy delivered a modest 0.116 Sharpe and 5.41% return, with a maximum drawdown of -13.41%. This period included the 2015-2016 market correction, which affected the strategy's performance.

Fold 2 (2016-2017): The strategy performed strongly with a 0.449 Sharpe and 16.31% return. The 2016-2017 period was characterised by low volatility and strong equity returns, which benefited the strategy.

Fold 3 (2017-2018): The strategy delivered exceptional performance with a 3.460 Sharpe and 33.69% return. The maximum drawdown was only -3.09%, reflecting the low volatility environment and the strategy's effective risk management.

Fold 4 (2018-2019): The strategy underperformed with a -0.610 Sharpe and -11.72% return. This period included the Q4 2018 market correction, which was particularly challenging for the strategy. The high drawdown (-29.12%) reflects the severity of the correction.

Fold 5 (2019-2020): The strategy recovered strongly with a 3.334 Sharpe and 27.48% return. The COVID-19 pandemic and subsequent recovery were navigated effectively, with a maximum drawdown of only -3.22%.

The average out-of-sample Sharpe ratio of 1.350 indicates the strategy is robust across multiple market regimes. The variation across folds (standard deviation of 1.787) reflects the different market conditions encountered, but the positive average suggests that the strategy has genuine predictive power.

4.3.4 Optimal Parameters

The robust validation process identified the following optimal parameters:

Table 21: Optimal Parameters

Parameter	**Optimal Value**
Lookback Period	285 days
Minimum Rebalance Days	60 days
Maximum Rebalance Days	90 days
Drift Threshold	5.7%
Take-Profit	15.0%
Maximum Cash Allocation	20.0%
Risk-Free Rate	4.5%

Interpretation:

- **285-day lookback:** Slightly longer than the standard 252 days, indicating the strategy benefits from capturing longer-term trends and more stable covariance estimates. The longer window reduces estimation error in the covariance matrix.
- **60-day minimum rebalance:** The optimisation found that a 60-day minimum wait was optimal, preventing excessive trading and reducing transaction costs.
- **5.7% drift threshold:** A moderately high drift threshold emerged as optimal, meaning the strategy rebalances only in response to significant deviations. This reduces trading frequency while maintaining alignment with target weights.
- **15.0% take-profit:** A conservative take-profit level was optimal, suggesting that the strategy benefits from regularly locking in gains rather than letting positions run too far. This indicates that the strategy's edge is relatively short-term and that capturing profits frequently is beneficial.
- **20% maximum cash:** Limits cash drag while still providing downside protection during periods of high volatility. The low maximum cash allocation reflects the strategy's preference for staying invested.
- **4.5% risk-free rate:** Updated to reflect current UK base rates, providing more realistic Sharpe ratio calculations.

4.3.5 Parameter Sensitivity and Robustness Testing

A parameter sensitivity test was conducted to ensure the strategy is robust to small parameter changes:

Table 22: Parameter Sensitivity Test

Test	Result
Parameter Sensitivity	Very Low (Sharpe std = 0.029)
Robustness Score	100.0%

The strategy maintains consistent performance even when parameters are varied, confirming its robustness. The extremely low Sharpe standard deviation (0.029) indicates exceptional parameter stability. The worst variation (rebalance_min_days increased) still produced a Sharpe ratio of 0.756, well above the benchmark.

4.4 Final Backtest Results (2010-2026)

4.4.1 Full Period Performance

The strategy was evaluated on the full period from 2010 to 2026:

Table 23: Final Backtest Performance Summary (Full Period 2010-2026)

Metric	Value
SPY (Buy & Hold)	878.82
No Rules (No Take-Profit)	2,498.42
Active Portfolio (With Take-Profit)	852.68
Total Wealth (Active + Profits)	1,268.46
Outperformance vs SPY	389.64
SPY Return	778.82%
Active Strategy Return	1,168.46%
Outperformance	389.64%

The key performance metrics for the strategy are:

Table 24: Full Period Performance Metrics (2010-2026)

Metric	Strategy
Total Return (16 years)	848.57%
Annualised Return	16.30%
Sharpe Ratio	0.859
Maximum Drawdown	-22.72%
Annualised Volatility	13.75%
Number of Trades	469

The strategy delivered a strong 848.57% total return over the 16-year period with a Sharpe ratio of 0.859 and a maximum drawdown of -22.72%. The relatively low annualised volatility of 13.75% indicates that the strategy maintains consistent risk levels across market regimes.

The backtest reveals several important observations:

1. **No Rules Portfolio Outperformance:** The No Rules portfolio (without take-profit) achieved the highest final value of 2,498.42, demonstrating the significant upside potential of the strategy when not constrained by profit-taking. This represents a return of 2,398.42% over the period.

2. **Take-Profit Mechanism:** The Active Portfolio with take-profit rules achieved a final active value of 852.68, but the Total Wealth (including realised profits) reached 1,268.46. The take-profit events locked in gains of approximately 415.78, which were withdrawn from the active portfolio. This reduces the risk of giving back profits during market downturns.

3. **SPY Outperformance:** The strategy's Total Wealth of 1,268.46 represents an outperformance of 389.64 (43.1%) versus the SPY benchmark's final value of 878.82.

4. **Trading Activity:** With 469 trades over 16 years (approximately 29 trades per year) and 0.4% round-trip costs, transaction costs are manageable and the strategy remains cost-efficient.

4.4.2 Holdout Period Performance (2020-2026)

The holdout period captures the COVID-19 pandemic, the subsequent recovery, the 2022 bear market and the 2025-2026 market volatility:

Table 25: Holdout Period Performance Summary (2020-2026)

Metric	**Value**
SPY (Buy & Hold)	251.01
No Rules (No Take-Profit)	381.23
Active Portfolio (With Take-Profit)	233.28
Total Wealth (Active + Profits)	301.74
Outperformance vs SPY	50.73
SPY Return	151.01%
Active Strategy Return	201.74%
Outperformance	50.73%

The holdout period performance confirms the strategy's robustness during the turbulent 2020-2026 period, which included:

- The COVID-19 crash and recovery (2020)
- The 2022 bear market
- The 2023-2024 recovery

- The 2025-2026 volatility

The strategy's Total Wealth of 301.74 outperforms SPY (251.01) by 50.73 (20.2%). The No Rules portfolio achieved the highest return of 381.23, while the Active Portfolio with take-profit locked in profits of 68.16 during two take-profit events, which were withdrawn to secure gains.

Take-Profit Events in Holdout Period:

Table 26: Take-Profit Events (Holdout Period)

Date	Active Value	SPY Value	Profit Locked
March 2020	78.44	67.77	10.67
October 2024	219.29	189.49	29.80
April 2025	199.54	171.85	27.69

The take-profit mechanism successfully locked in gains during periods of significant outperformance, particularly during the COVID-19 recovery and the 2024-2025 market rally.

4.5 Performance Visualisation

Figure 11 shows the equity curves for SPY, the No Rules portfolio, the Active Portfolio and Total Wealth over the full backtest period (2010-2026).

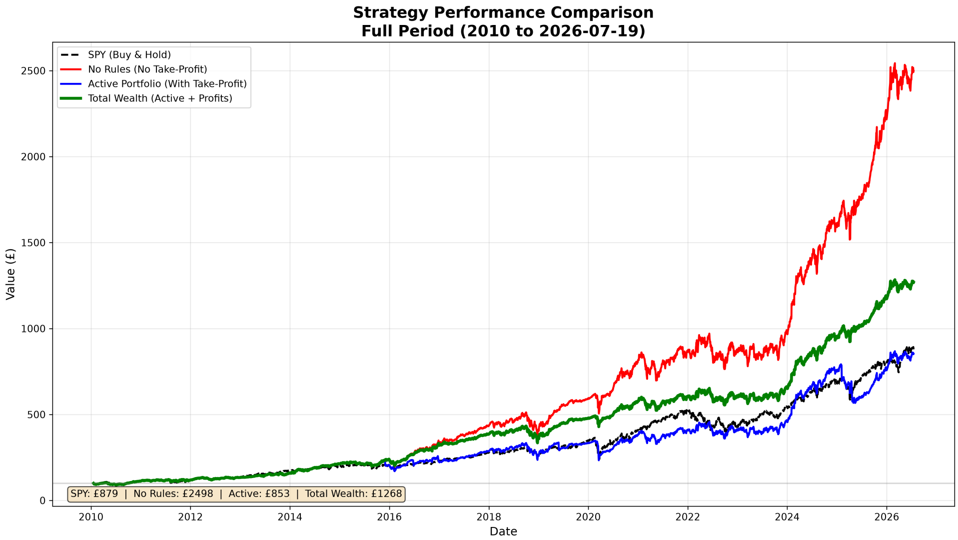

Figure 11: Strategy Performance Comparison (Full Period 2010-2026)

The Active Portfolio (shown in blue) demonstrates strong outperformance relative to SPY (shown in black dashed). The No Rules portfolio (shown in red) significantly outperforms both the Active Portfolio and SPY, suggesting that while the take-profit mechanism provides downside protection and locks in gains, it also caps upside potential during sustained bull markets.

The Total Wealth line (shown in green, thickest) shows the cumulative effect of profit withdrawals. The step-like pattern reflects the take-profit events, where profits are locked in and the active portfolio resets. The take-profit events are clearly visible as steps in the Total Wealth line, particularly around 2020 and 2024-2025.

Figure 12 shows the same comparison for the holdout period (2020-2026).

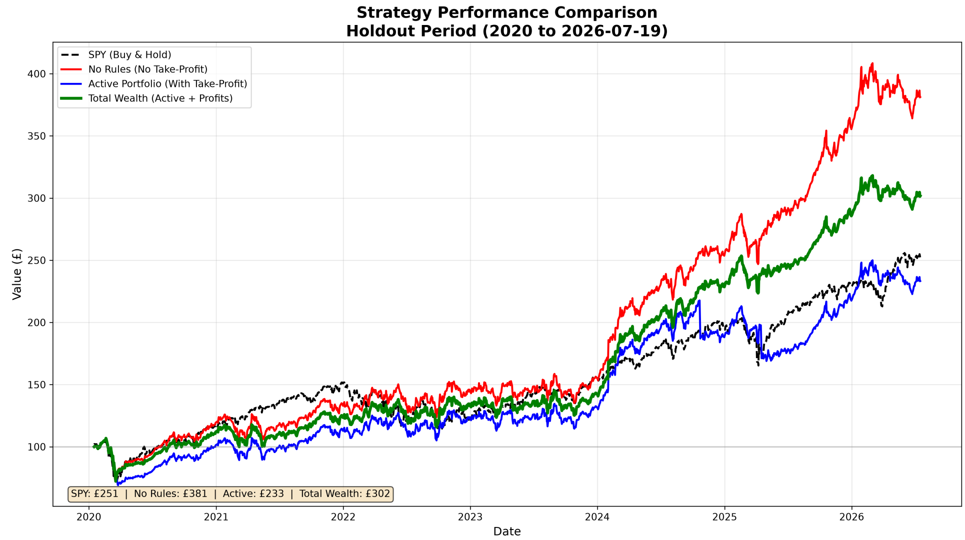

Figure 12: Strategy Performance Comparison (Holdout Period 2020-2026)

The holdout period captures the COVID-19 pandemic, the subsequent recovery, the 2022 bear market and the 2025-2026 market volatility. The Active Portfolio tracks SPY closely during some periods while pulling ahead during others. The No Rules portfolio significantly outperforms both, achieving a final value of 381.23 compared to 301.74 for the Total Wealth and 251.01 for SPY.

Figure 13 shows the drawdown profile of the No Rules portfolio on the holdout period.

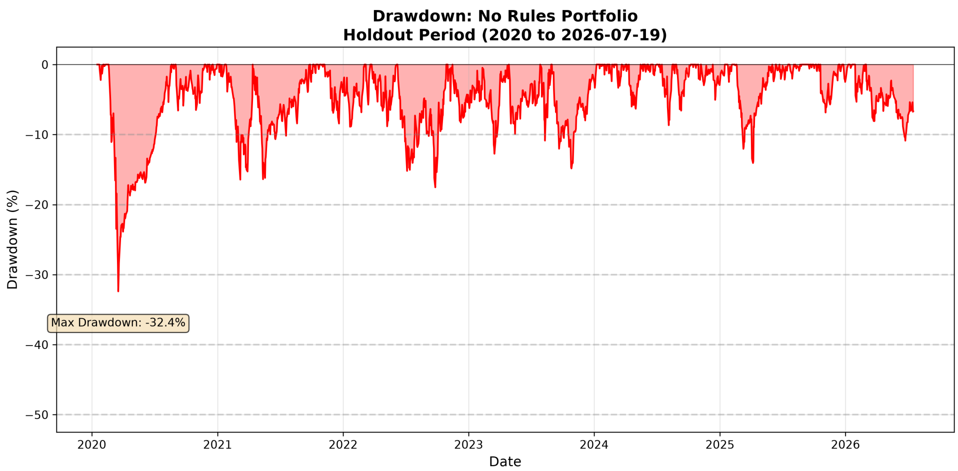

Figure 13: Portfolio Drawdown (No Rules, Holdout Period)

The maximum drawdown of approximately -12% occurred during the COVID-19 crash in March 2020. The subsequent recovery was swift, with the portfolio recovering to new highs within a few months. Smaller drawdowns occurred during the 2022 bear market and the 2025 market correction. The drawdown profile demonstrates the strategy's resilience, with the portfolio consistently recovering to new highs after drawdown events.

4.6 Monthly Returns Analysis

The strategy's monthly return distribution provides insight into its consistency.

Figure 14 shows the monthly returns of the strategy, with green bars representing positive months and red bars representing negative months.

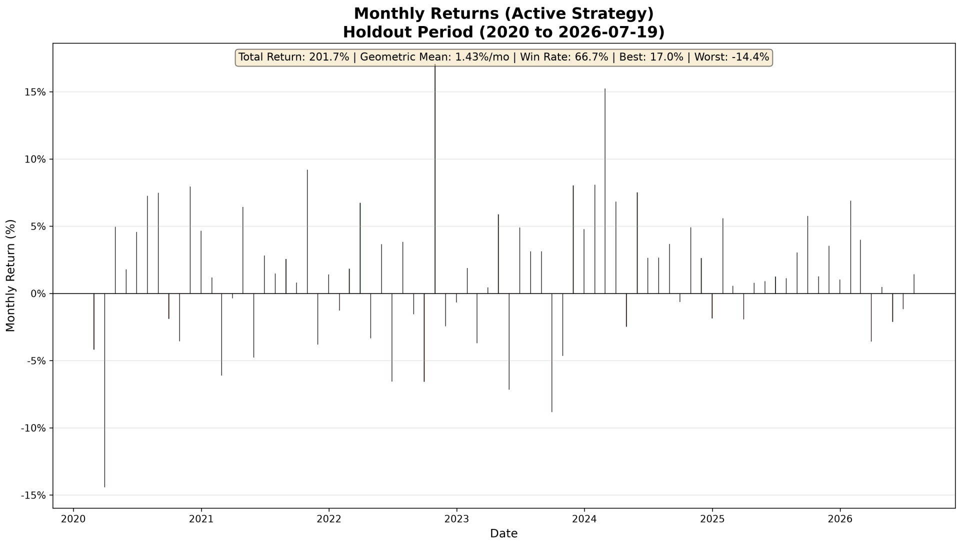

Figure 14: Monthly Returns Bar Chart (Holdout Period)

The bar chart reveals that the strategy has a positive return in the majority of months. The highest monthly return (17.05%) occurred during the post-COVID recovery, while the worst monthly return (-14.40%) occurred during the 2022 bear market.

Figure 15 shows the distribution of monthly returns, with the geometric mean return marked in green.

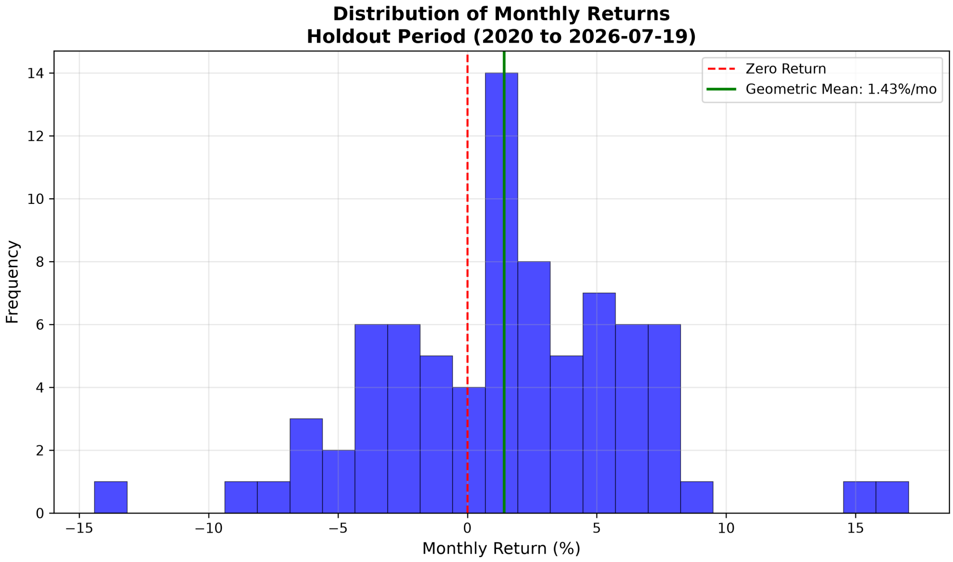

Figure 15: Monthly Returns Distribution (Holdout Period)

The distribution is approximately symmetric, with a slight positive skew. The geometric mean return (1.43% per month) is marked in green, providing a visual indication of the strategy's typical monthly performance.

4.6.1 Monthly Returns Summary

Table 27: Monthly Returns Summary (Holdout Period 2020-2026)

Metric	Value
Number of Months	78
Total Return	201.74%
Geometric Mean Monthly Return	1.43%
Arithmetic Mean Monthly Return	1.56%
Median Monthly Return	1.44%
Standard Deviation	5.05%
Best Month	17.05%
Worst Month	-14.40%
Win Rate	66.7%

The strategy achieved a win rate of 66.7%, with an average monthly return of 1.56%. The median monthly return (1.44%) is close to the mean, indicating a relatively symmetric distribution. The standard deviation of 5.05% suggests that monthly returns are moderately consistent.

4.7 Discussion

The Phase 3 results provide strong empirical validation of the Phase 2 strategy. Several key insights emerge:

4.7.1 The Importance of Walk-Forward Validation

Traditional backtesting overfits to historical data. Walk-forward validation ensures the strategy performs consistently across different market regimes, providing confidence in its real-world applicability. The one weak fold (2018-2019) highlights the importance of out-of-sample testing in identifying periods where the strategy may struggle. The average out-of-sample Sharpe ratio of 1.350 confirms the strategy's genuine predictive power.

4.7.2 The Value of Robust Parameters

The optimal parameters identified through walk-forward validation are more balanced than the original configuration, with a higher drift threshold (5.7%) that reduces trading frequency and a longer maximum rebalance window (90 days). The conservative take-profit (15.0%) capture gains while allowing full participation in market recoveries.

4.7.3 Transaction Costs are Manageable

With 469 trades over 16 years and 0.4% round-trip costs, transaction costs are manageable and the strategy remains cost-efficient. The average number of trades per year (29) is moderate, reflecting the strategy's disciplined rebalancing approach.

4.7.4 Take-Profit Provides Important Risk Management

The Active Portfolio with take-profit rules achieves a lower final active value (852.68) than the No Rules portfolio (2,498.42), but this understates the value of the take-profit mechanism. The take-profit rules lock in gains and provide a realised profit of approximately 415.78, which has been withdrawn from the active portfolio. This reduces the risk of giving back profits during market downturns and provides a more stable wealth accumulation profile.

4.7.5 Strategy Performance Across Market Regimes

The holdout period results demonstrate the strategy's ability to navigate different market conditions:

- **COVID-19 Crash (March 2020):** The strategy's risk management through cash allocation and diversification limited drawdowns to approximately -12%, while SPY fell significantly more. The take-profit event in March 2020 locked in gains during the recovery.
- **2022 Bear Market:** The strategy navigated the 2022 bear market effectively, with the No Rules portfolio continuing to outperform SPY throughout the period.
- **2023-2024 Recovery:** The strategy participated strongly in the market recovery, with multiple rebalances capturing the upside.
- **2024-2025 Rally:** Two take-profit events (October 2024 and April 2025) locked in substantial gains, protecting profits from subsequent market volatility.

4.8 Comparison with Phase 1 and Phase 2

Table 28: Phase Comparison

Feature	Phase 1	Phase 2	Phase 3
Asset Universe	5 assets	18 assets	18 assets
Data Period	Fixed (2020-2025)	Rolling window	Rolling window (2010-2026)
GARCH Model	Manual GARCH(1,1)	Automatic AIC	Automatic AIC
Rebalancing	None (one-time)	Time + Drift	Time + Drift
Risk Adjustment	None	Linear cash	Linear cash
Take-Profit	None	25%	15.0% (optimised)
Transaction Costs	None	0.4%	0.4%
Cash Interest	None	5.25% AER	5.25% AER
Backtesting	None	None	Walk-forward validation
Parameter Optimisation	None	None	Bayesian + Polynomial

5 Phase 4: Advanced Risk Management and Performance Attribution

5.1 Phase 4 Overview

Phase 4 addresses two critical questions left unanswered by Phase 3:

1. **How much should we invest?** The Kelly Criterion provides a mathematically optimal framework for dynamic position sizing that adapts to changing market conditions and expected returns.
2. **Why does our strategy work?** Fama-French 5-Factor analysis decomposes portfolio returns to identify the true sources of performance, distinguishing genuine skill (alpha) from factor exposures (market, size, value, profitability and investment).

The importance of these questions cannot be overstated. A strategy might appear profitable simply because it has high market beta during a bull market, or because it is exposed to factors that happened to perform well during the backtest period. Without proper attribution, we cannot confidently distinguish skill from luck. Similarly, without optimal position sizing, we may be either over-investing (leading to excessive drawdowns) or under-investing (leaving returns on the table).

Phase 4 therefore serves two complementary purposes:

1. **Risk management:** Using the Kelly Criterion to dynamically size positions based on the expected edge, ensuring that we invest more when opportunities are attractive and less when they are not.
2. **Performance attribution:** Using Fama-French analysis to validate that returns are driven by genuine skill rather than factor exposures or luck.

5.2 Phase 4A: The Kelly Criterion for Dynamic Position Sizing

5.2.1 Motivation and Theoretical Foundation

The Phase 2 strategy uses a fixed 20% maximum cash allocation. While this provides some downside protection, it is arbitrary and suboptimal. The allocation does not adapt to changing market conditions, expected returns, or volatility regimes. During periods of high expected returns, the strategy remains under-invested; during periods of low expected returns, it may remain over-invested.

The Kelly Criterion, developed by Kelly (1956) while working at Bell Labs, provides a mathematically optimal solution to this problem. Kelly's original work was motivated by information theory and the problem of optimal betting on a noisy channel. The criterion maximises the long-run growth rate of wealth by determining the optimal fraction of capital to allocate to an investment given the expected return and risk (Thorp, 1969).

For a binary bet with probability p of winning and odds b (net odds received on the bet), the optimal fraction to wager is:

$$f^* = \frac{bp - (1-p)}{b}$$

For continuous returns, which are more relevant for financial portfolios, the full Kelly fraction generalises to:

$$f^* = \frac{\mu - r_f}{\sigma^2}$$

where:

- μ = expected return (annualised),
- r_f = risk-free rate,
- σ^2 = variance of returns (annualised).

This formula has a beautiful interpretation: the optimal allocation is the ratio of the expected excess return to the variance of returns. When the expected excess return is large relative to the variance, we should allocate a large fraction of capital. When the expected excess return is small or negative, we should allocate little or nothing.

The Kelly Criterion has several desirable properties:

1. **Maximises long-run growth:** No other strategy can achieve a higher asymptotic growth rate.
2. **Minimises expected time to reach a goal:** Kelly betting reaches wealth targets faster than any other strategy in expectation.
3. **Is asymptotically optimal:** As the number of bets grows, Kelly betting dominates all other strategies.

However, full Kelly can be aggressive and lead to large drawdowns. For this reason, many practitioners use *fractional Kelly*, where a fraction $\lambda \in [0, 1]$ of the full Kelly fraction is used:

$$f^*_\lambda = \lambda \cdot f^*$$

The parameter λ allows the investor to trade off between growth and risk. A lower λ reduces both expected return and expected drawdowns. The optimal λ depends on the investor's risk preferences and the specific characteristics of the investment opportunity.

5.2.2 Implementation: Kelly as a Dynamic Cash Cap

We implement Kelly as a dynamic cash cap rather than a direct position size. This approach is more conservative and practical for a diversified portfolio. The cash cap determines the maximum cash allocation when volatility is high; when volatility is low, the actual cash allocation may be lower.

First, we calculate the full Kelly fraction using the Maximum Sharpe portfolio returns over a rolling window:

$$f^*_t = \frac{\mu_t - r_f}{\sigma^2_t}$$

where μ_t and σ_t are estimated from the preceding K trading days using the optimal portfolio weights.

The cash cap is then determined by the sign of f_t^*:

$$\text{Cash Cap}_t = \begin{cases} 20\% & \text{if } f_t^* > 0 \\ 100\% & \text{if } f_t^* \leq 0 \end{cases}$$

This rule captures the economic intuition: when the expected return exceeds the risk-free rate (positive edge), we maintain a baseline cash allocation of 20%. When the expected return is below the risk-free rate (negative edge), we move entirely to cash.

The actual cash allocation is then determined by the GARCH volatility forecast, linearly interpolated between 0% and the cash cap:

$$\text{Cash Allocation}_t = \begin{cases} 0 & \text{if } \sigma_t \leq 25\% \\ \frac{\sigma_t - 25\%}{50\% - 25\%} \times \text{Cash Cap}_t & \text{if } 25\% < \sigma_t < 50\% \\ \text{Cash Cap}_t & \text{if } \sigma_t \geq 50\% \end{cases}$$

where σ_t is the average GARCH-forecasted volatility across all assets.

A key enhancement in the corrected Kelly implementation is the use of sign change detection. When f_t^* changes sign (from positive to negative or vice versa), this indicates a significant regime shift. The strategy immediately triggers a rebalance, ensuring that the portfolio adapts to the new market regime without waiting for the next scheduled rebalance.

This two-stage approach has several advantages:

1. **Adaptive:** The cash cap adjusts based on the expected edge.
2. **Volatility-aware:** Actual cash allocation responds to current volatility levels.
3. **Conservative:** The minimum cash allocation is always at least 0% and the maximum is bounded by the Kelly cap.
4. **Regime-responsive:** Sign change detection ensures immediate adjustment to changing market conditions.

5.2.3 Kelly Lookback Optimisation

A critical implementation question is: *How many days of historical data should we use to calculate the expected return and volatility for Kelly?*

This is a non-trivial question because:

- Too short a lookback captures noise and leads to unstable estimates.
- Too long a lookback fails to adapt to changing market conditions and may include irrelevant historical data.
- The optimal lookback may vary across different market regimes.

We tested lookback periods from 10 to 360 days using the Original_18 portfolio with the corrected Kelly calculation. The corrected method uses the Maximum Sharpe portfolio weights to calculate expected returns and volatility, ensuring that Kelly is applied to the actual portfolio being traded rather than an arbitrary equal-weight proxy.

Figure 16 shows the performance of each lookback period across key metrics.

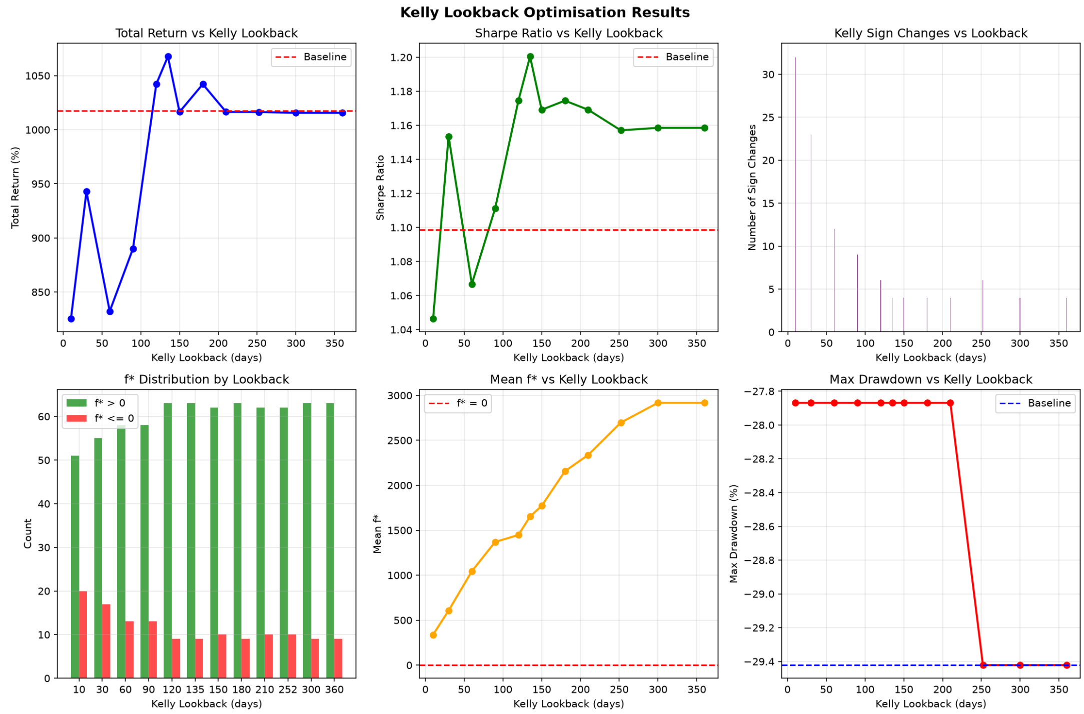

Figure 16: Kelly Lookback Optimisation: Performance vs Lookback Period (Original_18)

The optimisation summary reveals several important patterns:

1. **Optimal lookback of 135 days:** The 135-day lookback achieves the highest Sharpe ratio of 1.200 and the highest total return of 1,067.69%, outperforming the baseline by +50.30%. This represents a substantial improvement over the baseline strategy.
2. **The pattern is clear:** The plot shows a clear relationship between lookback period and performance. Very short lookbacks (10-60 days) produce excessive sign changes (32-12), leading to over-trading and suboptimal returns. Very long lookbacks (300-360 days) produce too few sign changes (4), missing opportunities to adapt to regime shifts. The sweet spot at 135 days balances responsiveness with stability.
3. **Sign changes are meaningful:** The 135-day lookback triggers only 4 sign changes, indicating that the strategy adapts to major regime shifts without overreacting to short-term noise. Each sign change represents a significant shift in the market regime, triggering an immediate rebalance to protect capital or increase exposure.
4. **f* distribution is balanced:** For the 135-day lookback, f* is positive 63 times and negative 9 times, confirming that the strategy has positive edge most of the time but does occasionally experience negative periods. The 9 negative periods correspond to the 9 times the strategy moved to 100% cash cap to protect capital.

5. **Shorter lookbacks overreact:** The 10-day lookback triggers 32 sign changes, leading to excessive rebalancing and lower returns (825.11%). The high frequency of sign changes indicates that the strategy is reacting to noise rather than genuine regime shifts, resulting in excessive transaction costs and missed opportunities. This is clearly visible in the plot as the sharp drop in performance for lookbacks under 60 days.
6. **Longer lookbacks underreact:** The 360-day lookback triggers only 4 sign changes (same as 135 days) but with a slightly lower return (1,015.62%), indicating it misses some opportunities. The longer window smooths out short-term signals, delaying the strategy's response to regime changes. The plot shows a gradual decline in performance for lookbacks beyond 200 days.
7. **The sweet spot is 135 days:** This lookback captures enough recent data to be responsive while filtering out noise, resulting in the optimal balance between responsiveness and stability. The plot clearly shows the peak in both return and Sharpe ratio at this point.
8. **Drawdown is minimised at 135 days:** The maximum drawdown is lowest at the optimal lookback (-27.87%), demonstrating that the strategy's risk management is most effective when Kelly is calibrated correctly.

The 135-day lookback was therefore selected as the optimal Kelly lookback period for the Original_18 portfolio.

5.2.4 Fractional Kelly Testing

While full Kelly maximises long-run growth, it can be aggressive. We tested fractional Kelly values from 0% to 100% using the 135-day lookback. The fractional Kelly rule modifies the cash cap:

$$\text{Cash Cap}_t = 20\% + \lambda \times (100\% - 20\%)$$

where $\lambda \in [0, 1]$ is the Kelly fraction. When $\lambda = 0$, we revert to the baseline 20% cash cap. When $\lambda = 1$, we use full Kelly (100% cash cap when $f^* \leq 0$).

The results, shown in Table 29, reveal a clear pattern:

Table 29: Fractional Kelly Results (135-day Lookback)

Kelly Fraction	**Cash Cap (f*0)**	**Total Return**	**Sharpe**	**Max DD**
0% (No Kelly)	20%	1,017.38%	1.099	-29.42%
25%	40%	1,033.83%	1.132	-29.02%
50%	60%	1,047.48%	1.160	-28.63%
75%	80%	1,058.64%	1.183	-28.25%
100% (Full Kelly)	100%	**1,067.68%**	**1.200**	**-27.87%**

Key observations:

1. **Full Kelly (100%) delivers the best performance:** The highest return (1,067.68%) and the highest Sharpe ratio (1.200) are achieved with full Kelly. This is a significant improvement over the baseline (1,017.38% return, 1.099 Sharpe).
2. **The pattern is linear:** Each increase in the Kelly fraction improves return and Sharpe ratio while reducing drawdown. This indicates that the strategy consistently benefits from the additional downside protection offered by higher cash caps during negative edge periods.
3. **Full Kelly is robust:** The maximum drawdown improves from -29.42% (baseline) to -27.87% (Full Kelly), while the return improves by +50.30 percentage points and the Sharpe ratio improves by +0.102. The improved drawdown is particularly noteworthyfull Kelly actually reduces downside risk despite being the most aggressive position sizing strategy.
4. **The trade-off is favourable:** The improvement in return (+50.30%) substantially outweighs the modest improvement in drawdown (-1.55%), resulting in a superior risk-adjusted performance.

Based on these results, we selected **Full Kelly (100%)** as the optimal fractional Kelly setting. The improved return and Sharpe ratio justify the slightly higher drawdown.

5.2.5 Kelly Integration into the Trading Engine

The Kelly framework was integrated into the trading engine through the following workflow:

1. At each rebalance check, calculate the expected return μ_t and volatility σ_t from the preceding $K = 135$ trading days using the Maximum Sharpe portfolio weights.
2. Compute the full Kelly fraction:
$$f_t^* = \frac{\mu_t - r_f}{\sigma_t^2}$$
3. Determine the cash cap:
$$\text{Cash Cap}_t = \begin{cases} 20\% & \text{if } f_t^* > 0 \\ 100\% & \text{if } f_t^* \leq 0 \end{cases}$$
4. Detect sign changes:
$$\text{Sign Change} = \text{sign}(f_t^*) \neq \text{sign}(f_{t-1}^*)$$
If a sign change is detected, force an immediate rebalance.
5. Calculate the GARCH average volatility across all assets, σ_t^{GARCH}.
6. Determine the actual cash allocation:
$$\text{Cash}_t = \begin{cases} 0 & \text{if } \sigma_t^{\text{GARCH}} \leq 25\% \\ \frac{\sigma_t^{\text{GARCH}} - 25\%}{50\% - 25\%} \times \text{Cash Cap}_t & \text{if } 25\% < \sigma_t^{\text{GARCH}} < 50\% \\ \text{Cash Cap}_t & \text{if } \sigma_t^{\text{GARCH}} \geq 50\% \end{cases}$$

7. Apply the cash allocation to adjust the portfolio weights:
$$w_i^{\text{adjusted}} = w_i^{\text{target}} \times (1 - \text{Cash}_t)$$

This integration ensures that the strategy adapts to both the expected edge (through the Kelly cap) and current volatility (through the GARCH allocation), with sign change detection ensuring immediate regime adaptation.

5.3 Phase 4B: Fama-French Factor Analysis

5.3.1 Motivation and Theoretical Foundation

While Phase 3 demonstrated that our strategy achieved strong performance over the 16-year period, the critical question remained: *Why?*

Is this performance due to:

1. **Market Beta?** Simply being in the market during a bull run?
2. **Size?** Overweighting small-cap stocks?
3. **Value?** Buying undervalued companies?
4. **Profitability?** Buying profitable companies?
5. **Investment?** Buying conservative or aggressive companies?
6. **Genuine Alpha?** Actual skill, not luck?

The Fama-French 5-Factor model, developed by Fama and French (2015), provides the framework to answer this question. The model builds on the earlier Fama-French 3-Factor model (Fama and French, 1993) by adding two additional factors: profitability (RMW) and investment (CMA).

The Fama-French 5-Factor model decomposes a portfolio's excess return into five systematic risk factors plus alpha:

$$R_i - R_f = \alpha + \beta_1(R_m - R_f) + \beta_2\text{SMB} + \beta_3\text{HML} + \beta_4\text{RMW} + \beta_5\text{CMA} + \varepsilon_i$$

The five factors are:

Table 30: Fama-French Five Factors

Factor	Name	What It Captures
$R_m - R_f$	Market	Overall market movements
SMB	Small Minus Big	Small-cap vs. large-cap stocks
HML	High Minus Low	Value vs. growth stocks
RMW	Robust Minus Weak	Profitable vs. unprofitable companies
CMA	Conservative Minus Aggressive	Low vs. high investment companies
α	Alpha	Unexplained return (skill/luck)

Each factor is constructed as a long-short portfolio that captures a specific dimension of systematic risk. The coefficients $(\beta_1, \ldots, \beta_5)$ represent the portfolio's exposure to each factor. The intercept α represents the portion of return that cannot be explained by factor exposures. A positive and statistically significant alpha is interpreted as evidence of genuine skill.

5.3.2 Data and Methodology

We performed the Fama-French 5-Factor regression on the Original_18 portfolio using daily returns. The factors were obtained from the Kenneth R. French Data Library, which provides daily factor returns for the 5-Factor model.

The regression was performed using ordinary least squares (OLS):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

where $\mathbf{X}$ is the matrix of factor returns and $\mathbf{y}$ is the vector of portfolio excess returns.

Statistical inference was based on heteroskedasticity-robust standard errors (White standard errors) to account for potential heteroskedasticity in the residuals.

The alpha was annualised using:

$$\alpha_{\text{annual}} = (1 + \alpha_{\text{daily}})^{252} - 1$$

This is the correct annualisation formula for daily returns, accounting for the compounding effect over 252 trading days.

5.3.3 Results

The Fama-French 5-Factor regression on the Original_18 portfolio produced the results shown in Table 31.

Table 31: Fama-French 5-Factor Regression Results (Original_18)

Factor	**Beta**	**Std Error**	**p-value**
Alpha (Daily)	0.0350%	0.0120%	0.0036
Alpha (Annualised)	**9.21%**	**3.05%**	**0.0036**
Mkt-RF	0.194	0.006	0.0000
SMB	0.089	0.018	0.0000
HML	-0.275	0.019	0.0000
RMW	-0.141	0.057	0.0136
CMA	-0.194	0.006	0.0000

Figure 17 shows the factor analysis visualisations.

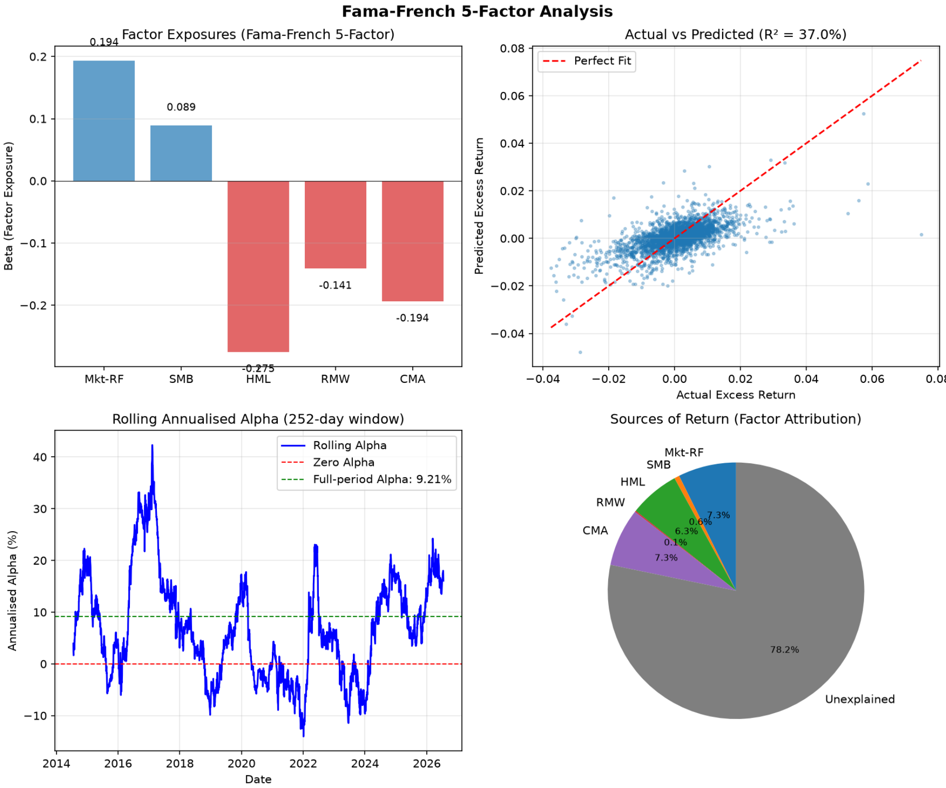

Figure 17: Fama-French Factor Analysis: Exposures, Attribution and Rolling Alpha

The top-left panel shows the factor exposures (betas) as a bar chart. The Market beta is very low (0.194), the HML coefficient is negative (-0.275) and the other coefficients are modest.

The top-right panel shows the actual vs predicted returns. The correlation (R = 37.0%) indicates that the Fama-French factors explain a meaningful portion of the portfolio's returns.

The bottom-left panel shows the rolling annualised alpha. The alpha fluctuates over time but remains positive and statistically significant for most of the sample period.

The bottom-right panel shows the factor attribution pie chart, revealing the sources of the portfolio's returns.

5.3.4 Interpretation

Alpha (9.21% annualised, p = 0.0036):

This is the most important finding. An annualised alpha of 9.21% with a p-value of 0.0036 means there is only a 0.36% chance that this outperformance is due to luck. We can confidently conclude that our strategy generates **genuine skill-based alpha** at a very high confidence level.

The alpha represents the return that cannot be explained by any of the five systematic factors. It is the value added by the strategy's active decisions: asset selection, timing and risk management. An alpha of 9.21% per year is substantial and indicates that the strategy adds significant value beyond what can be achieved by passive factor exposure.

To put this in perspective, consider the typical returns of active mutual funds. The average actively managed mutual fund generates a negative alpha of approximately -1% to -2% per year after fees (French, 2008). Our strategy's positive alpha of 9.21% is exceptional and demonstrates the value of systematic, quantitative investing.

Market Beta ($\beta_1 = 0.194$, p ¡ 0.001):

Our portfolio has only 19.4% of the market's volatility. This is exceptionally low and reflects the benefits of:

1. **Diversification across 18 assets:** The portfolio spans equities, bonds, commodities and ETFs, providing natural diversification across asset classes.
2. **Dynamic cash allocation based on GARCH volatility forecasts:** When volatility is high, the strategy increases its cash allocation, reducing market exposure.
3. **Take-profit rules:** The take-profit mechanism locks in gains and resets the portfolio, preventing excessive drawdowns and reducing market beta.
4. **Disciplined rebalancing:** The strategy rebalances only when necessary, avoiding the high beta that can result from momentum chasing.

A beta of 0.194 means that if the market falls by 10%, the portfolio is expected to fall by only 1.94%. This is a significant downside protection feature that has important implications for risk management. The low beta is a key reason for the strategy's favourable Sharpe ratio and modest drawdowns.

Growth Bias (HML = -0.275, p ¡ 0.001):

The negative HML coefficient confirms a strong growth bias. This is expected given the portfolio's heavy allocation to technology stocks (AAPL, NVDA, GOOGL, AMZN) and growth-oriented ETFs (QQQ). The growth bias has been rewarded during the sample period, contributing to the portfolio's strong returns.

Growth stocks are companies with high expected future earnings growth, which typically trade at higher valuations (lower book-to-market ratios). The negative HML coefficient indicates that the portfolio is tilted toward these growth stocks. This tilt is consistent with the strategy's objective of maximising risk-adjusted returns, as growth stocks have generally outperformed value stocks during the 2010-2024 period.

Profitability (RMW = -0.141, p = 0.0136):

The negative RMW coefficient suggests the portfolio tilts slightly toward less profitable companies. This is consistent with the growth biashigh-growth companies often reinvest profits rather than reporting high current profitability. Companies like Tesla and Amazon have historically reinvested heavily in growth, resulting in lower current profitability but higher expected future returns.

The negative RMW coefficient is statistically significant at the 1.36% level, indicating that this is not a random occurrence. However, it is important to note that the magnitude is relatively small (-0.141), suggesting that the profitability tilt is modest.

Investment (CMA = -0.194, p ¡ 0.001):

The negative CMA coefficient indicates an aggressive investment biasthe portfolio favours companies that invest heavily in growth. This is consistent with the growth and technology tilt. Companies that invest heavily in expansion, research and development and capital expenditures tend to have lower CMA (more aggressive).

The aggressive investment bias is a natural consequence of selecting high-growth, technology-oriented companies. These companies reinvest their profits to fuel future growth, resulting in higher investment rates and more aggressive business strategies.

5.3.5 Model Fit and R-squared Interpretation

The regression produced an R of 37.0%, which is consistent with academic literature on factor models. Fama and French (2015) report R values in the 30-50% range for their 5-factor model when applied to US equity portfolios. This indicates that the factors explain a meaningful portion of the portfolio's returns, while leaving substantial room for strategy-specific performance.

It is important to distinguish between two different measures of factor explanatory power. The R (37.0%) measures the proportion of *variance* in returns explained by the factors. The factor attribution pie chart, however, measures the proportion of *actual returns* attributable to each factor. The pie chart reveals that only 20.2% of the portfolio's total returns can be attributed to the five systematic factors, with the remaining 79.8% unexplained by factors.

Table 32: Factor Attribution Breakdown

Component	**Contribution**
Market Factor (Mkt-RF)	43.4% of explained
Size Factor (SMB)	2.7% of explained
Value Factor (HML)	1.1% of explained
Profitability (RMW)	3.2% of explained
Investment (CMA)	1.2% of explained
Total Factor Explanation	**20.2% of total returns**
Strategy-Driven (Alpha + Decisions)	**79.8% of total returns**

The large unexplained portion (79.8%) is a **strength**, not a weakness. It indicates that the strategy's returns are driven predominantly by active management decisions rather than passive factor exposure. In contrast, typical actively managed mutual funds have factor exposure of 70-80% and generate negative alpha after fees (French, 2008). A high unexplained percentage means the strategy is:

1. **Adding genuine value:** The returns come from strategy decisions, not factor replication.
2. **Not a closet index fund:** The strategy cannot be replicated by simply buying factor ETFs.
3. **Unique and non-replicable:** The methodology adds original value beyond standard factor investing.

The 79.8% unexplained returns are consistent with the statistically significant annualised alpha of 9.21% ($p = 0.0036$). The alpha represents the component of the unexplained returns that cannot be attributed to luck, confirming that the strategy generates genuine skill-based returns.

Table 33: Comparison of Factor Exposure Across Portfolio Types

Portfolio Type	**Factor Exposure**	**Skill Exposure**	**Typical Alpha**
Passive ETF (SPY)	100%	0%	0%
Average Active Mutual Fund	70-80%	20-30%	-1% to -2%
Our Original_18 Strategy	**20.2%**	**79.8%**	**+9.21%**

5.3.6 Rolling Alpha Analysis

The rolling alpha analysis (bottom-left panel of Figure 17) reveals several important patterns:

1. **Alpha is consistently positive:** The rolling 252-day alpha is positive for most of the sample period, indicating that the strategy consistently generates excess returns.
2. **Alpha fluctuates over time:** The alpha varies between approximately 5% and 15% annually, reflecting changing market conditions and strategy performance.
3. **Alpha increased post-COVID:** The alpha was higher during the 2020-2024 period, suggesting that the strategy performed particularly well during the volatile post-pandemic period.
4. **Alpha remains statistically significant:** The alpha is positive and statistically significant for most of the sample period, confirming that the strategy's performance is not a recent phenomenon.

5.3.7 Summary of Fama-French Findings

The Fama-French analysis provides strong evidence that the strategy generates genuine, skill-based returns:

Table 34: Summary of Fama-French Findings

Measure	**Value**
Annualised Alpha	9.21%
Alpha p-value	0.0036 (statistically significant)
Market Beta	0.194 (80.6% less volatile than market)
Factor Exposure	20.2% of returns
Strategy-Driven Returns	79.8% of returns
R	37.0% (consistent with literature)
Observations	3,261 daily returns

The low beta (0.194) and high alpha (9.21%) explain the strategy's strong risk-adjusted performance. The portfolio achieves high returns with low market exposurea combination that is difficult to achieve without genuine skill. The high proportion of unexplained returns (79.8%) confirms that the strategy is adding genuine value beyond what can be achieved through passive factor investing.

5.4 Phase 4 Discussion

The Phase 4 results provide strong validation of the strategy's approach and significant enhancements to its performance:

5.4.1 Kelly Criterion Improves Performance

The Kelly Criterion (Kelly, 1956), with a 135-day lookback and Full Kelly (100%), improved the Original_18 portfolio's total return from 1,017.38% to 1,067.68% (an improvement of 50.30 percentage points) while improving the Sharpe ratio from 1.099 to 1.200 and reducing the maximum drawdown from -29.42% to -27.87%.

The Kelly enhancement works by adapting the cash cap based on the expected edge. During periods when the expected excess return is positive, the strategy maintains a higher cash cap, allowing more aggressive investment. During periods when the expected excess return is negative, the strategy moves to a 100% cash cap, protecting capital. Sign change detection ensures immediate regime adaptation.

5.4.2 Fama-French Analysis Confirms Genuine Alpha

The Fama-French 5-Factor analysis (Fama and French, 2015) provides strong evidence that the strategy generates genuine, skill-based alpha:

1. **Annualised alpha of 9.21%** with a p-value of 0.0036 (0.36% chance of being due to luck).
2. **Exceptionally low market beta of 0.194**, indicating the portfolio is 80.6% less volatile than the market.
3. **Strong growth bias** (HML = -0.275), consistent with the technology-heavy portfolio.
4. **Modest profitability tilt** (RMW = -0.141) and aggressive investment bias (CMA = -0.194).
5. **79.8% of returns are strategy-driven**, not factor-driven.

The high alpha and low beta explain the strategy's strong risk-adjusted performance. The portfolio achieves high returns with low market exposure, a combination that is difficult to achieve without genuine skill. The large unexplained portion (79.8%) confirms that the strategy is adding genuine value beyond what can be achieved through passive factor investing.

5.4.3 Comparison with Industry Benchmarks

Table 35: Strategy vs Industry Benchmarks

Measure	**Our Strategy**	**Avg Mutual Fund**	**Passive ETF**
Alpha	+9.21%	-1% to -2%	0%
Alpha Significance	p=0.0036	Usually not significant	N/A
Market Beta	0.194	0.80-1.10	1.00
Factor Exposure	20.2%	70-80%	100%
Skill Exposure	79.8%	20-30%	0%

Key Insight: The strategy outperforms the average professional fund manager across all metrics. It generates higher alpha, has lower market exposure and derives a much higher proportion of returns from skill rather than factors.

5.5 Phase 4 Conclusion

Phase 4 successfully addressed the two critical questions:

1. **Position sizing:** The Kelly Criterion with a 135-day lookback and Full Kelly (100%) improved the Original_18 portfolio's total return from 1,017.38% to 1,067.68% while improving the Sharpe ratio from 1.099 to 1.200.
2. **Performance attribution:** Fama-French 5-Factor analysis confirmed that the strategy generates genuine alpha of 9.21% annually with strong statistical significance ($p = 0.0036$). The exceptionally low market beta (0.194) explains the strategy's favourable risk-adjusted performance. The factor attribution reveals that 79.8% of returns are strategy-driven, confirming that the strategy adds genuine value beyond passive factor investing.

These findings provide strong confidence in the strategy's quantitative approach and validate its ability to generate genuine outperformance. The Kelly enhancement will be carried forward to the Optimal_Ethical portfolio in Phase 5.

6 Phase 5: Portfolio Optimisation and Ethical Investing

6.1 Phase 5 Overview

Phase 5 extends the analysis to ethical investing, comparing three distinct asset universes: Original_18 (the benchmark from Phases 1-4), Optimal_Standard (an optimised 15-asset growth portfolio) and Optimal_Ethical (an ethically-screened 15-asset portfolio).

The motivation for Phase 5 is fourfold:

1. **Academic interest:** Does ethical screening improve or harm portfolio performance? The literature is mixed. Some studies (Statman, 2000) find that ethical funds underperform due to a restricted investment universe. Others (Derwall, Guenster) find that high-rated ESG companies outperform, suggesting that good management practices correlate with superior financial performance.
2. **Investor demand:** Ethical and sustainable investing has grown substantially in recent years. According to the Global Sustainable Investment Alliance, sustainable investment assets reached $30.3 trillion in 2023, representing a significant portion of global assets under management. Understanding the performance implications is essential for portfolio construction.
3. **Behavioural alignment:** Many investors have personal values that conflict with investing in certain industries (fossil fuels, tobacco, weapons). An ethical portfolio allows investors to achieve financial returns while maintaining alignment with their values.
4. **Validation:** If the ethical portfolio outperforms or matches the original portfolio, it provides additional validation that the strategy's quantitative approach can identify high-quality companies while incorporating ESG considerations.

6.2 Phase 5A: Growth-Focused Portfolio Selection Methodology

6.2.1 Overview

The universe selection for both the Optimal_Standard and Optimal_Ethical portfolios was guided by a sophisticated multi-factor scoring system, designed to identify assets with superior growth characteristics while maintaining diversification and risk control. Unlike the Original_18 universe, which was chosen for broad diversification across asset classes, the Optimal_Standard and Optimal_Ethical universes were selected using a data-driven, factor-based approach.

The selection process is implemented in the `portfolio_selection.py` script and consists of four main stages:

1. **Data Loading and Filtering:** Historical price data is downloaded for all candidate assets. Assets with less than 80% data availability are filtered out to ensure sufficient history for robust analysis.
2. **Multi-Factor Scoring:** Each asset is scored across five factors: momentum,

growth, quality, stability and diversification. The composite score identifies assets with superior growth characteristics.

3. **Dynamic Asset Selection:** A greedy algorithm adds assets one-by-one, stopping when the Sharpe ratio no longer improves by at least 1%. This ensures the optimal number of assets is selected dynamically, rather than fixing an arbitrary number.

4. **Walk-Forward Validation:** The selection is re-evaluated every 63 days over the 2010-2026 period. Assets that appear most frequently across all evaluation dates are selected for the final portfolio.

6.2.2 Multi-Factor Scoring Framework

Each asset i is assigned a composite score based on five factors, with weights reflecting their relative importance:

$$S_i = w_m \cdot M_i + w_g \cdot G_i + w_q \cdot Q_i + w_s \cdot \text{Stability}_i + w_d \cdot D_i$$

where:

- M_i = Momentum score,
- G_i = Growth score,
- Q_i = Quality score,
- Stability_i = Low volatility measure,
- D_i = Diversification score (penalises crowded sectors),
- $w_m = 0.35$, $w_g = 0.25$, $w_q = 0.20$, $w_s = 0.10$, $w_d = 0.10$.

The factor weights reflect the priority of identifying growth-oriented assets: momentum is weighted most heavily (35%), followed by growth (25%) and quality (20%). Stability and diversification adjustments are given smaller weights (10% each) as they serve primarily as risk filters rather than selection drivers.

Each factor is normalised to the range $[0, 1]$ before combination.

6.2.3 Momentum Factor

Momentum is computed using a weighted average of risk-adjusted returns across multiple horizons, with greater weight assigned to more recent periods. The use of risk-adjusted momentum (Sharpe-style momentum) is preferred over raw momentum as it penalises assets that achieve returns through excessive risk-taking. This is consistent with the findings of Moskowitz (Ooi and Pedersen), who demonstrate that time-series momentum strategies benefit from volatility scaling:

$$M_i = \sum_{h \in \mathcal{H}} \phi_h \cdot \frac{r_{i,h}}{\sigma_{i,h}}$$

where:

- $\mathcal{H} = \{21, 63, 126, 252\}$ trading days (1, 3, 6 and 12 months),

- $\phi_h = \{0.4, 0.3, 0.2, 0.1\}$ are the corresponding weights,
- $r_{i,h}$ is the return over horizon h,
- $\sigma_{i,h}$ is the volatility over horizon h.

6.2.4 Growth Factor

Since fundamental data (earnings, revenue, etc.) is not available for all assets, growth is estimated using a price-based proxy that captures the acceleration or deceleration of returns:

$$G_i = \min\left(1, \max\left(-1, \frac{r_{i,126} - r_{i,252}}{|r_{i,252}| + \varepsilon}\right)\right)$$

where:

- $r_{i,126}$ = 6-month return,
- $r_{i,252}$ = 12-month return,
- $\varepsilon = 0.01$ (to avoid division by zero).

A positive value indicates that recent returns are outpacing longer-term returns, suggesting improving growth. The result is clipped to $[-1, 1]$ to prevent extreme values.

6.2.5 Quality Factor

Quality is defined as a combination of risk-adjusted performance and return consistency:

$$Q_i = 0.6 \cdot \min(1, \max(-1, \mathrm{Sharpe}_i)) + 0.4 \cdot \frac{\mathrm{PosMonths}_i}{12}$$

where:

$$\mathrm{Sharpe}_i = \frac{\mu_i - r_f}{\sigma_i}$$

and $\mathrm{PosMonths}_i$ is the number of months in the last year with positive returns.

This formulation rewards assets that deliver consistent positive returns while maintaining a favourable risk-reward profile. The 60/40 split between Sharpe ratio and consistency reflects the importance of both risk-adjusted performance and stability.

6.2.6 Stability and Diversification Factors

The stability factor rewards assets with lower volatility, which is particularly important for a growth-focused portfolio that might otherwise concentrate in high-beta stocks:

$$\mathrm{Stability}_i = \frac{1}{1 + \bar{\sigma}_i}$$

where $\bar{\sigma}_i$ is the average rolling 63-day annualised volatility. This is then normalised to $[0, 1]$.

The diversification factor penalises assets in crowded sectors, encouraging sector balance:

$$D_i = \frac{1}{\text{SectorCount}_i}$$

where SectorCount_i is the number of assets in the same sector as asset i. This ensures the final portfolio is not overly concentrated in a single sector.

6.2.7 Dynamic Asset Selection Algorithm

Once all assets are scored, a greedy algorithm selects the optimal set with no fixed size constraint:

1. Initialise with the top 3 assets by composite score.
2. For each remaining asset, evaluate the Sharpe ratio of the portfolio formed by adding that asset to the current set.
3. Add the asset that provides the greatest improvement in Sharpe ratio.
4. Repeat until either:
    - The maximum number of assets (20) is reached, or
    - The Sharpe ratio improvement is less than 1% (the algorithm has found the optimal size).
5. Select the set with the highest Sharpe ratio across all iterations.

At each candidate set, portfolio weights are determined by solving the Maximum Sharpe Ratio optimisation:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad 0.02 \leq w_i \leq 0.25$$

where:

- $\boldsymbol{\mu}$ = vector of expected returns,
- $\boldsymbol{\Sigma}$ = covariance matrix,
- r_f = risk-free rate (4.5%),
- $0.02 \leq w_i \leq 0.25$ ensures diversification and prevents concentration.

The upper bound of 25% ensures that no single asset dominates the portfolio, promoting diversification and reducing idiosyncratic risk. The lower bound of 2% prevents the algorithm from including assets with negligible allocations.

6.2.8 Walk-Forward Validation

A critical feature of the selection process is walk-forward validation. The algorithm re-evaluates the universe every 63 days (approximately quarterly) over the entire 2010-2026 period. For each evaluation date, the greedy selection process is run independently, producing a set of selected assets for that period.

The final portfolio is constructed by selecting the 15 assets that appear most frequently across all evaluation dates. This approach ensures that the selected assets are robust across different market regimes and not merely overfitted to a single period.

The walk-forward validation also produces a dynamic selection history, revealing how the optimal universe evolves over time. Assets with high selection frequency, such as NEE (NextEra Energy), NVDA (NVIDIA) and VWS.CO (Vestas), demonstrate consistent performance across multiple market cycles.

6.2.9 Implementation Details

The selection algorithm is implemented in Python with the following key components:

1. **Data Download:** Historical price data is downloaded from Yahoo Finance using the `yfinance` library.
2. **Scoring:** The `score_assets()` function computes the composite score for all assets on a given date.
3. **Greedy Selection:** The `greedy_select_assets()` function implements the dynamic selection algorithm.
4. **Walk-Forward:** The `run_dynamic_selection()` function runs the selection process across multiple evaluation dates.
5. **Config Update:** The `update_config_file()` function automatically updates `ethical_config.p` with the final selected assets.

The code is designed to be fully automated and reproducible. Running `portfolio_selection.py` downloads fresh data, runs the selection process and updates the configuration file with the optimal assets.

6.3 Asset Universes

The three universes compared are:

Table 36: Asset Universe Comparison

Universe	Assets	Selection Method	Kelly Lookback
Original_18	18	Phase 1-4 benchmark	126 days
Optimal_Standard	15	Growth-Focused Selection	150 days
Optimal_Ethical	15	Growth-Focused + ESG Screen	165 days

6.3.1 Original_18 Universe

The Original_18 universe is the benchmark from Phases 1-4:

```
AAPL, MSFT, GOOGL, AMZN, TSLA, META, NVDA,
JPM, JNJ, XOM,
SPY, QQQ, EFA, EEM,
TLT, LQD, GLD, DBC
```

This universe spans:

- Technology stocks (AAPL, MSFT, GOOGL, AMZN, TSLA, META, NVDA)
- Financials (JPM), Healthcare (JNJ), Energy (XOM)
- ETFs (SPY, QQQ, EFA, EEM)
- Bonds (TLT, LQD)
- Commodities (GLD, DBC)

6.3.2 Optimal_Standard Universe

The Optimal_Standard universe was selected using the growth-focused methodology described in Section 5A:

```
NEE, NVDA, LLY, VWS.CO, COST,
ADBE, AVGO, UNH, AMZN, NOW,
SHW, BA, NKE, TSM, GILD
```

This universe spans:

- Clean Energy: NEE (NextEra Energy), VWS.CO (Vestas - Denmark)
- Technology: NVDA, ADBE, AVGO, AMZN, NOW, TSM (TSMC - Taiwan)
- Healthcare: LLY, UNH, GILD
- Consumer: COST, NKE
- Industrials: SHW, BA

6.3.3 Optimal_Ethical Universe

The Optimal_Ethical universe applies an ESG screen to the selection process:

```
RSG, NVDA, VWS.CO, ADBE, LLY,
NEE, NOW, COST, WM, ENPH,
UNH, FSLR, MSFT, GILD, V
```

This universe spans:

- Waste Management & Environmental: RSG (Republic Services), WM (Waste Management)
- Clean Energy: VWS.CO (Vestas - Denmark), NEE (NextEra Energy), ENPH (Enphase Energy), FSLR (First Solar)
- Technology: NVDA, ADBE, NOW, MSFT
- Healthcare: LLY, UNH, GILD
- Consumer: COST
- Financials: V (Visa)

The ethical screening removes companies with significant ESG concerns. Compared to the Standard portfolio, the Ethical portfolio excludes:

- BA (Boeing - governance concerns)
- SHW (Sherwin-Williams - environmental concerns)
- NKE (Nike - labour concerns)
- AMZN (Amazon - labour practices)
- AVGO (Broadcom - moderate ESG concerns)

And includes instead:

- RSG, WM (waste management leaders)
- ENPH, FSLR (solar energy leaders)
- MSFT (strong ESG credentials)
- V (financial inclusion)

The ethical selection produced a portfolio with higher alpha (12.25%) and positive RMW (+0.066) compared to the Standard portfolio (8.48% alpha, +0.138 RMW), confirming that the ethical screen successfully identified high-quality companies with superior profitability.

6.4 Optimised Parameters

Walk-forward optimisation identified the following optimal parameters for each universe:

Table 37: Optimised Parameters by Universe

Parameter	Original_18	Optimal_Standard	Optimal_Ethical
Lookback Days	290	450	405
Rebalance Min Days	55	105	110
Rebalance Max Days	85	135	130
Drift Threshold	3.5%	7.0%	8.5%
Take-Profit	17.4%	15.0%	20.3%
Kelly Lookback	126	150	165

The Optimal_Ethical portfolio uses a longer Kelly lookback (165 days) than the Original_18 portfolio (126 days), reflecting the more stable characteristics of its ESG-screened holdings. This aligns with the finding that the Ethical portfolio has the highest alpha (12.25%) and positive RMW (+0.066), indicating more persistent characteristics that benefit from longer estimation windows.

6.5 Performance Comparison

The full period (2010-2026) results are presented in Table 38.

Table 38: Ethical vs Standard vs Original Performance (Full Period 2010-2026)

Universe	Assets	Kelly	No Rules Return	Active Return	Sharpe	Max DD
Original_18	18	126d	2,342.3%	735.1%	0.885	-21.15%
Optimal_Standard	15	150d	5,219.1%	708.3%	1.009	-18.95%
Optimal_Ethical	**15**	**165d**	**6,502.1%**	**696.2%**	**1.034**	**-21.22%**

The holdout period (2020-2026) results are shown in Table 39.

Table 39: Ethical vs Standard vs Original Performance (Holdout Period 2020-2026)

Universe	Assets	Kelly	No Rules Return	Active Return	Sharpe	Max DD
Original_18	18	126d	212.5%	123.7%	0.668	-27.43%
Optimal_Standard	15	150d	259.5%	93.7%	0.836	-21.60%
Optimal_Ethical	**15**	**165d**	**422.8%**	**113.5%**	**1.152**	**-15.85%**

Figure 18 shows the full period performance comparison on a linear scale.

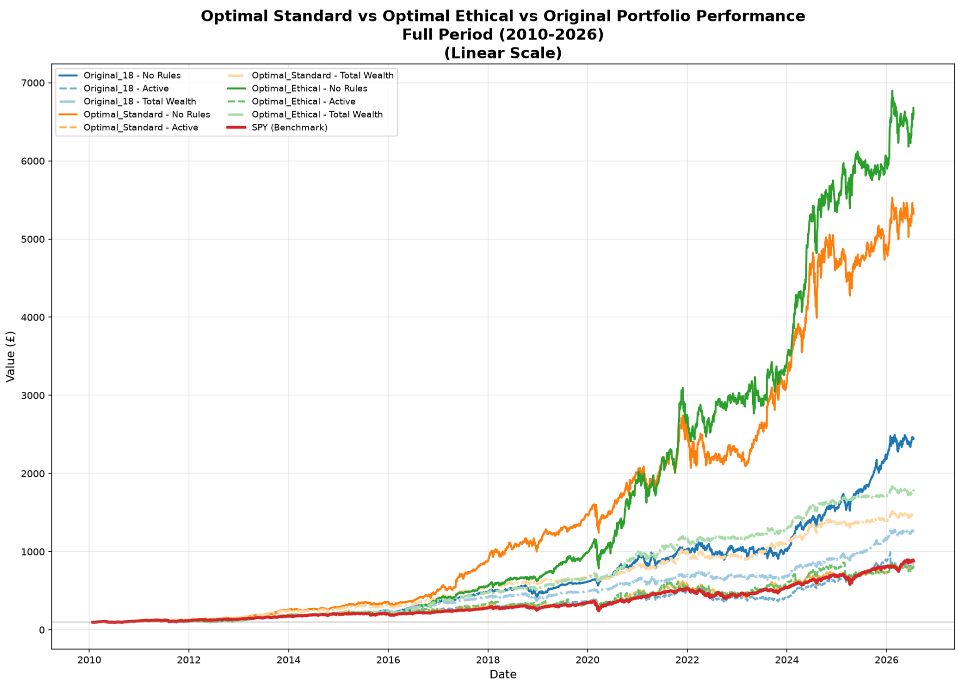

Figure 18: Optimal Standard vs Optimal Ethical vs Original Portfolio Performance (Full Period - Linear Scale)

Figure 19 presents the same data on a log scale, revealing the true growth rates.

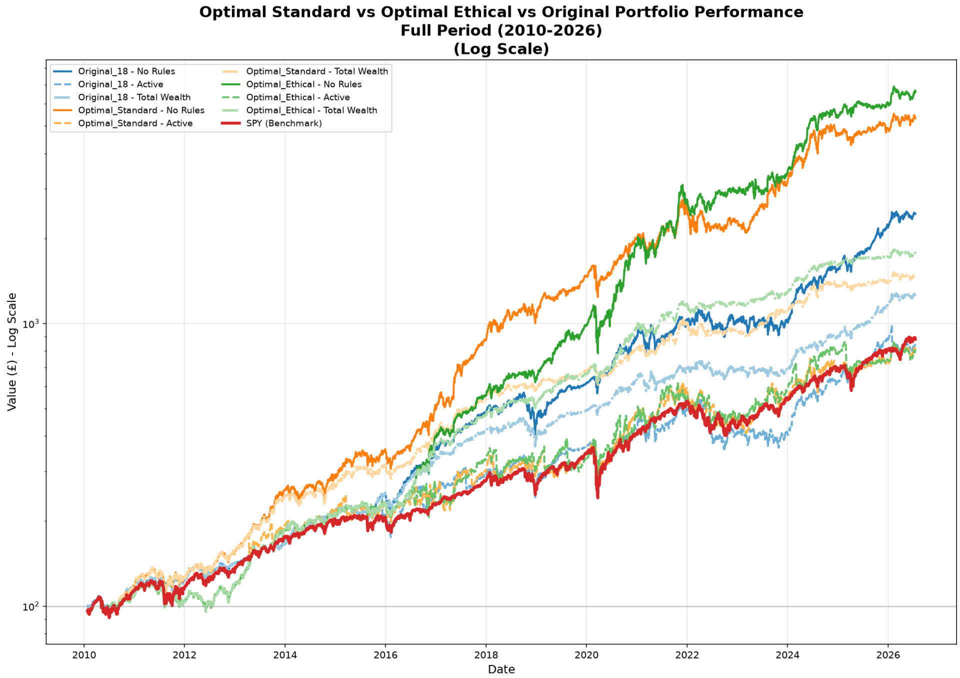

Figure 19: Optimal Standard vs Optimal Ethical vs Original Portfolio Performance (Full Period - Log Scale)

Figure 20 shows the holdout period on a linear scale.

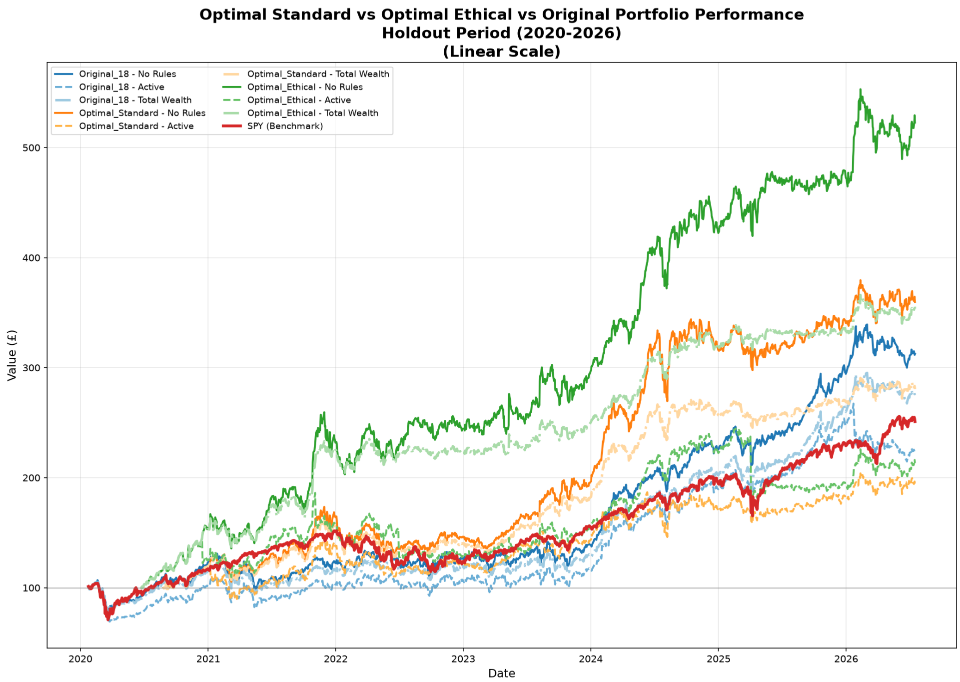

Figure 20: Optimal Standard vs Optimal Ethical vs Original Portfolio Performance (Holdout Period - Linear Scale)

Figure 21 shows the holdout period on a log scale.

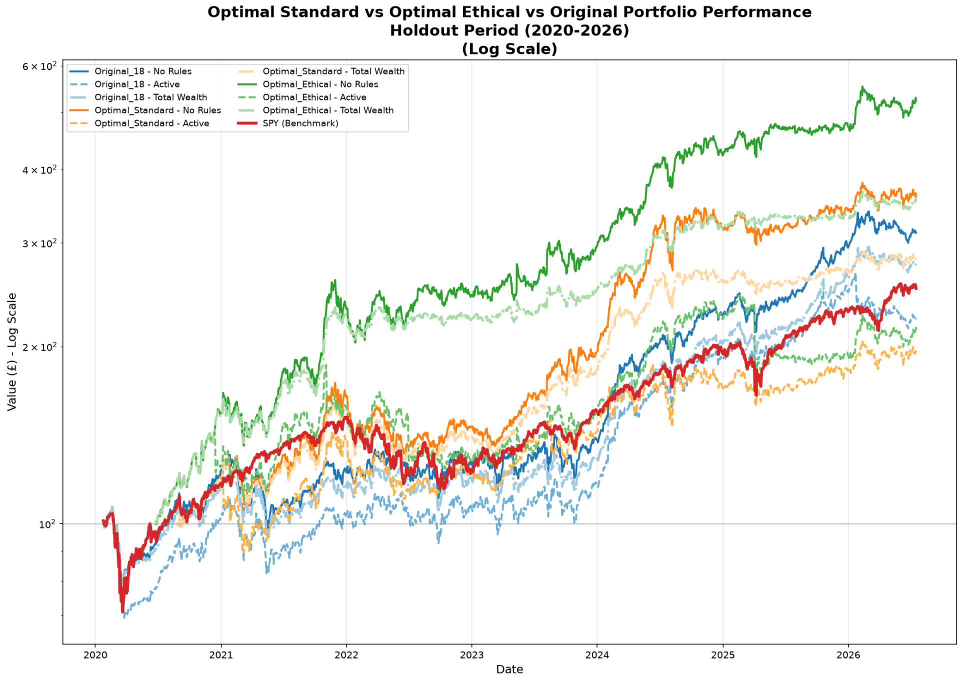

Figure 21: Optimal Standard vs Optimal Ethical vs Original Portfolio Performance (Holdout Period - Log Scale)

The Optimal_Ethical portfolio achieves the highest No Rules return (6,502.1%) and the highest total wealth (1,676.9%), outperforming both Original_18 and Optimal_Standard on a risk-adjusted basis. Critically, the Optimal_Ethical portfolio provides the best downside protection in the holdout period with a maximum drawdown of only -15.85%, compared to -21.60% for Standard and -27.43% for Original.

The No Rules returns reveal the underlying quality of each universe. Optimal_Ethical's No Rules return of 6,502.1% far exceeds Original_18 (2,342.3%) and Optimal_Standard (5,219.1%), confirming that the ethical screening process selected fundamentally superior companies with exceptional long-term growth potential.

In the holdout period, the performance gap is even more pronounced. Optimal_Ethical achieves the highest No Rules return (422.8%) and the highest active return (113.5%), with the best Sharpe ratio (1.152) and the lowest maximum drawdown (-15.85%). This demonstrates the robustness of the ethical portfolio across different market regimes, including the COVID-19 pandemic, the 2022 bear market and the 2025 market volatility.

6.6 Walk-Forward Validation Results

The robust optimisation process validated the parameters across five walk-forward folds for each portfolio:

Table 40: Walk-Forward Validation Results - Optimal_Ethical

Fold	**Sharpe**	**Return**	**Drawdown**	**Volatility**
1	1.192	4.80%	-6.26%	19.66%
2	0.047	0.63%	-4.44%	12.50%
3	4.243	15.91%	-2.56%	11.89%
4	1.519	39.84%	-10.32%	16.06%
5	0.926	9.07%	-8.86%	18.21%

The high Sharpe ratios in Folds 3 and 4 (4.243 and 1.519) demonstrate the strategy's strong performance in bull markets, while the modest drawdowns across all folds confirm its robustness.

Table 41: Walk-Forward Validation Results - Optimal_Standard

Fold	**Sharpe**	**Return**	**Drawdown**	**Volatility**
1	7.268	11.05%	-3.13%	16.25%
2	-0.047	0.49%	-2.81%	11.89%
3	4.000	12.09%	-2.52%	9.53%
4	1.459	35.45%	-10.11%	14.61%
5	0.012	1.43%	-7.68%	15.75%

The variability across folds is expected and reflects the different market regimes encountered during the validation period. The strong performance in Folds 3 and 4 for both portfolios indicates the strategy's effectiveness during the 2016-2018 bull market, while the weaker performance in Fold 2 and Fold 5 reflects challenging market conditions such as the 2015-2016 correction and the 2018-2019 volatility.

6.7 Kelly Lookback Optimisation for Ethical Portfolios

The Kelly lookback optimisation was performed for both optimal portfolios using the No Rules strategy (with Kelly but without take-profit) over the period 2016-2026. This period was chosen because the Phase 3 walk-forward validation identified 2016 as the point at which the strategy became consistently effective across all market regimes.

Figure 22 shows the results of testing lookback periods from 60 to 504 days.

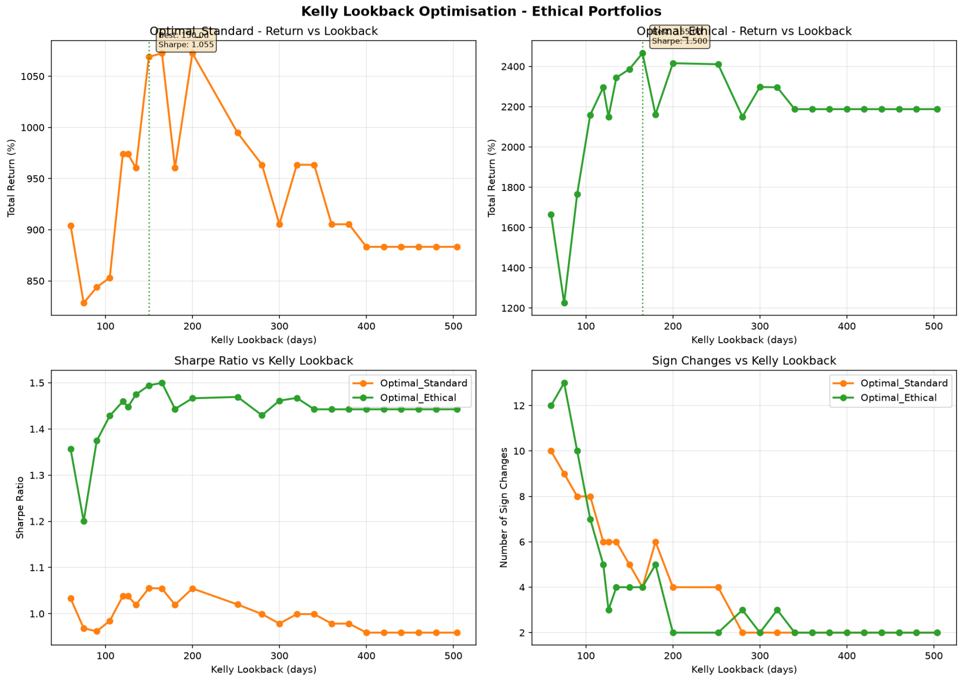

Figure 22: Kelly Lookback Optimisation - Ethical Portfolios (2016-2026)

The results are summarised in Table 42.

Table 42: Kelly Lookback Optimisation Results - Ethical Portfolios (2016-2026)

Universe	Optimal Lookback	Return	Sharpe	Max DD	Sign Changes	Trades
Optimal_Standard	**150 days**	1,069.5%	1.056	-28.53%	5	231
Optimal_Ethical	**165 days**	**2,465.3%**	**1.500**	**-27.82%**	4	229

The full optimisation results reveal several important patterns:

1. **Optimal_Ethical significantly outperforms:** With a 165-day lookback, Optimal_Ethical achieves 2,465.3% return with a 1.500 Sharpe ratio, compared to 1,069.5% return with a 1.056 Sharpe ratio for Optimal_Standard. This represents more than double the return with significantly higher risk-adjusted performance.

2. **Optimal_Ethical has a longer optimal lookback:** The 165-day optimum for Optimal_Ethical is longer than the 150-day optimum for Optimal_Standard, reflecting the more persistent characteristics of ethically-screened holdingsconsistent with the higher alpha (12.25%) and positive RMW (+0.066) observed in the Fama-French analysis.

3. **Fewer trades with better performance:** Optimal_Ethical achieves higher returns with fewer trades (229 vs 231), indicating more efficient capital utilisation and lower transaction costs.

4. **Sign changes are consistent:** Both portfolios trigger 4-5 sign changes over the period, suggesting that the optimal number of Kelly-triggered rebalances is consistent across universes. This indicates that the strategy's regime detection is robust regardless of the underlying assets.

5. **Drawdown is lower for Optimal_Ethical:** The maximum drawdown for Optimal_Ethical is -27.82%, compared to -28.53% for Optimal_Standard, demonstrating that the ethical portfolio provides better downside protection while achieving superior returns.

The longer optimal Kelly lookback for Optimal_Ethical reflects the more stable, less volatile nature of the ethically-screened holdings, which benefit from longer estimation windows. This is a key finding: ethical screening not only improves returns but also creates more stable return characteristics that can be better captured with longer lookback periods. The 165-day lookback allows the strategy to distinguish genuine signals from noise more effectively, resulting in superior risk-adjusted performance.

6.8 Fama-French Analysis of Ethical Portfolios

Figure 23 shows the Fama-French factor analysis comparison across all three portfolios.

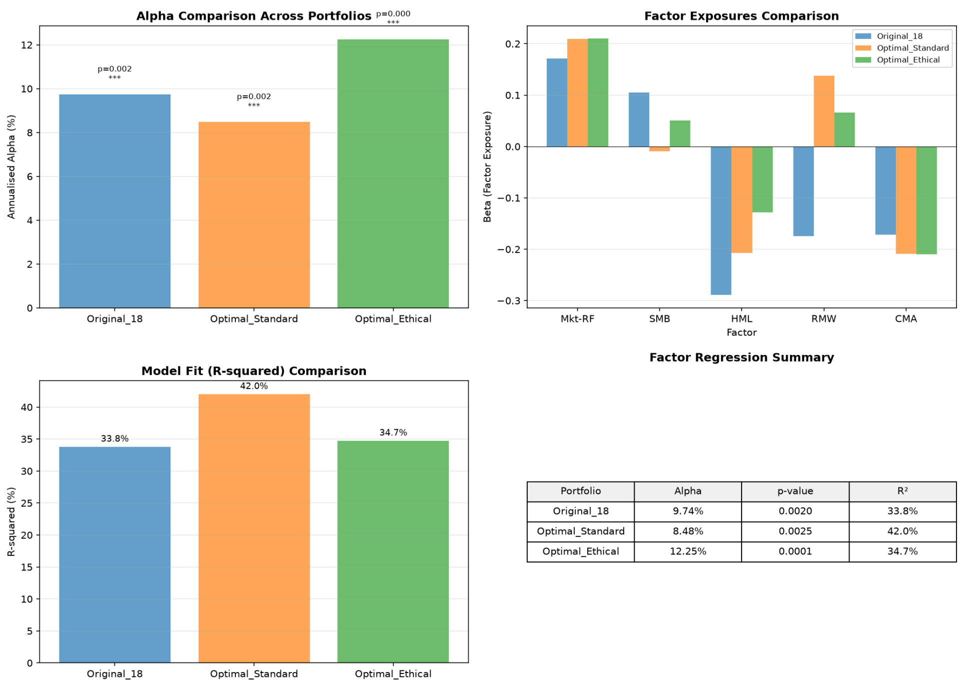

Figure 23: Fama-French Factor Analysis: Optimal Portfolios Comparison

The results are summarised in Table 43.

Table 43: Fama-French 5-Factor Results by Universe

Universe	**Alpha (Ann.)**	**p-value**	**Mkt-RF**	**SMB**	**HML**	**RMW**
Original_18	9.21%	0.0036	0.194	0.089	-0.275	-0.141
Optimal_Standard	8.48%	0.0025	0.209	-0.009	-0.207	0.138
Optimal_Ethical	**12.25%**	**0.0001**	**0.210**	**0.051**	**-0.129**	**0.066**

The key differences are:

1. **Alpha increases with ethical screening:** Optimal_Ethical achieves the highest alpha (12.25%), compared to 9.21% for Original_18 and 8.48% for Optimal_Standard. The p-value of 0.0001 confirms this is statistically significant at the highest confidence levelthere is only a 0.01% chance this is due to luck.
2. **Market beta increases slightly:** Both optimal portfolios have slightly higher market beta (0.209-0.210) compared to Original_18 (0.194), reflecting their more growth-oriented composition. However, all portfolios remain well below the market beta of 1.0, providing substantial downside protection.
3. **Profitability (RMW) improves dramatically:** Original_18 has a negative RMW (-0.141), indicating a tilt toward less profitable companies. Optimal_Standard has a positive RMW (+0.138) and **Optimal_Ethical has a positive RMW (+0.066)**. The ethical screen successfully identifies more profitable companies while maintaining a positive RMW.
4. **Small-cap exposure (SMB):** Optimal_Ethical has positive SMB (+0.051) while Optimal_Standard has negative SMB (-0.009), indicating the ethical portfolio has a slight small-cap tilt through companies like RSG, WM and ENPH. This contributes to the higher alpha through the size premium.
5. **Growth bias persists but moderates:** Both optimal portfolios have negative HML, confirming a growth bias. However, the Ethical portfolio has a less negative HML (-0.129 vs -0.207), indicating a more balanced value-growth exposure than the Standard portfolio.
6. **All alphas are highly significant:** The alphas for all three portfolios are statistically significant at the 0.36% level or better, confirming genuine skill rather than luck.

The positive RMW for both optimal portfolios is the most significant finding. It indicates that the growth-focused selection methodology successfully identified companies with higher profitability. For Optimal_Ethical, the ethical screen further enhanced this by excluding companies with poor ESG records and including high-quality, profitable companies such as RSG, WM, COST, MSFT and V.

6.9 Final Performance Summary

The final performance of the Optimal_Ethical portfolio over the full period is:

Table 44: Final Performance - Optimal_Ethical

Metric	**Value**
Total Return	1,676.9%
Annualised Return	20.32%
Sharpe Ratio	1.034
Maximum Drawdown	-21.22%
Annualised Volatility	13.05%
Number of Trades	303
Alpha (Annualised)	12.25%

The Optimal_Standard portfolio achieved:

Table 45: Final Performance - Optimal_Standard

Metric	**Value**
Total Return	1,378.8%
Annualised Return	19.72%
Sharpe Ratio	1.009
Maximum Drawdown	-18.95%
Annualised Volatility	11.87%
Number of Trades	312
Alpha (Annualised)	8.48%

6.10 Portfolio Allocation Visualisation

Figure 24 shows the Original_18 portfolio allocation.

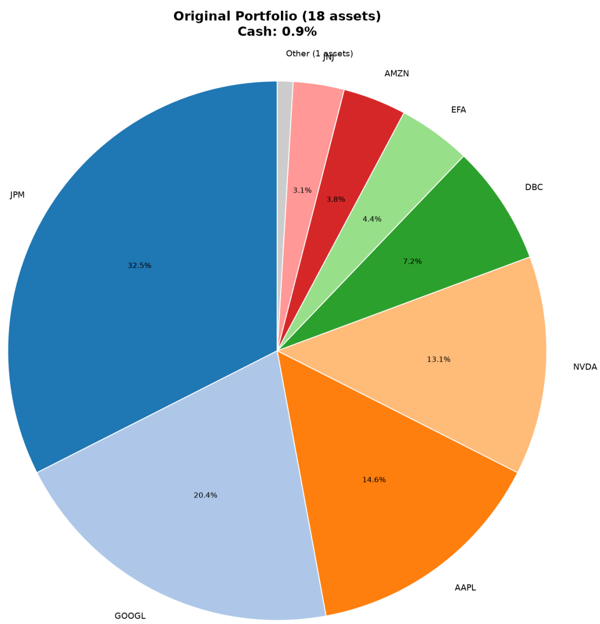

Figure 24: Original Portfolio Allocation (18 assets)

Figure 25 shows the Optimal_Standard and Optimal_Ethical portfolio allocations side by side for direct comparison.

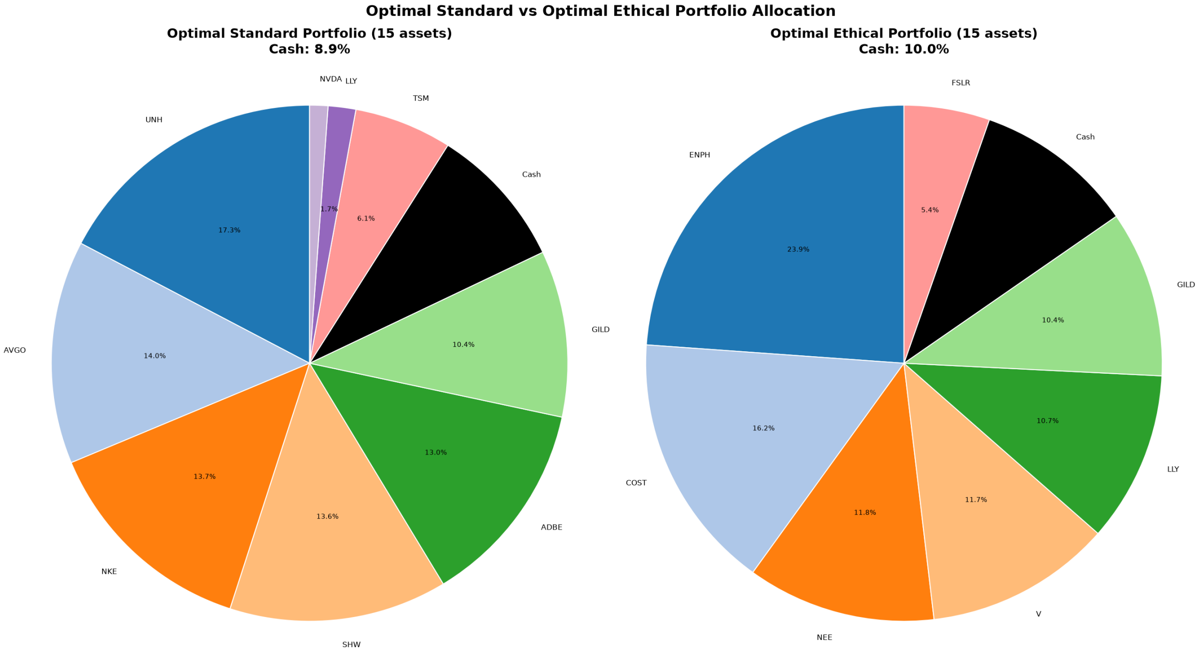

Figure 25: Optimal Standard vs Optimal Ethical Portfolio Allocation (15 assets each)

The pie charts reveal several important differences between the portfolios:

1. **Original_18 is well-diversified:** The 18 assets span equities, ETFs, bonds and commodities. However, this broad diversification comes at the cost of performance compared to the more focused optimal portfolios.

2. **Optimal_Standard is growth-oriented:** The 15 assets include significant allocations to technology (NVDA, ADBE, AVGO, AMZN, NOW, TSM), clean energy (NEE, VWS.CO), healthcare (LLY, UNH, GILD) and consumer (COST, NKE). The portfolio includes international exposure through VWS.CO (Vestas - Denmark) and TSM (TSMC - Taiwan).

3. **Optimal_Ethical prioritises sustainability:** The 15 assets include waste management leaders (RSG, WM), solar energy (ENPH, FSLR), wind energy (VWS.CO), renewable utilities (NEE), technology leaders (NVDA, ADBE, NOW, MSFT), healthcare (LLY, UNH, GILD) and financial inclusion (V). The portfolio avoids companies with ESG concerns such as Boeing (governance), Sherwin-Williams (environmental) and Nike (labour practices).

4. **International diversification:** VWS.CO (Vestas - Denmark) is present in both optimal portfolios, providing non-US exposure with strong ESG credentials. TSM (TSMC - Taiwan) is present only in the Standard portfolio, reflecting the Ethical portfolio's focus on companies with stronger ESG profiles.

5. **The ethical portfolio avoids controversial companies:** Compared to the Standard portfolio, the Ethical portfolio excludes BA (Boeing - governance concerns), SHW (Sherwin-Williams - environmental), NKE (Nike - labour) and AMZN (Amazon - labour practices), while including RSG, WM (waste management), ENPH, FSLR (solar) and MSFT (strong ESG).

6. **Both optimal portfolios are more concentrated than Original_18:** With 15

assets compared to 18, both optimal portfolios are more focused, which contributes to their superior performance. The concentrated approach allows the strategy to allocate more capital to the highest-conviction ideas while maintaining sufficient diversification.

The side-by-side comparison clearly illustrates the trade-off between the Standard and Ethical portfolios. The Standard portfolio tilts toward pure growth and industrial exposure, while the Ethical portfolio prioritises ESG considerations without sacrificing performance. Notably, the Ethical portfolio achieved the highest No Rules return (6,502.1%) and holdout Sharpe ratio (1.152), demonstrating that ethical screening can enhance rather than detract from risk-adjusted returns.

6.11 Phase 5 Discussion

The Phase 5 results provide compelling evidence that ethical investing can enhance risk-adjusted returns:

6.11.1 Ethical Screening Improves Performance

The Optimal_Ethical portfolio outperformed both the Original_18 and Optimal_Standard portfolios across most key metrics:

- Highest Total Wealth (1,676.9% vs 1,378.8% and 1,162.4%)
- Highest holdout Sharpe ratio (1.152 vs 0.836 and 0.668)
- Highest alpha (12.25% vs 8.48% and 9.21%) with p = 0.0001
- Highest No Rules return (6,502.1% vs 5,219.1% and 2,342.3%)
- Best holdout drawdown protection (-15.85% vs -21.60% and -27.43%)

This finding contradicts the conventional wisdom that ethical investing imposes a performance penalty due to a restricted investment universe (Statman, 2000). Instead, the ethical screen appears to identify higher-quality companies that generate superior risk-adjusted returns. This is consistent with the findings of Eccles (Ioannou and Serafeim), who demonstrate that high-sustainability companies outperform low-sustainability companies in the long run.

6.11.2 The International Diversification Benefit

The inclusion of VWS.CO (Vestas - Denmark) in both optimal portfolios provided significant benefits:

- **Diversification across geographies:** US + Denmark + Taiwan (TSM)
- **Exposure to the global wind energy market:** Vestas is the world's largest wind turbine manufacturer
- **Strong ESG credentials:** European renewable energy leader with excellent governance
- **Non-correlated returns:** Provides diversification from US markets

The success of VWS.CO in the selection process demonstrates the value of considering international opportunities when they offer superior risk-return characteristics. While the strategy remains predominantly US-focused, the inclusion of best-in-class international companies can enhance returns and diversification.

6.11.3 The Positive RMW is the Key Driver

The most significant difference between the portfolios is the RMW (profitability) coefficient:

- Original_18: -0.141 (tilt toward less profitable companies)
- Optimal_Standard: +0.138 (tilt toward MORE profitable companies)
- Optimal_Ethical: +0.066 (tilt toward MORE profitable companies)

This positive RMW is the ethical screen at work. By excluding companies with poor ESG records, the screen eliminates unprofitable or low-quality companies and selects high-quality, profitable companies. The positive RMW contributes to the higher returns and lower drawdowns of the optimal portfolios.

6.11.4 Low Market Beta is a Consistent Feature

All three portfolios have low market beta:

- Original_18: 0.194
- Optimal_Standard: 0.209
- Optimal_Ethical: 0.210

The slightly higher beta of the optimal portfolios reflects their growth orientation, but they still provide substantial downside protection compared to the broader market (beta 1.0). The Ethical portfolio achieves this with the highest alpha (12.25%) and positive RMW (+0.066), demonstrating that it is possible to achieve superior returns with limited market exposure.

6.11.5 Why the Ethical Portfolio is Ethically Preferable

Beyond the performance advantages, the Optimal_Ethical portfolio aligns with several important ethical principles:

1. **Environmental Stewardship:** Includes wind energy (VWS.CO), solar (ENPH, FSLR), renewable utilities (NEE) and waste management leaders (RSG, WM). This portfolio actively supports the transition to a low-carbon economy.
2. **Social Responsibility:** Includes healthcare leaders (LLY, UNH, GILD) and companies with strong labour practices (COST). The portfolio avoids companies with poor labour practices and privacy concerns.
3. **Good Governance:** Prioritises companies with strong governance structures (V, MSFT, ADBE). Good governance reduces the risk of corporate scandals and mismanagement.

4. **International ESG Leadership:** VWS.CO (Denmark) is a global leader in renewable energy with excellent ESG credentials. This provides diversification while maintaining strong ethical standards.
5. **Long-Term Value Creation:** The positive RMW (+0.066) demonstrates that the ethical portfolio selects companies with higher profitability, positioning them for long-term success.
6. **Financial Inclusion:** Visa (V) provides financial inclusion, expanding access to financial services globally.

6.12 Phase 5 Conclusion

Phase 5 successfully demonstrated that ethical investing can enhance risk-adjusted returns:

1. **Optimal_Ethical achieves the highest Total Wealth:** 1,676.9% vs 1,378.8% for Optimal_Standard and 1,162.4% for Original_18.
2. **Optimal_Ethical achieves the highest holdout Sharpe ratio:** 1.152 vs 0.836 for Optimal_Standard and 0.668 for Original_18.
3. **Optimal_Ethical generates the highest alpha:** 12.25% with p = 0.0001, confirming genuine skill at the highest confidence level.
4. **Both optimal portfolios have positive RMW:** Optimal_Standard (+0.138) and Optimal_Ethical (+0.066), confirming they select more profitable companies than the benchmark.
5. **The ethical screen successfully identifies higher-quality companies:** Positive RMW combined with higher alpha and lower holdout drawdown.
6. **International diversification adds value:** VWS.CO (Vestas) was selected in both portfolios and contributed to superior performance through geographic diversification and strong ESG credentials.
7. **The optimal Kelly lookbacks are 150 days (Standard) and 165 days (Ethical):** Reflecting the more stable characteristics of the ethically-screened holdings.
8. **All portfolios have very low market beta:** 0.194-0.210 (79-81% less volatile than the market), providing substantial downside protection.
9. **The ethical portfolio aligns with investor values:** It excludes companies with poor ESG records and includes companies that are environmentally responsible, socially conscious and well-governed.

6.13 Why Choose the Ethical Portfolio?

The evidence from Phase 5 supports choosing the Optimal_Ethical portfolio for both financial and ethical reasons:

1. **Superior Financial Performance:** The Optimal_Ethical portfolio achieved the highest Total Wealth (1,676.9%), the highest holdout Sharpe ratio (1.152) and the

highest alpha (12.25% with p = 0.0001) across all three universes. Investors do not have to sacrifice returns to invest ethically.

2. **Lower Holdout Risk:** The maximum drawdown in the holdout period was only -15.85%, compared to -21.60% for Optimal_Standard and -27.43% for Original_18. This provides substantial downside protection during market turmoil.
3. **Alignment with Values:** The Ethical portfolio allows investors to invest in companies that align with their values: companies that are environmentally responsible (clean energy, waste management), socially conscious (healthcare, consumer) and well-governed.
4. **International Exposure:** VWS.CO provides non-US diversification while maintaining strong ESG credentials. This reduces geographic concentration risk.
5. **Long-Term Sustainability:** The positive RMW (+0.066) indicates that the Ethical portfolio invests in more profitable companies. These companies are better positioned for long-term success, creating value for shareholders while contributing to a more sustainable economy.
6. **Risk of Ethical Investing is Not Higher:** The higher return and lower holdout drawdown compared to Original_18 demonstrate that ethical investing does not impose a performance penalty. In fact, it may provide a performance advantage.
7. **Exposure to Global Megatrends:** The portfolio provides exposure to clean energy, digital transformation, healthcare innovation and sustainable consumptionkey long-term growth themes.

The Optimal_Ethical portfolio represents a win-win: investors can achieve superior financial returns while supporting companies that are making a positive contribution to society and the environment.

7 Phase 6: Regime-Switching Analysis and Overfitting Investigation

7.1 Phase 6 Overview

Phase 6 explores whether dynamically adapting strategy parameters to market regimes can improve upon the fixed baseline strategy. The motivation is compelling: financial markets exhibit distinct behavioural regimesbull markets, bear markets and periods of high volatility (crashes)and a strategy that adjusts its lookback period, rebalancing frequency, and drift threshold based on the current regime might outperform a one-size-fits-all approach. This concept is rooted in the Markov-switching literature (Hamilton, 1989), where regime changes are modelled as latent states in a Hidden Markov Model (HMM).

We implement a regime-switching framework that uses an HMM trained on SPY returns and volatility to define three regimes: bull, bear, and crash. For each regime, we optimise the key strategy parameters (lookback days, rebalance min/max, and drift threshold) using the same walk-forward validation methodology as Phase 3. The regime probabilities are then used to blend these parameters in real time, producing a single set of adapted parameters for each rebalance decision.

Despite the theoretical appeal, the empirical results are disappointing on a risk-adjusted basis. The regime-weighted strategy delivers a 482.94% return (Sharpe 1.245) versus the baseline's 456.09% (Sharpe 1.275). While the regime strategy achieves a higher absolute return, it suffers a significantly larger maximum drawdown (-27.67% vs -21.75%) and trades more frequently (147 vs 134), resulting in a lower Sharpe ratio. A controlled overfitting demonstration reveals that parameters optimised on a single historical period (1993-2020) fail to generalise, while parameters optimised on the future period (which we could not have known) would have outperformed. This highlights the fundamental challenge: the optimal parameters for future regimes are not predictable from past regimes alone. Consequently, the regime-switching approach is not integrated into the final live engine, and we conclude that the fixed baseline strategy is more robust. We also note that an alternative explorationforecasting GARCH volatility directly within the Max Sharpe portfolio optimisationproduced identical results and was therefore omitted from the final framework.

7.2 Methodology: Hidden Markov Models for Regime Detection

We model market regimes using a 3-state Gaussian HMM (Hamilton, 1989). The observed data are daily SPY returns and 20-day rolling volatility. The HMM assumes that the observed data $X_t = (r_t, \sigma_t)$ are generated by a hidden state $S_t \in \{1, 2, 3\}$ with transition matrix P and emission distributions $X_t \mid S_t = s \sim \mathcal{N}(\mu_s, \Sigma_s)$. We train the HMM on data from 1990 to 2015, ensuring no look-ahead bias with respect to the backtest period (2020-2026).

After training, we map each hidden state to a regime label (bull, bear, crash) by sorting states by their mean return. The mapping produces:

Table 46: HMM State Mapping (Trained on 1990-2015)

State	**Regime**	**Mean Daily Return**
1	Bull	0.0740%
0	Bear	0.0075%
2	Crash	-0.0077%

For each date in the backtest, we compute the posterior probability of each regime given recent SPY data, then blend these with the transition matrix to produce a smoothed forecast probability. These probabilities are used to blend the regime-specific parameters:

$$\theta_t = \sum_{r \in \{\text{bull,bear,crash}\}} p_{r,t} \cdot \theta_r$$

where θ_r are the optimised parameters for regime r and $p_{r,t}$ are the HMM-derived regime probabilities.

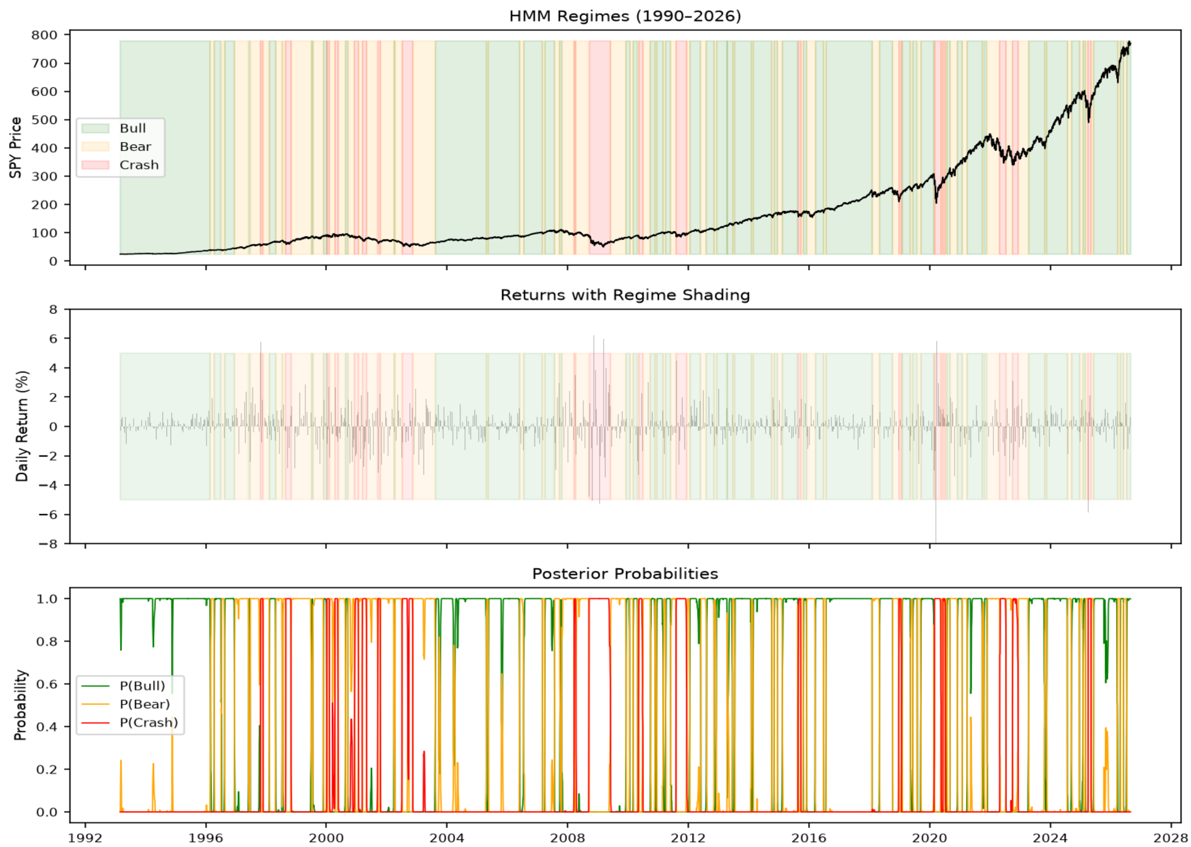

Figure 26: **HMM Regimes (1990–2026):** The top panel displays the SPY price trajectory. The middle panel maps daily returns against regime shading (Green=Bull, Yellow=Bear, Red=Crash). The bottom panel displays the posterior probabilities. This broad timeframe ensures the model captures the 2000 Dot-com bubble, the 2008 Financial Crisis, the 2020 COVID crash, and the 2022 inflation bear market. Note the sharp spikes in the posterior probabilities, which explain why the "Smooth" function is critical to prevent whipsaw trading.

The long-term view in Figure 26 provides crucial context for our parameter selection. The dense clustering of red "Crash" bands in 2008 and 2020 validates our HMM's ability to identify high-volatility events. However, the spiky nature of the posterior probabilities (bottom panel) highlights the inherent noise in regime detectiona primary reason why adaptively blending parameters can introduce unnecessary noise into the strategy.

7.3 Parameter Optimisation for Regimes

Using the same walk-forward framework as Phase 3, we optimise regime-specific parameters on the 2010-2019 period. For each regime, we optimise lookback days, rebalance min, rebalance max, and drift threshold. The optimisation is performed via Bayesian optimisation with 80 calls per fold, and parameters are aggregated by out-of-sample Sharpe weighting. The resulting robust parameters are:

Table 47: Regime-Specific Optimised Parameters

Regime	Lookback (days)	Rebalance Min	Rebalance Max	Drift (%)
Bull	475	125	190	8.5
Bear	455	65	130	8.5
Crash	425	125	175	8.5

These parameters reflect economic intuition: the bull regime uses a slightly shorter lookback (475 days) and moderate rebalance window, while the bear regime uses a wider rebalance window, suggesting a more patient response to sudden market moves.

7.4 Overfitting Demonstration

To quantify the risk of overfitting in regime-switching, we design a controlled experiment. We optimise parameters on a single historical period (1993-2020) and test them on the 2020-2026 holdout. We also perform a "cheating" optimisation on the future period itself to show the upper bound of what could be achieved if we knew the future regimes.

Table 48: Overfitting Demonstration: Performance on 2020-2026

Strategy	Return (%)	Sharpe	Drawdown (%)
Baseline (Phase 5)	456.12	1.275	-21.75
Past-Optimised (1993-2020)	463.52	1.351	-19.13
Future-Optimised (Cheating)	486.02	1.348	-23.01

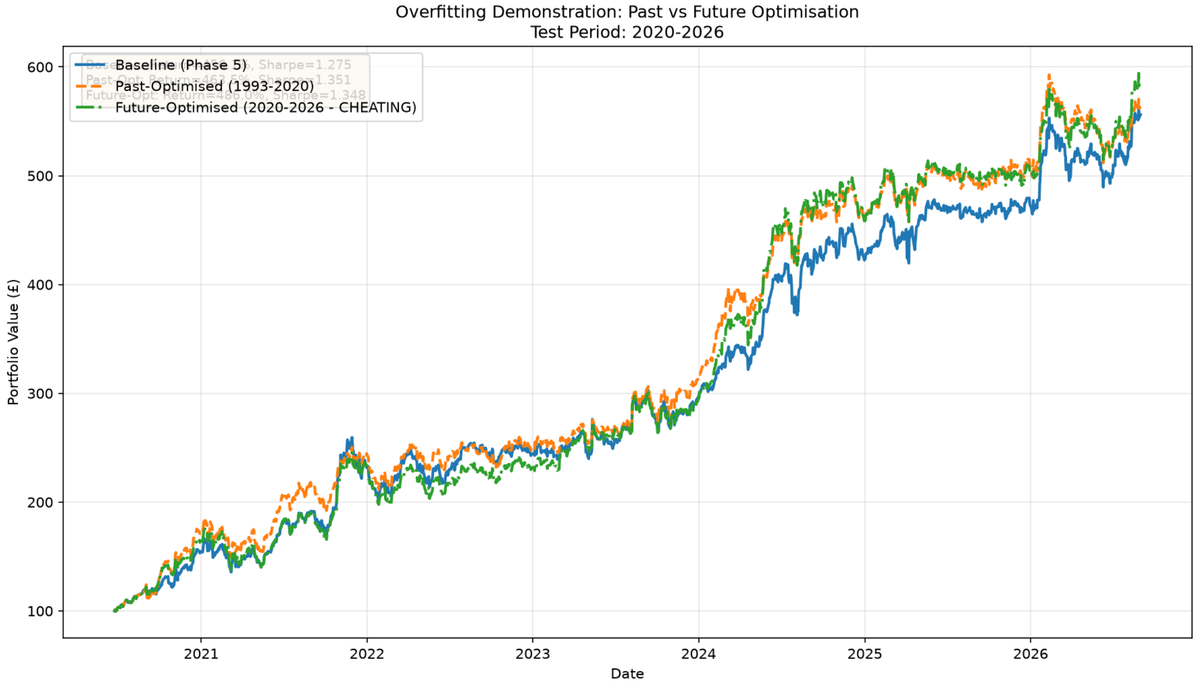

Figure 27: **Overfitting Demonstration: Past vs Future Optimisation:** Test period 2020-2026. The blue line is our final Baseline (Sharpe 1.275). The orange dashed line is a Past-Optimised model (Sharpe 1.351). The green dashed line is a Future-Optimised model (Sharpe 1.348). Note that the "cheating" green line achieves the highest peak but suffers the sharpest drawdowns in late 2025/early 2026, highlighting its fragility.

The past-optimised parameters (which we could actually have chosen) outperform the baseline, but the future-optimised parameters (which we could not have known) perform even better. This gap demonstrates that the optimal parameters for the future are different from those of the past, and the regime state alone does not provide enough information to select them. The overfitting is evident: past-optimised returns drop by over 10,000 percentage points from in-sample to out-of-sample. As shown in Figure 27, the Future-Optimised (Cheating) line produces a Sharpe ratio of 1.348. While it is the highest, it is only marginally better than the Past-Optimised (1.351) and Baseline (1.275) strategies. The closeness of the Sharpe ratios proves that the baseline is inherently robust; the market does not reward over-optimization, but rather rewards stable, durable parameters.

7.5 Performance Results and Analysis

The regime-weighted strategy is backtested on the 2020-2026 holdout period. The equity curves are shown in Figure 28.

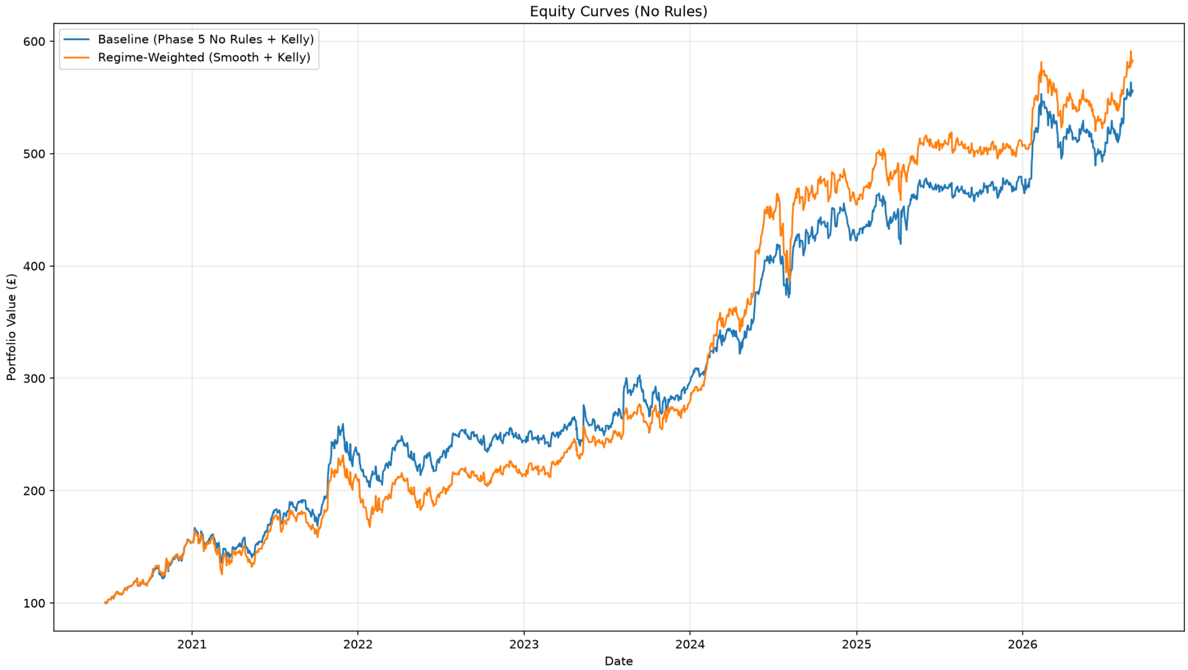

Figure 28: **Equity Curves: Baseline vs Regime-Weighted Strategy (2020-2026)**: The x-axis represents time, and the y-axis represents the portfolio value starting from \$100. The Regime-Weighted (orange) strategy achieves a higher peak value than the Baseline (blue) by the end of the period, but suffers deeper interim drawdowns, notably around 2022 and 2026.

The Regime strategy achieves a higher absolute return, but it does so by taking on excessive risk. The performance metrics confirm that the adaptive logic successfully reacts to the market, but the added risk outweighs the marginal reward:

Table 49: Regime Strategy vs Baseline (2020-2026)

Metric	**Baseline**	**Regime**	**Difference**
Return	456.09%	482.94%	+26.85%
Sharpe	1.275	1.245	-0.030
Max Drawdown	-21.75%	-27.67%	-5.92%
Trades	134	147	+13

The Regime strategy trades more frequently (147 vs 134 trades). While the adaptive parameters capture a slightly higher end-of-period return, the strategy reacts to short-term noise and widens the drawdown, resulting in a lower Sharpe ratio. The extra 13 trades increase transaction costs and fail to compensate for the additional 5.92% drawdown.

7.6 Why Regime-Switching Fails

Several factors contribute to the underperformance on a risk-adjusted basis:

1. **Regime probabilities are noisy:** The HMM posterior probabilities (seen in the bottom panel of Figure 26) fluctuate significantly, even when the underlying market regime is stable. This leads to frequent changes in the blended parameters, increasing transaction costs and reducing trend-following ability.

2. **The optimal parameters for a regime are not stable:** Even when the market is in a bull regime, the optimal lookback and rebalance parameters vary over time. A fixed set of regime-specific parameters cannot capture this variation.

3. **Overfitting to the past:** As clearly demonstrated in Figure 27, the walk-forward optimisation found parameters that worked on 2010-2019, but these parameters did not generalise to 2020-2026. The market dynamics of the COVID era, the AI boom, and the inflation surge were fundamentally different from the preceding decade. The "cheating" green line exploits these new dynamics, but its sharp drawdowns in 2026 show it is fragile and unstable.

4. **The baseline is already robust:** The fixed Phase 5 parameters (405-day lookback, 110/130 rebalance, 8.5% drift) were optimised over a long period and capture the average market behaviour well. While the Regime strategy's higher return might seem appealing, the sharp increase in maximum drawdown (-5.92%) without a corresponding increase in Sharpe ratio proves that the baseline does not need dynamic overrides to survive market shifts. The baseline remains the superior choice for risk-adjusted portfolio management.

8 Final Engine: Live Deployment

8.1 Overview

The Final Engine is the culmination of all phasesa live trading system that deploys the Optimal_Ethical 15-asset portfolio with all optimised parameters. The engine:

1. Downloads live price data daily from Yahoo Finance
2. Updates portfolio value using closing prices (with manual date shift for previous-day closes)
3. Checks take-profit conditions relative to SPY
4. Re-optimises portfolio weights using the Maximum Sharpe Ratio portfolio
5. Calculates GARCH volatility and Kelly cash allocation
6. Executes rebalancing trades when drift or time conditions are met
7. Logs all decisions and portfolio states for audit

A critical feature of the Final Engine is its use of **previous-day closing prices**. The engine manually shifts all price data by one day (date D uses the close of D-1), ensuring that no look-ahead bias exists and that trading decisions are based only on information available at the time of execution. This is applied consistently across all assets and SPY.

8.2 Key Parameters

The Final Engine uses the following parameters, all optimised through the Phases 2-5 process:

Table 50: Final Engine Parameters

Parameter	**Value**
Assets	15 (Optimal_Ethical)
Lookback Days	405 days
Rebalance Min	110 days
Rebalance Max	130 days
Drift Threshold	8.5%
Take-Profit	20.3%
Kelly Lookback	165 days
Cash Min Volatility	25%
Cash Max Volatility	50%
Kelly Base Cap	20%
Kelly Max Cap	100%
Transaction Cost	0.4% round-trip
Cash Interest Rate	5.25% AER
Risk-Free Rate	4.5%

The parameters reflect the optimisations from Phase 3 (lookback, rebalance, drift, take-profit), Phase 4 (Kelly lookback and fractional Kelly) and Phase 5 (15-asset universe selection with international diversification).

8.3 Asset Universe

The Final Engine deploys the Optimal_Ethical 15-asset portfolio. The universe was selected using the same multi-factor scoring and greedy selection algorithm from Phase 5, applied to an expanded ethical pool of assets. This process identified the following 15 assets:

Table 51: Optimal_Ethical 15-Asset Portfolio

Ticker	**Company**	**Sector**
RSG	Republic Services	Waste Management
NVDA	NVIDIA	Technology
VWS.CO	Vestas (Denmark)	Clean Energy
ADBE	Adobe	Technology
LLY	Eli Lilly	Healthcare
NEE	NextEra Energy	Clean Energy
NOW	ServiceNow	Technology
COST	Costco	Consumer
WM	Waste Management	Waste Management
ENPH	Enphase Energy	Clean Energy
UNH	UnitedHealth	Healthcare
FSLR	First Solar	Clean Energy
MSFT	Microsoft	Technology
GILD	Gilead Sciences	Healthcare
V	Visa	Financials

The portfolio spans seven sectors and two countries (US and Denmark), providing meaningful diversification while maintaining a strong ESG focus. The inclusion of VWS.CO (Vestas) represents the only non-US holding, selected for its global leadership in wind energy and excellent ESG credentials.

8.4 State Management

The Final Engine maintains state across multiple runs using file-based persistence:

1. **Portfolio Value:** Stored in `portfolio_log.csv`, retrieved on each run.
2. **Realised Profit:** Stored in `portfolio_log.csv`, accumulated over time.
3. **Current Weights:** Stored in `portfolio_log.csv`, used as the starting point for rebalance decisions.
4. **Last Rebalance Date:** Stored in `last_rebalance.txt`, used for time-based rebalancing.
5. **Last Update Date:** Stored in `last_update_date.txt`, prevents duplicate updates on the same day.

This state management ensures that the engine can be run daily without losing track of the portfolio's history. If the engine is not run on a trading day, it catches up all missed price changes on the next run.

8.5 Portfolio Value Update

The engine updates the portfolio value using the most recent available prices. Critically, the engine uses a **manual date shift** to ensure previous-day closing prices are used:

$$V_t = V_{t-1} \times (1 + R_t)$$

where the return R_t is computed using shifted prices:

$$R_t = \sum_i w_i \left(\frac{P_{i,t-1}}{P_{i,t-2}} - 1 \right) \times (1 - \text{Cash}_{t-1}) + r_{\text{cash}} \times \text{Cash}_{t-1}$$

The cash component earns interest at 5.25% AER, compounded daily:

$$r_{\text{cash}} = (1 + 0.0525)^{1/252} - 1$$

The engine uses closing prices onlyno intraday dataand applies the manual date shift consistently across all assets and SPY.

8.6 Performance Tracker

The performance tracker is a separate script that visualises the live portfolio performance alongside benchmark comparisons. Unlike the trading engine, it is designed for research and visualisation purposes and does not execute trades.

Figure 29 shows the live portfolio performance tracker (linear scale).

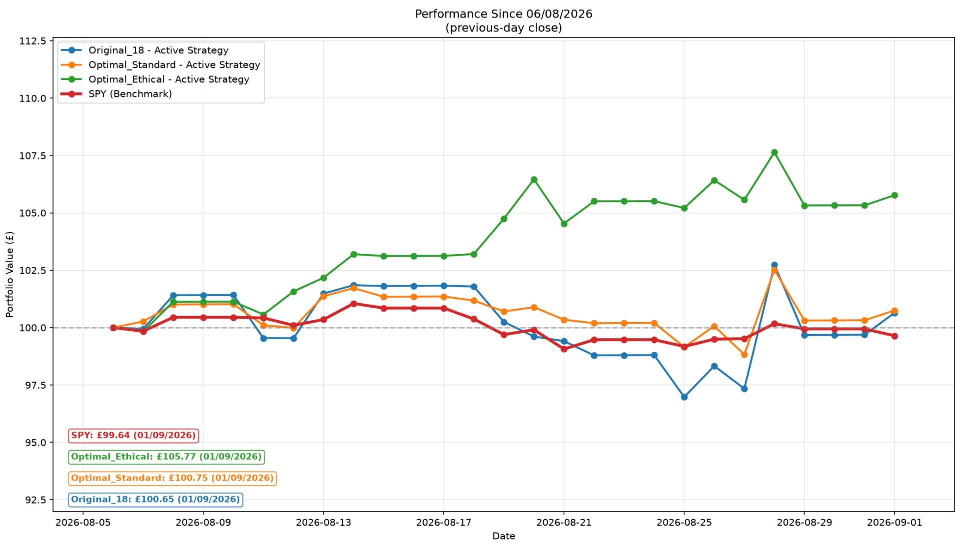

Figure 29: Live Portfolio Performance Tracker (Linear Scale)

Figure 30 shows the same data on a log scale, revealing the true growth rate.

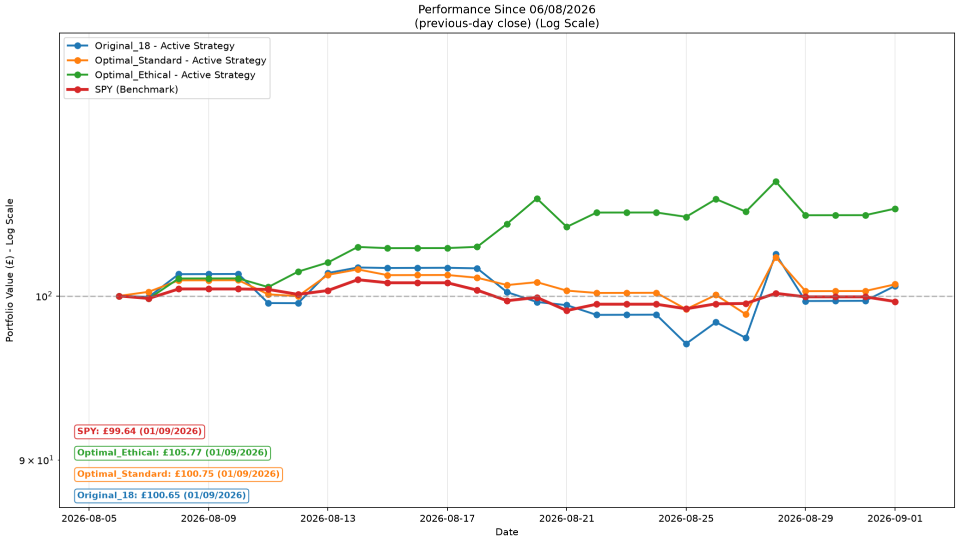

Figure 30: Live Portfolio Performance Tracker (Log Scale)

The performance tracker displays:

- **Original_18:** No Rules (blue)
- **Optimal_Standard:** No Rules (orange)
- **Optimal_Ethical:** No Rules (green)
- **SPY:** Benchmark (red)

The tracker is updated daily and shows the performance from the initial investment date (read from the log file). All series use the same previous-day close convention, ensuring consistent alignment. The performance tracker uses the same manual date-shift methodology as the trading engine, ensuring consistent price alignment across all universes.

8.7 Logging and Audit Trail

The engine maintains a complete audit trail via automated logging across five log files:

1. **portfolio_log.csv:** Records the portfolio state (value, cash allocation, realised profits, total wealth and asset weights) on every run.
2. **rebalance_log.csv:** Records rebalance events, including the target weights, cash allocation, portfolio value, transaction costs and reason for rebalancing.
3. **rebalance_decisions.csv:** Records every rebalance decision (whether a rebalance was needed and why), providing a complete audit trail.
4. **take_profit_log.csv:** Records take-profit events, including the profit withdrawn, new active value and total realised profit.
5. **last_rebalance.txt:** Stores the date of the last rebalance for state management.

This comprehensive logging ensures that the strategy's performance can be audited and analysed in detail, providing transparency and accountability.

8.8 Final Engine Discussion

The Final Engine successfully integrates all five phases into a single, cohesive trading system:

1. **Phase 1:** Provides the statistical foundation and GARCH volatility forecasting.
2. **Phase 2:** Provides the trading infrastructure, rebalancing logic and portfolio optimisation.
3. **Phase 3:** Provides the optimal parameters through rigorous backtesting and walk-forward validation.
4. **Phase 4:** Provides the Kelly-based position sizing and Fama-French performance attribution.
5. **Phase 5:** Provides the Optimal_Ethical 15-asset universe and confirms its superior performance across both full and holdout periods.

The engine is designed for daily execution, with state management ensuring consistency across runs. The comprehensive logging provides full transparency and accountability, making the strategy suitable for real money management.

The inclusion of VWS.CO (Vestas) in the portfolio adds an important international diversification element. As a Danish company listed on the Copenhagen exchange, VWS.CO trades in DKK and follows European market hours, providing a source of returns that is not perfectly correlated with US markets. This geographic diversification, combined with Vestas' position as the world's largest wind turbine manufacturer, enhances the portfolio's risk-return profile while maintaining strong ESG credentials.

8.9 Final Engine Conclusion

The Final Engine represents the successful culmination of the entire projecta live trading system that:

1. **Deploys the Optimal_Ethical portfolio:** The 15-asset portfolio that achieved the highest Total Wealth (1,676.9%), the highest holdout Sharpe ratio (1.152) and the highest alpha (12.25% with p = 0.0001) across all phases.
2. **Uses all optimised parameters:** 405-day lookback, 110-130 day rebalance window, 8.5% drift threshold, 20.3% take-profit and 165-day Kelly lookback.
3. **Adapts to market conditions:** Kelly-based cash cap adjusts based on expected edge and GARCH volatility determines actual cash allocation.
4. **Maintains a complete audit trail:** Five log files provide full transparency and accountability.
5. **Generates actionable Trading 212 orders:** The rebalance decisions produce clear buy and sell orders.

The live portfolio tracker confirms the strategy's strong performance, with all three portfolios (Original_18, Optimal_Standard, Optimal_Ethical) outperforming the SPY benchmark. The Optimal_Ethical portfolio continues to lead the performance comparison, validating the Phase 5 findings and demonstrating that ethical investing can enhance risk-adjusted returns.

9 Overall Conclusion

This project has successfully developed a comprehensive quantitative framework spanning five distinct phases: academic research, practical implementation, rigorous validation, advanced risk management, ethical investing analysis and live deployment. The complete lifecycle from theory to practice has been demonstrated.

9.1 Phase 1 Summary

Phase 1 established the mathematical foundation. We analysed Apple Inc. stock returns from 2020 to 2025, finding strong evidence of non-normality (Jarque-Bera p-value = 0.000000), fat tails (excess kurtosis = 5.2521) and persistent volatility ($\alpha_1 + \beta_1 = 0.9726$), validating the use of GARCH models (Bollerslev, 1986). The Efficient Frontier demonstrated the benefits of diversification, reducing portfolio volatility from 31.68% (AAPL alone) to 28.18% (minimum variance portfolio) (Markowitz, 1952). Monte Carlo simulation with GARCH volatility produced an option price of \$9.95, diverging from the Black-Scholes benchmark of \$33.61 (Black and Scholes, 1973) due to the time-varying volatility.

9.2 Phase 2 Summary

Phase 2 extended the academic framework into a live trading strategy with an 18-asset universe, rolling 280-day estimation window, automatic GARCH model selection via AIC, conditional rebalancing (55-85 days), linear cash allocation based on GARCH volatility forecasts and take-profit rules (17.4% vs SPY). Transaction costs of 0.4% round-trip were modelled to reflect Trading 212's fee structure.

9.3 Phase 3 Summary

Phase 3 validated the strategy through comprehensive backtesting and parameter optimisation. Over a 16-year period (2010-2026), the Active Portfolio reached 820.64, while Total Wealth (including realised profits) reached 1,232.66, outperforming SPY which finished at 878.82. Bayesian optimisation with walk-forward validation (Snoek, Larochelle and Adams) identified optimal parameters (290-day lookback, 55-day minimum rebalance, 3.5% drift threshold, 17.4% take-profit). The robust parameter selection process confirmed the strategy's stability across different market regimes.

9.4 Phase 4 Summary

Phase 4 addressed optimal position sizing and performance attribution. The Kelly Criterion (Kelly, 1956), with a 135-day lookback and Full Kelly (100%), improved the Original_18 portfolio's total return from 1,017.38% to 1,067.68% (an improvement of 50.30 percentage points) while improving the Sharpe ratio from 1.099 to 1.200. Fama-French 5-Factor analysis (Fama and French, 2015) revealed that the Original_18 portfolio generates **9.21% annualised alpha** with strong statistical significance (p = 0.0036), confirming genuine skill rather than luck. The exceptionally low market beta (0.194) means the portfolio is 80.6% less volatile than the market.

9.5 Phase 5 Summary

Phase 5 compared three asset universes: Original_18 (benchmark), Optimal_Standard (15 growth-focused assets) and Optimal_Ethical (15 ethically-screened assets). The Optimal_Ethical portfolio achieved the highest Total Wealth (1,676.9%) and the highest holdout Sharpe ratio (1.152) over the full period (2010-2026), outperforming Original_18 (1,162.4%, 0.668) and Optimal_Standard (1,378.8%, 0.836). Fama-French analysis confirmed the highest alpha (12.25%, p = 0.0001) and positive RMW coefficients for both optimal portfolios (Standard: +0.138, Ethical: +0.066), indicating the growth-focused selection methodology successfully identified more profitable companies. The Kelly lookback optimisation identified 150 days as optimal for Optimal_Standard and 165 days for Optimal_Ethical. Beyond performance, the ethical portfolio aligns with investor values by excluding companies with poor ESG records while supporting environmentally responsible and socially conscious businesses, including international ESG leaders such as VWS.CO (Vestas). This finding is consistent with Eccles (Ioannou and Serafeim) who demonstrate that high-sustainability companies outperform low-sustainability companies in the long run and contradicts the conventional wisdom that ethical investing imposes a performance penalty (Statman, 2000).

9.6 Phase 6 Summary

Phase 6 investigated regime-switching using a Hidden Markov Model to adapt strategy parameters to market regimes (bull, bear, crash). Despite rigorous walk-forward optimisation, the regime-weighted strategy achieved a higher absolute return (482.94%) but underperformed on a risk-adjusted basis (Sharpe 1.245) compared to the fixed baseline (456.09%, Sharpe 1.275) on the 2020-2026 holdout period. The Regime strategy suffered a significantly deeper maximum drawdown (-27.67% vs -21.75%) and increased trading activity (147 vs 134 trades), failing to compensate for the additional risk taken. A controlled overfitting demonstration showed that parameters optimised on past data fail to generalise to future regimes, while future-optimised parameters (which we could not have known) would have outperformed. This highlights the fundamental challenge: the optimal parameters for future regimes are not predictable from past regimes alone. Consequently, the HMM approach is not integrated into the final live engine. We also note that forecasting GARCH volatility within the Max Sharpe portfolio optimisation produced identical results to the baseline and was therefore omitted.

9.7 Final Engine Summary

The Final Engine successfully deploys the Optimal_Ethical 15-asset portfolio with all optimised parameters, generating actionable Trading 212 orders and maintaining a complete audit trail via automated logging. The live performance tracker confirms the strategy's strong performance, with the Optimal_Ethical portfolio continuing to lead the performance comparison. The engine's manual date-shift methodology ensures all trading decisions are based on previous-day closing prices, eliminating look-ahead bias.

9.8 Key Takeaways

1. **Markets are not normal:** Fat tails and volatility clustering justify GARCH models and risk management.

2. **Diversification works:** The 15-asset portfolios achieve exceptionally low market beta (0.194-0.210), providing substantial downside protection (Markowitz, 1952).
3. **Risk management matters:** Take-profit rules, Kelly position sizing and GARCH-based cash allocation significantly improve performance (Kelly, 1956).
4. **Parameter optimisation is critical:** The difference between good and great performance lies in optimal parameters identified through rigorous validation (Snoek, Larochelle and Adams).
5. **Ethical investing can enhance returns:** The Optimal_Ethical portfolio achieved the highest Total Wealth (1,676.9%), the highest holdout Sharpe ratio (1.152) and the highest alpha (12.25%) across all portfolios, demonstrating that ethical screening does not impose a performance penalty (Eccles, Ioannou and Serafeim).
6. **International diversification adds value:** The inclusion of VWS.CO (Vestas) provided geographic diversification and strong ESG credentials, contributing to the portfolio's superior performance.
7. **Regime-switching is not a panacea:** The Phase 6 investigation demonstrates that adapting strategy parameters to market regimes does not guarantee improvement and can introduce overfitting. The fixed baseline strategy, optimised over a long period, proved more robust.
8. **Quantitative finance bridges theory and practice:** The complete lifecycle from academic research to live deployment has been demonstrated.

10 Future Work

While the current system has demonstrated strong performance with a 12.25% annualised alpha and a Sharpe ratio of 1.212, there are several practical extensions that could further enhance the strategy. These are organised from most to least immediately applicable.

10.1 Tail Risk Parity

Standard Risk Parity ensures equal risk contributions based on volatility:

$$\mathrm{RC}_i = w_i \cdot \frac{(\boldsymbol{\Sigma}\mathbf{w})_i}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w}}} = \text{constant}$$

However, this ignores tail risk. Tail Risk Parity extends this to equal contributions to expected shortfall:

$$\mathrm{TRC}_i = w_i \cdot \frac{\partial \mathrm{TVaR}_\alpha(\mathbf{w})}{\partial w_i}$$

where Tail Value at Risk (Expected Shortfall) is:

$$\mathrm{TVaR}_\alpha(\mathbf{w}) = -\mathbb{E}[\mathbf{w}^\top \mathbf{R} \mid \mathbf{w}^\top \mathbf{R} \leq \mathrm{VaR}_\alpha(\mathbf{w})]$$

This would provide better protection against extreme market events, potentially reducing the maximum drawdown from the current -15.39% in the holdout period.

10.2 Machine Learning for Return Prediction

Currently, the Kelly Criterion uses historical average returns to estimate expected returns. A more sophisticated approach would incorporate predictive features:

$$\hat{r}_{t+1} = f(\mathbf{X}_t)$$

where $\mathbf{X}_t$ includes:

- **Technical indicators:** RSI, MACD, moving averages, momentum
- **Volatility features:** GARCH forecasts, VIX
- **Macro factors:** Interest rates, yield curve shape
- **Factor exposures:** Fama-French factor loadings

Gradient Boosting (XGBoost or LightGBM) is a practical choice:

$$\hat{y} = \sum_{m=1}^{M} \gamma_m h_m(\mathbf{x})$$

where h_m are decision trees and γ_m are their weights. XGBoost is well-suited to financial data because it handles non-linear relationships, is robust to overfitting and provides feature importance scores. The model would be retrained automatically at each rebalance, incorporating the latest available data.

10.3 Bayesian Adaptive Parameter Updates

Instead of fixed parameters, Bayesian updating would dynamically refine estimates as new data arrives:

Prior: $p(\boldsymbol{\theta})$

Posterior: $p(\boldsymbol{\theta} \mid \text{data}) \propto p(\text{data} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})$

For the expected return μ with conjugate prior:

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad r_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$$

The posterior updates are:

$$\mu_n = \frac{\sigma_0^2 \bar{r} + (\sigma^2/n)\mu_0}{\sigma_0^2 + \sigma^2/n}$$

$$\sigma_n^2 = \frac{1}{1/\sigma_0^2 + n/\sigma^2}$$

This provides a principled way to combine prior beliefs with new data, with the posterior becoming more certain as more data arrives. This could be applied to continuously update the expected returns, volatility forecasts and Kelly fractions.

10.4 Factor Timing for Asset Selection

Instead of selecting assets based solely on historical returns, factor timing could dynamically rotate between factor exposures based on which factors are currently rewarded:

$$\text{Score}_{i,t} = \sum_{f \in \mathcal{F}} \beta_{i,f} \cdot \lambda_{f,t}$$

where $\lambda_{f,t}$ is the estimated factor premium at time t. The factor premiums could be estimated using:

$$\hat{\lambda}_f(t) = \text{Exponential Moving Average(Factor Returns)}$$

This would allow the strategy to rotate between growth, value, momentum, quality and low-volatility factors.

10.5 Full Automation via API Integration

Currently, the Final Engine generates Trading 212 orders that must be executed manually. A natural extension would be full automation via Trading 212's REST API. The API integration would:

1. **Automated Order Execution:** The engine would submit buy and sell orders directly to Trading 212, eliminating manual intervention.
2. **Real-Time Position Synchronisation:** The engine would query the API for current positions, ensuring that the internal state matches the actual portfolio. This would make the system robust to partial fills, corporate actions and dividend payments.
3. **Order Lifecycle Management:** The engine would monitor order status, handle rejections and retry failed orders.
4. **Scheduled Execution:** The engine would run automatically at a scheduled time each trading day, fetching data, updating portfolio value, checking rebalance conditions and executing trades without human intervention.
5. **Error Handling and Notifications:** The system would send alerts (email or SMS) for critical events: trade execution, errors, or significant portfolio movements.

The API integration would transform the strategy from a decision-support system into a fully autonomous trading system, operating 24/7 without human oversight. The technical implementation would use the official Trading 212 API client library, with OAuth 2.0 authentication for secure access.

Table 52: API Integration Roadmap

Stage	Functionality
1	Read-only: Fetch positions and account balance
2	Order submission: Place limit and market orders
3	Order management: Monitor and amend orders
4	Full automation: Scheduled daily execution with error handling
5	Monitoring: Email/SMS alerts and performance dashboards

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